QMX+ Panadapter — User Guide (v1.20.2)

By Steffen Lav (OZ1LAV). Generated from the project README — for the live page, issues, and releases see https://github.com/SteffenLav/qmx-panadapter

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The full documentation also lives online at **tab5.lav.dk** — the user guide, quick-start, and reference as searchable, cross-linked web pages, kept in sync with every release.

Features

Everything below is in the firmware **today**. Nothing here needs a PC, and only the items marked *(needs WiFi)* need a network.

See the band

Real-time panadapter — Spectrum and waterfall across a 48 kHz window centred on the QMX VFO, at 30 Hz, with 12 kHz IF-offset compensation. Tap or drag to tune (mode-aware snapping), pinch to zoom ×1–×24 with a true zoom-FFT, one-finger drag to pan and retune. Adaptive per-bin noise floor, flat-spectrum mode, adjustable waterfall black level and contrast, selectable FFT window, and a graphical S-meter calibrated in dBm. A colour band-plan strip tracks the VFO (IARU region 1/2/3, auto-selected from your grid) and can be dragged to scrub across the band.

A spectrum that holds still — From ×2 zoom up, the spectrum and waterfall stay where they are and the VFO marker moves across them, so a signal stays put on screen while you tune towards it and the waterfall's history stays lined up under the frequency it belongs to. It really does hold still: the picture never slides along with the dial. Tune far enough and your filter passband reaches the screen edge, keeps going, and once about half of it has slid off the display pages in one step, carrying part of the old screen across. The trigger is your filter passband rather than a percentage of the view, so it behaves the same in every mode. Tapping a spot moves only the marker when the spot is already on screen, and re-frames when it is not. On by default, and switchable. At ×1 the view stays centred on the dial, because the view is already the whole 48 kHz the radio sends. The band above dial+12 kHz is hatched on both screens — the radio genuinely cannot hear it, and drawing something there would be inventing signal.

Hear the band

Audio, and binaural CW *(beta)* — The I/Q the radio already sends is turned into sound on the device. The **panoramic CW split** places two stations on different frequencies to the left and right of the stereo image instead of letting them land on top of each other, with **pan width**, **pan blend** and **overlap** adjustable live while you listen. This plays to something your ears do far better than any decoder: separating two sources by where they are coming from. A single-channel reader has to pull both signals out of one waveform; binaural CW hands the job to the part of your hearing that evolved for it, and two stations calling together stop being a jumble. Best at **300 Hz bandwidth and up** — **below 200 Hz there is little separation to be had**, because the signals then share too much of the audio passband for any amount of panning to pull them apart. Audio competes for resources, so open **Audio Settings** from the settings drawer, where it is switched on and off. This is a first implementation and is **beta**, with the same features planned for the web interface — sending only the I/Q to the PC and doing the work there. See Audio.

Receive from the 3.5 mm jack *(beta — Tony, @tony1tf, #17)* — The panadapter can take its signal from the Tab5's own headphone jack instead of the radio's USB audio. Tony asked for an audio input for digital modes and then ran a test build on his own Tab5 to find where the jack lands: the mic ring arrives on the audio chip's third channel, which is the fact the feature is built on. The jack is one real audio channel, not I/Q, so the second channel is synthesised by a 255-tap filter that builds the quadrature pair the decoders need. There is a level meter, a clip indicator and the jack wiring drawn on screen. ⚠ **No signal has been decoded through it yet** — the path runs and the levels read correctly, and that is the whole of what is proven. ⛔ In this mode the USB radio connection is off by design, so CAT, the S-meter and the band/mode display go quiet. Changing the source needs a restart, which the Tab5 performs itself on **Save**. See Audio.

Decoded CW ** — In CW or CW-R, a line along the bottom of the panadapter shows the Morse the radio is decoding, with an estimate of the sending speed. The **QMX does the decoding itself** and hands the text over CAT, so it costs the panadapter no processing at all and does not touch the spectrum or FT8. Works on QMX firmware 1.03 and later, on the Tab5 and in the browser, and switches off in the settings drawer under Radio *(suggested by Uwe DL8UG)*. With a card in the slot it is also written to `qmx-panadapter/cw-decode.txt` with timestamps, so a half-caught callsign can be resolved afterwards *(Michael K Johnson KZ4LY)*.

Tune and control the radio

QMX CAT control — Live frequency, mode (USB/LSB/CW/DiGi, plus AM on QMX firmware 1.04+), SSB filter bandwidth, CW passband, passband overlay, TX power and SWR readout, QMX volume in decibels matching the radio's own display, and an Antenna Tune button on 1.04+. Round-trip CAT latency under 50 ms. Band presets with per-band frequency recall, and 32 memory channels in a 4×8 grid holding any frequency and mode.

RIT — Pull in a caller answering off your frequency without moving transmit: arm it, tap the caller, and a marker shows where you are listening while the gold line goes on meaning the dial. A long press parks the offset and restores it.

Radio menus — The QMX's own 80×24 menu system, on the Tab5 and in the browser, over the radio's *second* USB serial port so the panadapter keeps decoding while you are in there. For a QMX+ with no control panel it is the only way in. Leaving hands the radio back properly and restores your frequency and mode.

CW keyer speed *(Ralph Hellmig)* — A slider in the settings drawer directly above IF calibration, and the same control on the web page. It sets the radio's keyer speed over CAT and then shows what the radio actually took, because the radio is the authority here and does not store the value itself. ⛔ **A speed of 0 means a straight key**, not "no speed set". See Radio Menus.

Touch, keyboard, mouse — Edge-swipe navigation with breathing grip handles, an on-screen keyboard, and optional support for the M5Stack Tab5 70-key snap-on keyboard — which drives the radio's own menus, works in every window with buttons, scrolls the settings drawer and the manual, and carries Ctrl shortcuts you can reassign yourself. It can be attached or reattached at any time, not only at boot. A **Bluetooth keyboard** does the same: every text box, Enter and Esc in every window, Tab between fields and the arrow keys, with the on-screen keyboard stepping aside while one is connected. A keyboard and a mouse can be connected at once, in either order. ⚠ Bluetooth keyboards are **US layout only** — a Bluetooth keyboard reports key *positions* rather than characters, so letters, digits, Enter and the arrows are right on any keyboard while punctuation on a non-US layout will not be. Plus USB mouse support. *(The mouse and the QMX cannot share the single USB host port — the ESP32-P4's USB stack lacks the Transaction Translator a hub would need — so the mouse is for setup, log review and reading the manual with the radio unplugged.)*

Bluetooth mouse — A pointer that works *while the QMX stays plugged in*, which a USB mouse cannot do on this hardware. Pair once; it reconnects by itself. Move, click, and a wheel that scrolls whatever is under the pointer — **and over the spectrum or waterfall, that wheel tunes**: 10 Hz a click in CW and the digital modes, so a signal can be zero-beaten by ear, and 100 Hz in SSB *(Roy KI0ER, John Dusek)*. It stops at the band edges, and anything on top of the panadapter — a modal, the settings drawer — takes the wheel instead, so the dial never moves under something you are reading. **The mouse must be Bluetooth 4.0 or later (Bluetooth Low Energy).** The Tab5's Bluetooth comes from a co-processor with no Bluetooth Classic radio, so an older Classic mouse cannot work and no firmware change can help — it never appears to the Tab5 at all. Most mice sold since about 2014 qualify; a dual-mode mouse works on its Low Energy channel.

Work the digital modes

FT8 and FT4, receive — Continuous on-device decoding in the same view as the panadapter: FT8 (15 s slots) and FT4 (7.5 s slots). **JS8** is a third sub-mode of this page - see below. The decode list shows callsign, country, signal report, slot-timing offset (DT), audio tone (HZ), distance and bearing, with the station you are working held at the top. Include/exclude filters match anything in the message text, worked-before stations can be excluded band-by-band, and a pileup tracker collects everyone calling you.

FT8 and FT4, transmit — Tap a station and the correct *next* message is sent (WSJT-X double-click style); or call CQ from one of three editable presets, with an optional stop-after-N-calls limit. Full automatic exchange through to `73` and an ADIF entry, a resend if your partner never heard your final, a polite hold for a station already working someone else, grey-listing for stations that never answer, and an optional unattended auto-answer robot. Your TX tone is a permanent on-screen button with a live occupancy strip, drag-to-pick tone selection, **TX Hold**, and a cycling **TXCQ ANY / EVEN / ODD** time-window choice. ARRL Field Day exchange mode included.

JS8 — a keyboard mode, on the device *(beta)* — JS8 carries **free text** on FT8's modulation, so stations hold short conversations rather than trading a fixed exchange. All of it runs on the Tab5: sync, LDPC error correction and the CRC check on receive, the encoder and transmitter on send. It is a sub-mode of the FT8/FT4 page - pick **JS8** in the preset picker - and adds a **Conversation** tab beside **Stations**, so you can read what has been said as well as who is on the band. To send, tap **Free text** under Call CQ, or use the composer on the web page; both report how many frames the message takes and how many seconds it holds the frequency before the radio is keyed. Directed messages and heartbeats are shown with the sender's callsign and grid. Full detail in JS8.

WSPR — A third page, reached by the same swipe: **Panadapter** → **FT8/FT4/JS8** → **WSPR**. WSPR is a propagation beacon rather than a contact mode — a very slow, very weak signal carrying only your callsign, grid and power, which stations worldwide report hearing, so over an evening you get a picture of where your antenna and your band actually reach at levels where nothing else would be heard at all. Nobody replies and nothing goes in your log. The page lists what was heard each two-minute cycle with the band, distance and bearing, and the furthest of the session. Receiving is the default; transmitting is opt-in, refuses to key without your callsign and grid, and has a transmit schedule and optional band hopping. **Calibrate Power** sweeps *Max. PA voltage* on a dummy load and measures real RF output, and **Never exceed PA voltage** keeps that sweep away from a limit you set yourself at each step, per band. **Declared power** then only offers the standard WSPR dBm steps your calibration actually reaches — each one labelled with its real measured wattage, not a textbook figure — so what you declare (published worldwide with every spot) matches what actually goes out. A WSPR transmission keys the radio for about 110 seconds in every 120, so picking a low declared level for long beacon runs is how you protect the finals now, rather than a fixed voltage cut applied underneath you. **Output power**, a separate slider for every other mode (FT8, CW, SSB, ...), shares the same calibration but stays out of WSPR's way entirely — Declared power always owns the radio there. What you hear can be published to wsprnet.org *(needs WiFi)*.

Who is out there, and who hears you

Live spots on the spectrum *(needs WiFi)* — **POTA** park activations, and optionally **RBN** CW skimmer spots and **DX cluster** spots, drawn onto the trace at the frequency the station is actually using. The cluster is where phone activity comes from — skimmers are machines, and no machine recognises a callsign spoken into a microphone. One station is one entry, however many sources report it. Grey means you have already worked them on that band. Press and drag to pick one, lift to tune it *with the right mode*. Spots fade with age and are gone after 30 minutes; corner counts take you to the ones just off-screen.

Knowing you are getting out *(needs WiFi)* — **Who is hearing me** asks PSK Reporter which receivers have copied *your* call, with distance, bearing and the report they gave you. **SWR protection** cuts a transmission short and latches the transmitter off above your chosen limit, because an FT8 burst is thirteen seconds of key-down into whatever the antenna is doing.

PSK Reporter spotting *(needs WiFi)* — The stations you decode are reported to PSK Reporter the way WSJT-X does, so you appear on the map as a monitoring station. **On by default**; sends your call and grid plus each decoded station's call, grid, frequency, report and mode, batched at most once every five minutes — over the internet only, **never on the air**. One checkbox turns it off, and it is inert until your callsign and grid are set, and in simulation mode.

Log it and confirm it

QSO logging and upload — Every contact is written to an ADIF log on the device, with timestamp, callsign, grid, frequency, band, mode, both signal reports and distance (never a report that was not actually exchanged). Read the log on the Tab5 or in the browser, delete single records or all of them, and upload *(needs WiFi)* to **QRZ Logbook**, **eQSL**, **ARRL LoTW** and **your own Cloudlog or Wavelog** — LoTW QSOs are signed on the device itself with your own callsign certificate. A signal report or a park/summit reference can be corrected after the fact from the browser's log table. The browser's **ADIF restore** merges a previously downloaded (or any logger's) ADIF file back in — contacts already logged are skipped — so an erase-and-reinstall does not cost you your worked-station history. Or **restore straight from the microSD card**, on the Tab5 itself with no computer at all. **Search the log** on either screen — by callsign, country, band, mode, date or park — and in the browser **tick rows and export just those** as their own ADIF file.

Activation logging — Start a **POTA** or **SOTA** activation and every contact is stamped with the reference as it is logged, counted against the threshold, with chases tagged from spots already seen and a single-reference ADIF export for uploading.

Run it from somewhere else

Web UI *(needs WiFi)* — The whole panadapter in any browser on the LAN: live spectrum and waterfall at ~10 fps, click or drag to tune, mouse-wheel pan and tune, band/mode/bandwidth/zoom control, a graphical S-meter and the whole-band plan strip. In FT8/FT4 mode it shows a live TX status banner and a **Call CQ** button instead of the stream. Also: the QSO log as a sortable table, config download/upload as an editable text file, a microSD file browser, screenshots, and the diagnostic log.

Know what time it is

A GPS of your own, and a sky view — Two receivers work, and both are confirmed on hardware: an **M5Stack Unit GPS v1.1** on PORT.A and a **Module GPS v2.1** on the 30-pin M-Bus. Either gives you UTC **and the full date** with no internet and no radio — every other offline source on this device gives a time of day only, so the day itself was always a guess. Your station grid fills itself from the fix, and never overwrites a grid you typed by hand. **GPS Stats** is a page of its own under Radio in the drawer and on the web: the satellites in use drawn on an elevation-ring sky view, with the fix, the satellite count and which receiver is answering. The clock now also says whether the time came from a GPS on the Tab5 or from the radio. See Time sync.

Time and date, with or without a network — SNTP when WiFi is up *(needs WiFi)*, the Tab5's own supercap-backed RTC across power-off, the QMX's clock as an offline fallback, GPS detected and phase-locked automatically if your QMX has one, and a manual set-and-sync panel. FT8/FT4 timing also self-corrects from the decoded band consensus when offline.

Keep it working

Updating from the device *(needs WiFi)* — A new release is fetched quietly in the background and offered once, in a window in the middle of the screen, with **Restart now** or **Later**. Nothing is ever installed without you asking, and the automatic download can be switched off if you are on a metered connection. The spectrum, waterfall and FT8 decoding keep running throughout.

Built-in user manual — This entire guide is compiled into the firmware, so it opens instantly with no WiFi, no card and no download, and can never describe a different version than the one you are running. It opens at the chapter for the screen you are on, warning banners are tappable, and a **Need guidance?** panel lets you pick your symptom in plain words. See Getting help.

microSD station backup — Insert a card (a plain FAT32 32 GB card is ideal) and your whole station is mirrored to `/qmx-panadapter/`: the ADIF log, a full config export, your LoTW certificate and key, the diagnostic log and a self-describing `README.txt`. Continuous with WiFi off; one complete backup per start-up with WiFi on, because the card and the WiFi co-processor share a bus. The **SD** dot is green while writes are reaching the card and yellow when nothing has landed lately. It restores too, which is the half that was missing: **Restore from SD** in the Tab5's own log window, or from the browser. And because the card mirrors the *present*, the copy from just before the log last got **smaller** is kept beside it as `qso.prev.adi` — so a deletion you notice two restarts later is still recoverable. **After inserting the SD card your Tab5 needs a restart** — the card is claimed during a short window early in boot, so one pushed in while it is running is not used until then. *(The card holds credentials — WiFi password, QRZ/eQSL logins, LoTW private key — so keep it physically secure.)*

Diagnostics — An always-on diagnostic log, nothing to enable: 5 MB in RAM, a rolling copy in flash that survives a power cut, and a full mirror to microSD if a card is in. Downloadable from the browser or over USB serial. **A crash survives the reboot** — if the Tab5 ever restarts unexpectedly, the next boot records what happened, in which part of the firmware and how far into the session, so a diagnostic download is enough to answer it.

Practice mode — A simulator with phantom stations that call CQ and reply through the real encode/decode pipeline, so you can rehearse a full QSO with **no radio connected** — and with a hard interlock that never keys a QMX even if one is attached.

Settings you decide the shape of — Every setting appears on the Tab5 and in the browser, and the drawer has a **Basic** and an **Advanced** view. Which sections appear in which is yours to set, from a **Tab5 config** button in the web UI's Settings window; a firmware update that adds a setting shows it rather than hiding it behind a layout saved before it existed.

Frequency format — The frequency readout, preset buttons, WSPR band picker, keypad and spectrum scale can be punctuated `14.074.000` (the default, and what the QMX shows on its own LCD) or `14,074,000` *(Don N2VGU)*. Settings drawer, under Advanced.

Built for the field — WiFi is entirely optional, and can be given a static address if you want the browser bookmark to keep working; battery percentage with a charge limit for battery care; display sleep and a 180°

flip for awkward mounting; config backup and restore as a text file; and a settings reset that does not need a reflash.

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1. Quick Guide

Step 0 — Check your QMX firmware first

Power the QMX on by itself and read the firmware version off its own display at boot. You need **1.03.002 or newer**. If yours is older, update the QMX *before* connecting the Tab5 — everything that follows depends on it. This takes 10 seconds to verify and saves hours of debugging. Everything from **1.03.002** up to and including **1.04.010** works with the panadapter; on 1.04 the Tab5 additionally offers **AM mode** and an **Antenna Tune** button (SWR tune with a live power/SWR readout) — both stay hidden on 1.03.002, so there's no downside either way.

Do not use 1.04.011 (19-Sep-2026) yet. One operator reports no transmit output when the radio is keyed over CAT from the Tab5, and others report the radio's own Tune SWR and Hardware Test screens freezing until power-cycled. All of it clears on 1.04.010. Reported to QRP Labs; this note will go when there is a fix.

Step 1 — Flash the Tab5 firmware

> ### Use this flasher, not the M5Stack launcher
>
> **Do not install the panadapter with the M5Launcher.** It writes its own partition layout, which has no place for the panadapter's saved settings and no room for its file storage. A unit installed that way can still receive, but your callsign, grid and WiFi will not save - and with no callsign stored, every transmit refuses. The cure is a clean flash (`flash.bat`, type **E**), which rewrites the partition table.

Use the one-click flasher in `tools/QMX-Panadapter flasher/` (also attached to each release):

1. Plug the Tab5 into your computer with a **USB-C data cable** — charge-only cables will not work.

2. Run the flasher:
    - **Windows** — double-click `flash.bat` (fetches esptool by itself on first run; nothing to install)
    - **macOS** — double-click `flash.command` (needs esptool once — `brew install esptool` recommended; `pip3 install esptool` often fails on recent macOS with an "externally-managed-environment" error). If macOS blocks the script, right-click → **Open**, or run `bash flash.command` in Terminal (no `chmod` needed).
    - **Linux** — `bash flash.command` (needs `pip3 install esptool`)
3. The flasher asks **normal or clean** (see below) — just press **Enter** for a normal flash.
4. Wait for `SUCCESS`. The Tab5 restarts on the new firmware.

The firmware is already in the zip — `bootloader.bin`, `partition-table.bin` and `qmx_panadapter.bin` sit next to the script, and those are what get flashed. Nothing is downloaded except `esptool` itself, once, if your computer doesn't already have it. So after the first successful run the flasher works with no internet at all.

Normal vs clean flash. Just before flashing, the flasher asks:

> *Type E for a CLEAN/ERASE flash, or just Enter for a normal flash*

- **Normal (press Enter)** — updates the firmware and **keeps** all your saved settings (WiFi, callsign, grid, memory channels, ADIF log). This is what you want almost every time.
- **Clean (type E)** — wipes the whole chip first, so **every saved setting is permanently erased**: WiFi name *and* password, callsign/grid, all memory channels, and the logged QSOs. Use it only if something is stuck or corrupted (e.g. WiFi refuses to turn on no matter what). Back up first with **Config** ↓ (below) so you can restore in seconds afterwards.

If the flash keeps failing, reboot the Tab5 and try again — before anything else. The Tab5's USB serial connection is produced by the ESP32-P4 itself rather than by a separate USB chip, so a busy running firmware can leave the COM port listed in Windows Device Manager while it no longer answers the flasher. A reboot clears it. After that, the usual causes are a charge-only cable or a serial monitor still holding the port. *(Samuel W7STF, after a 7-hour session.)*

Building from source? See Build from source.

Step 2 — Connect the cables

You need **two** USB connections, and the cable to the QMX is the one people get wrong:

Connection	Port on Tab5	Cable	Carries
Tab5 → QMX	**USB-A** (host)	**USB-A → USB-C, full data cable**	I/Q audio (UAC) + CAT (CDC-ACM)
Tab5 → power	**USB-C**	Any USB-C power cable (5 V / 2 A+)	Power

> **The #1 failure is a charge-only cable.** Many USB-C cables — especially thin ones bundled with phones and chargers — carry power only, no data lines. If you use one between the Tab5's USB-A port and the QMX, the Tab5 powers the QMX but sees no audio and no CAT: the spectrum stays flat and the top bar shows `Band: ---`. Use a cable you know does data (one that works for a USB stick or phone file transfer). When in doubt, swap the cable first.

Power-on order matters. Turn the **Tab5 on first** and let it finish loading, then turn the **QMX on**. Within a few seconds the top bar should populate Band / Mode / BW and the spectrum should come alive.

Once flashed you can power the Tab5 from any 5 V/2 A USB-C source or the internal battery — the laptop is only needed for flashing. For diagnostics the Tab5 also outputs a serial log over the USB-C data connection (useful if WiFi is not available), but for most users the built-in **diagnostic log** — always on, nothing to enable — is the easier path — see Step 7.

Step 3 — Find your way around

The whole app is driven by **one-finger swipes from the screen edges** and **taps on the top bar**. The settings drawer is always one swipe away — no hunting for hidden menus.

Gesture	From	Does
Swipe →	Left edge	Cycle Panadapter → FT8/FT4/JS8 → WSPR
Swipe ←	Right edge	Open settings drawer
Swipe ↑	Bottom edge	Open memory channel picker
Tap or swipe ↓	Any top-bar item	Open that item's selector

Slim "breathing" grip handles on each edge show where to swipe — faint enough not to clutter the display, visible if you look for them.

The top bar reads **Band · Mode · BW · Freq · Signal · Zoom** left to right. Tap any item to open a selector — **Freq** opens the frequency keypad; **Mode** switches USB/LSB/CW/DiGi; **BW** selects SSB filter width (USB/LSB only); **Band** jumps to a configured band; **Zoom** selects a zoom preset.

See the full gesture table in the Reference section.

Step 3b — Choose FT8 or FT4 mode

FT8 is available on all HF bands; FT4 is available on select bands with published frequencies (20 m, 30 m, 40 m, etc.). To switch between them:

1. **In Panadapter mode:** Tap the **Mode** selector in the top bar, then choose **DiGi** (the digital mode group).
2. **In FT8 screen:** Tap the **Preset** frequency button to open the band/mode picker — you'll see **two columns**: FT8 (gold header) on the left, FT4 (cyan header) on the right. Tap any frequency in the FT4 column to switch modes.

The decode list, TX controls, and slot timing (countdown bar, which-parity-next) all adapt automatically to the active mode. Switching modes clears stale decodes from the previous mode so you don't work old QSOs.

When to use each mode:

- **FT8** — Most populated band in SOTA/POTA; longer decode window gives more time to prepare replies on slow links or weak signals.

- **FT4** — Faster QSO cycle for busy bands (Field Day, major contests, 20 m pile-ups) and operators who prefer shorter waits between exchanges.

Step 4 — Fill in your settings

On first boot the Tab5 opens a setup prompt automatically — it will ask for your callsign, grid square, and WiFi credentials before you reach the main screen. You can skip any field and fill it in later via the settings drawer.

To open the settings drawer at any time: swipe in from the right edge, or tap the right grip handle.

- **WiFi** — enter your SSID and password. Recommended for everyone: you get accurate UTC time (needed for FT8 slot timing), the browser web UI, and remote diagnostics. The **WiFi initiated** checkbox below the WiFi section is an on/off master switch — uncheck it for POTA or field use with no network.
- **Identity** (callsign + grid square) — *only required for FT8/FT4 transmit*. Enter your callsign and Maidenhead grid (4 or 6 characters, e.g. `JO45` or `JO45ab`). The grid also drives the distance and bearing columns in the FT8 decode list.

Everything else — dB range, smoothing, colour map, brightness, IQ balance — has sane defaults. Adjust if you want to, but you don't need to on first use.

Step 5 — Using the panadapter

Switch the QMX to any band and watch it come alive. This is the foundation of the device regardless of which modes you operate.

Tap or drag to tune. Touch anywhere on the spectrum or waterfall to place the cyan cursor. Drag and it snaps to a mode-aware grid — 10 Hz steps in CW, 250 Hz in SSB, 500 Hz for FT8/FT4 — so you can land precisely on a signal before lifting your finger. Lift, and the QMX retunes.

Swipe fast to pan instead. A quick horizontal swipe (rather than a held drag) slides the whole view left or right and retunes to wherever you let go — the fastest way to slide along a band. See Touch-to-tune for exactly how the two gestures are told apart.

Or move precisely with the band-plan slider. For a controlled move rather than a quick swipe, drag the framed "visible window" on the band-plan strip (along the bottom) to exactly where you want to be on the band — the spectrum, waterfall, and VFO all follow it, so you can place yourself precisely on a segment before you even tap a signal.

Zoom in on a crowded band. Pinch with two fingers to zoom up to ×24. On a CW band at ×8 or higher you can resolve individual signals a few hundred Hz apart, read the spacing between callers, and pick your target before you tune. The frequency axis scales with you, down to Hz precision at high zoom. Double-tap anywhere to reset to the full 48 kHz view.

Find your place on the band. The thin band-plan strip along the bottom of the screen (just above the status bar) colour-codes the CW/Digi/Phone segments around you — pick your IARU region (or leave it on Auto) in the settings drawer → **Band-plan region**. The framed window on the strip is a **slider**: drag it (or tap anywhere on the strip) to move precisely to any spot on the band, and the spectrum, waterfall and VFO all follow along. You can also grab that slider from anywhere along the bottom status bar and drag sideways — a taller target — while a vertical up-swipe there still opens memory channels.

Read your passband. Two grey vertical lines on the spectrum mark your current filter edges. The BW label in the top bar shows the active width; tap it to choose 2.5 / 2.7 / 2.9 / 3.2 kHz in USB or LSB. A coloured tint fills the passband so you can always see exactly what slice of the band you're receiving.

Watch the S-meter. The Signal field in the top bar is a live tick-scale bar (S1 through S9+20), driven by the actual signal under your VFO cursor.

Flat-spectrum mode. Toggle **Flat spectrum** in the settings drawer to switch from absolute dBm to dB-above-local-noise-floor. Noise collapses to a calm baseline; real signals — including weak CW tones in the mud — pop sharply above it. Recommended on noisy bands.

Save your spots. Swipe up from the bottom edge to open the memory channel picker — a 4×8 grid of 32 channels. It doesn't greet you with blank cells: a fresh device arrives with a handful of **example channels already filled in**, and each one is **colour-coded by mode** (CW green, DiGi teal, USB brick-red, LSB purple) so you can read the whole grid at a glance without opening a single label. The very first time you open it, a quick **guided animation** walks you through the gestures — watch a channel slide to a new slot, then drop into the wastebin and vanish — so you learn "drag to move, drag to delete" in about ten seconds, once, and never again.

Then it's yours: **tap** a channel to recall it (the QMX retunes instantly), **tap an empty slot** to store the current frequency/mode, **long-press** to edit, **drag** a channel to reorganise the grid, and **drag onto the wastebin** (bottom-right cell) to delete it.

Monitor from another room — or operate remotely. Once WiFi is configured, open `http://<tab5-ip>` in any browser for a full-featured live panadapter with mouse-driven tuning, band/mode controls, and ADIF log download. The IP is shown in the bottom status bar. See Web UI.

Step 6 — FT8 (if that's your thing)

Swipe in from the **left edge** to switch to the FT8 screen. The Tab5 starts decoding 15-second slots immediately — no PC, no WSJT-X required.

1. Tap the **Preset** button and pick your band's conventional FT8 dial frequency (e.g. 14.074 MHz for 20 m).
2. Watch the decode list fill. CQ stations appear at the top sorted by SNR; exchanges and replies below.
3. To reply to a station: **hold your finger on their row** for ~250 ms. A dim highlight appears after ~80 ms so you can see which row you're targeting before the gate fires. Lift — a confirmation modal shows the exact message before anything is armed.
4. Tap **Auto Pounce** to hand the full QSO to the auto-engine (works through report → RR73 → 73 with patient retry), or **Transmit** for a single manual message.
5. To call CQ: tap **Call CQ**. The engine picks a clear audio slot, fires CQ, automatically answers the first caller, runs the full exchange, logs the QSO, and resumes calling CQ.

Every completed QSO is written to an ADIF log downloadable from the web UI. See FT8 Transmit for the full picture, including ARRL Field Day exchange mode and a no-radio-keyed practice/simulation mode.

Step 7 — Something not working?

Spectrum flat, top bar shows `---`:

- Check the cable between Tab5 and QMX — almost always a charge-only cable.
- Make sure the QMX is powered on and not stuck in its own bootloader.
- Power cycle in order: Tab5 first, then QMX.

For anything else:

1. The diagnostic log is **always on** — nothing to enable.
2. Reproduce the problem (let it run a minute; power-cycle the QMX if the issue is about connection).

3. Grab the log:
    - **Over WiFi:** in the web UI take **Files ▲** → **Diagnostic download** ↓. That one item fetches **both** logs — `qmx-log.txt`, the current session held in RAM, and `qmx-log-saved.txt`, the copy persisted to flash that survives a reboot or a flat battery. If the device restarted, the saved one is the interesting file. (Directly: `/api/log` and `/api/log/saved`.)
    - **microSD:** if a card is inserted, the log is mirrored to `/qmx-panadapter/qmx-log.txt` on the card — continuously with WiFi off, or up to the start-up backup with WiFi on (alongside the always-on flash-persisted copy, which is complete either way).
    - **Over USB (no WiFi needed):** capture the serial console with `tools/capture_serial_log.ps1`.
4. Open an issue and attach the log. It includes Tab5 and QMX firmware versions plus every CAT command exchanged — usually enough to pinpoint the problem immediately.

Getting help

This whole guide is inside the firmware. No WiFi, no microSD card, no download: it works on the first boot, in a field with no signal, and it can never describe a different version than the one you are running. Swipe ← from the right edge to open the settings drawer; the top two buttons are the two ways in.

User Manual — opens where you are. Not a contents page to search: it opens the chapter covering the screen you were on — the panadapter chapter, the FT8 **receive** chapter, or the FT8 **transmit** chapter if a transmission is armed or running (someone mid-transmission is asking a different question). Inside: **Contents** is a two-column list — press and slide, and lifting your finger opens the highlighted chapter; **Back** returns to the previous page (shown only when there is one); **Exit** puts you back where you came from. Edge swipes and top-bar taps are stood down while the manual is open, so a stray touch cannot retune behind it. Holding the **User Manual** button for 3 s resets the reader; the manual itself is part of the firmware and cannot be lost.

Need guidance? — describe the symptom, not the cause. The second drawer button opens a short list headed "*What do you need help with?*", written the way you would say it out loud — "My radio is not showing up", "Nothing appears in the decode list", "It never transmits", "How do I change what my CQ says?", "Where are my contacts logged?" Pick the one that fits and the manual opens at the section that answers it. The list holds questions as well as faults, and it scrolls — keep going past the first few rows.

Rows the Tab5 can see are happening now are highlighted and floated to the top: no CAT link to the QMX, IQ mode never confirmed, an empty decode list while FT8 is running with the radio present, or WiFi switched on but not connected. WiFi switched **off** deliberately — POTA, battery — is not reported as a fault; that row stays as a normal question. **The device ranks, you choose:** a highlighted row is a suggestion, and it will never navigate for you on inference. The rows offered are scoped to the screen you opened the panel from, so FT8 never offers you spectrum symptoms.

Warnings you can tap. A warning you cannot act on is half a warning. The red **"QMX IQ mode not confirmed"** banner across the top of the screen is a button — tap it for the section on that exact fault. The **"Waiting for QMX"** message carries a small **Need help?** button; it says *help* rather than *what is wrong* on purpose, since a radio that is off is often off deliberately.

If a chapter does not answer your question, that is a documentation bug worth reporting — the wording of these rows is written from what operators actually say.

---

2. Panadapter

The panadapter is your primary view — a real-time spectrum analyser and waterfall for the QMX.

2.1. Layout

```
type: stack
title: The panadapter screen, top to bottom
row: 60 Top bar | frequency, mode, filter width, S-meter, zoom
row: 200 Spectrum | green trace, amber VFO cursor, tinted passband
row: 32 Frequency axis | absolute MHz labels
row: 370 Waterfall | newest row at the top, SDR gradient
row: 22 Band-plan strip | CW / Digi / Phone segments, with the visible span marked
row: 36 Status bar | battery, UTC clock, WiFi
note: amber | the VFO cursor sits at the dial frequency; in USB the passband tint runs
note: dim | in a digital mode the tint spans 150-3200 Hz, the QMX's one fixed digital
note: dim | the top-right 200 x 120 of the spectrum is a deadzone, so a tap near the d
```

Status bar indicators:

Left to right: battery, the SD dot, the firmware version, the UTC clock, and the WiFi indicator with the network name and IP address.

Item	Meaning
Battery %	Charge level (e.g. `88%`, or `88%` ⚡ while charging). `(limit)` is appended when charging has been deliberately capped at your Battery Care setting - it reads identically to "not charging" otherwise, and this tells you it's on purpose (v1.9.4)
SD (green dot)	A card is mounted and **writes are reaching it** — something landed in the last 90 seconds (see Settings)
SD (yellow dot)	A card is mounted but **nothing has been written lately**. Either there is genuinely nothing to write, or writes are failing — see Troubleshooting
SD (grey dot, struck through)	No card in the slot. Not an error
`UTC(GPS)` / `UTC(NTP)` / `UTC(FT8)` / `UTC(FT4)`	The time source **currently** in charge — GPS-disciplined QMX, WiFi/SNTP, or FT8/FT4 offline sync (RTC / manual show as `UTC(RTC)` / `UTC(MAN)`)
Firmware version	Currently flashed firmware version
WiFi fan icon	Link strength, at the right-hand end with the network name and IP address. The dot alone means a weak link (above 25 %), plus the first bow above 50 %, plus the second above 80 %; all three faint means connected but very weak, or `off`. From v1.3.4 this replaces the old `-NN dBm` figure — the width went to the network name, which is far more often too long to fit

2.2. Touch Controls

Tap to Tune

Tap anywhere on the spectrum or waterfall to jump to that frequency. The tap snaps to a frequency grid so you land on a sensible frequency rather than wherever your fingertip happened to be.

With a mouse the rule is simply the colour of the pointer: **white means clicking tunes, green means you are over something that will act instead** — a callsign, a top-bar control, an edge handle. Press and hold, or click and hold, and drag if you want to place the cursor before committing; a finger and a pointer behave identically.

> **Note - Fixed in v1.8.1**
>
> Before v1.8.1 a band across the upper part of the spectrum quietly refused to tune — the touch areas behind the top bar had been made shallower without the tune code being told, so that strip belonged to nobody. It looked as though tuning only worked near the centre frequency, while the waterfall was fine.

> Callsigns from Live Spots are drawn over the spectrum and are the one exception: tapping a **callsign** tunes to that station and sets the mode, rather than tuning to the point you touched.

The snap step follows the **mode**, not the zoom level — a CW signal needs to be found to the hertz, an FT8 dial frequency does not:

Mode	Snap step
CW / CW-R	10 Hz
USB / LSB	500 Hz
DiGi / FT8 / FT4 / RTTY	500 Hz

> **Note - In CW the display allows for your CW offset — including when you change it**
>
> A CW signal you can hear at 700 Hz is not on the dial frequency, so the spectrum, the waterfall and tap-to-tune all allow for the radio's **CW offset**. That figure is read from the radio, and since v1.8.5 it is **re-read every few seconds while you are in CW**, so changing CW offset or CW centre on the radio itself is followed within a few seconds.
>
> Before that it was read only once when the Tab5 connected, so if you changed it afterwards the display kept compensating by the old value and tapping a signal tuned you slightly off frequency — you would then transmit off frequency as though XIT were on. Reported by Roy KI0ER.

| AM / FM | 1 kHz |

RIT — receiving off your transmit frequency

Someone answering your call a couple of hundred hertz off your frequency is the ordinary case this is for: you want to hear them without moving your own transmit frequency, which everyone else is listening on.

1. Tap **RIT** at the top right of the spectrum. It turns amber and reads *click a signal* (in the browser) or waits for a tap.
2. Tap the caller on the spectrum or waterfall. The dial does **not** move. A dashed magenta line appears where you are now listening, in the spectrum and down the waterfall, and the filter window moves onto them.
3. It stays armed, so the next caller is another tap.
4. Tap **RIT** again to clear the offset and switch it off.

The offset is also printed beside its own marker over the waterfall, as `+250 Hz`, so you can read how far off you are listening without looking at the corner.

Long-press RIT to park the offset instead of clearing it. The offset is remembered and RIT switches off; long-press again and it comes back unchanged. The button reads `RIT (+250)` in brackets while an offset is parked. This is for a round robin or a net where one station is off frequency: you drop the offset when the turn passes back to the stations who are on frequency, and pick it up again without re-dialling anything (suggested by Roy KI0ER). A parked offset is discarded when you retune — it belonged to that one station.

If you never use RIT, the button can be hidden: **Settings -> Radio -> Show RIT button** (also in the web settings). It is shown by default. If an offset is actually engaged the button appears regardless of that setting — the radio listening away from your dial is not something to leave unsaid on screen.

The gold line goes on marking the **dial** — which is where you transmit — so seeing both lines at once tells you transmit has not followed your receiver. Tuning anywhere by any means clears RIT, on the assumption that you have moved on.

While RIT is armed, taps use a 10 Hz grid whatever the mode. The normal grid above exists to land the dial on a tidy frequency; RIT is a fine adjustment onto one caller's tone, and SSB's 250 Hz step would leave only a handful of usable offsets. The range is ±500 Hz — tapping further away clamps to the limit and says so.

Available in the browser too, where a click does the same thing. The offset itself is shared, so setting it on either screen shows on both; the *armed* state is per-screen, because arming changes what a click does and a click in a browser is not a finger on the Tab5.

The grid is anchored to absolute frequency, not to where your finger first touched, so the cursor always lands on the same set of points no matter where the drag began.

Pinch to Zoom

Use two fingers to pinch in (zoom out) or pinch out (zoom in). The display centers on the passband width — in USB/LSB modes, the passband center stays on screen even when the VFO is off to the side.

> **Note - The darker band at each end of a zoomed view (improved in v1.8.3)**
>
> Zooming filters the signal before it is re-analysed, and that filter starts rolling off just inside the edge of what is drawn — so the outer part of a zoomed view is slightly attenuated. It is a property of the filter, not a fault, and it has behaved this way since zoom was added.
>
> v1.8.3 doubles the length of that filter, which roughly **halves the width of the darkened band** at each side. The alternative — widening the filter instead — was rejected deliberately: it would trade the dark edge for false signals appearing where nothing is transmitting. Measured and reported by Samuel W7STF, whose own estimate of the affected width was accurate.

Pan (One-Finger Drag)

Drag horizontally with one finger to scroll the spectrum left/right. Release to tune to the new center frequency.

Band-Plan Strip

Along the bottom of the screen — just above the status bar, below the waterfall — is a thin coloured strip showing where you are within the band at a glance. Each colour zone marks the conventional segment (CW, Digi, Phone) for your selected region (auto / ITU Region 1/2/3, set in settings).

The **visible-span block** inside the strip shows the exact portion of the band the spectrum and waterfall are currently displaying — it narrows as you zoom in and shifts as you pan.

A **passband sub-block** inside the visible-span block mirrors the current filter width at band scale (grey tint), so you can see how your passband sits within the CW/Digi/Phone zones.

Tap and drag do different things, on purpose:

- **Tap** anywhere on the strip to jump to that frequency. Tapping says *go there*.
- **Drag the visible-span block** to move the window along the band. Dragging says *look there*. The block, the VFO marker and the passband all travel together as one piece, keeping the dial exactly where it sits inside the window, and the radio follows. Let go and it stays where you put it.

The filter passband and the CW/Digi/Phone labels hide while you drag and fade back in when you let go, so you can see the VFO marker arrive at its new home.

The grab area is taller than the strip looks. The strip itself is only a few millimetres, which is not a finger, so touches are accepted for about 8 mm of the waterfall directly above it as well. Tap-to-tune gives up that last centimetre of the waterfall — nothing interesting lives there, and it is the difference between catching the handle first time and getting a tune cursor instead.

Drag from the bottom bar too. The visible-span block acts as a slider handle that reaches *below* the thin strip: touch on or just under it — anywhere along the bottom status bar — and **drag sideways** to scrub the band, exactly like dragging the strip itself. This gives you a much taller grab target. It coexists with the memory-picker gesture on the same row: a **sideways** drag retunes the band-plan, while a **vertical up-swipe** still opens the memory picker.

The strip updates live as you zoom, pan, or change bands.

Outside a band, the strip reads **"Out of band — drag to tune"** in one flat colour instead of the CW/Digi/Phone zones, and returns to normal as soon as you are back inside a band. The frequency marker is hidden while you are out, because there is no band plan to position it against.

Out of band the strip becomes a coarse tuner (v1.8.3). Inside a band the strip is a *map* — where you touch is the frequency you get. Outside one there is no map to touch, so it works the other way round: a handle sits in the middle of the strip, and you **drag it off centre to move the dial**. Let go and it springs back to the middle, ready for the next pull.

- Dragging all the way to either edge moves by **half of what is currently on screen**, so two drags in the same direction overlap rather than skipping a gap. Nothing can scroll past unseen.
- It follows the zoom: zoomed in, the same drag is a finer step.
- A plain **tap** out of band does nothing on purpose — with no band plan behind it, a tapped position has no frequency to mean.

Suggested by Samuel W7STF, who pointed out that a row saying only "out of band" earns nothing when `Band: ---` in the top-left already tells you that.

Memory Channels

Swipe ↑ from the bottom edge to open the memory picker — a 4×8 grid of 32 channels, each free to hold any frequency/mode (not tied to a band). Each channel stores:

- Frequency
- Mode (USB, LSB, CW, DiGi)
- Label (your own name for the channel)

Each button shows the mode colour (CW = green, DiGi = teal, USB = brick red, LSB = purple) so you can tell mode at a glance without reading the label.

Recall: Tap a filled channel to immediately retune to it.

Tap an empty slot to create a new channel there directly — the frequency keypad opens straight away.

Long-press + drag a filled channel to move it to a different empty slot — no editing needed, the data follows your finger. Release on an empty slot to drop.

Long-press in place to open the editor (change name, frequency, or mode).

Entering a frequency outside the legal amateur band edges is rejected immediately with an error — the pad stays open so you can correct it without starting over.

2.3. Top Bar

Tap any item to open its selector:

Item	Purpose
14.074	Frequency. Tap to open the frequency keypad.
USB	Mode. Tap to open the picker: USB, LSB, CW, DiGi — plus **AM** once the connected QMX reports 1.04 or newer firmware.
2.5 kHz	Bandwidth (SSB only). Tap to choose 2.5, 2.7, 2.9, or 3.2 kHz.
S7	S-meter. Shows received signal strength. Display only — there is nothing to tap.
1.0x	Zoom level. Tap to choose x1, x2, x4, x8, x16 or x24. Pinching sets any value in between.

2.4. Spectrum & Waterfall

The **spectrum** shows signal power in dBm (default range −130 to −30 dBm) as a green curve with a dim fill. Each frame is smoothed per-bin with an exponential moving average — α = 0.4 by default, adjustable in the drawer — which balances a stable picture against a snappy response to CW keying and SSB attack transients.

The scale labels down the right-hand edge are **worked out from the dB range you set** (v1.8.3): the firmware picks round values that fit inside your Min and Max, choosing a finer step for a narrow range. Before this they were fixed at −40 to −120 regardless, so any range other than the default was described by labels that did not belong to it (found by Samuel W7STF running −118/−13). At the default range the labels are unchanged.

The **frequency axis** shows absolute MHz labels across whatever span is on screen, refreshed on every CAT frequency update. At high zoom the labels resolve to kHz or Hz precision. With **Still Spectrum** on — the default above ×1 — the axis stays put as you tune and the VFO marker moves along it; with it off, the axis is centred on the QMX VFO.

The **waterfall** runs newest row at the top, in a thermal SDR palette (black -> dark blue -> teal -> green -> yellow -> red). Four colour maps are available in the drawer: **Thermal, Viridis, Turbo** and **Grayscale**.

The waterfall floor tracks the band automatically. Its black level follows a running median sampled only from bins *inside the passband*, EMA-smoothed, so the background colour follows conditions instead of sitting at a fixed anchor. Bins outside the passband are drawn darker and excluded from that calculation, so they cannot wash out dim in-band signals.

Flat-spectrum mode (drawer toggle, persisted) switches both spectrum and waterfall to a per-bin adaptive display: every bin renders as dB *above its own running noise floor*. Real signals — weak CW tones especially — stand out sharply against a calm baseline, which is why it is the mode to reach for on a noisy band. The dB-range sliders have no effect while it is on, and the axis shows relative dB above floor rather than absolute dBm.

The picture is cleared when it would otherwise mislead. Switching between Panadapter, FT8 and FT4, or changing band, clears the waterfall and resets the spectrum baseline, so stale signals from the old band cannot linger in the new one. The noise floor recalibrates and a new baseline settles in about a second.

Waterfall controls (in settings):

- **Black level** — how far above noise-floor to go black (default 9 dB)
- **Contrast** — dB span of the colour ramp (default 45 dB)
- **FFT window** — Blackman-Harris, Hann, or Nuttall (default Blackman-Harris)

> **Note - Adaptive floor was removed in v1.8.3**
>
> That slider could not change anything: the per-bin noise floor it blended towards is re-seeded many times a second, so both ends of it produced the same picture. It has been taken out of the drawer, as it already had been from the browser. The stored value is still kept and still exported with your configuration, so the control can come back the day the underlying floor tracking runs.

2.5. Frequency Keypad

Tap the **frequency** on the top bar to open the keypad. Enter frequency in MHz format:

- **14.074** for 14.074 MHz
- **1.832** for 1.832 MHz
- **14.074.250** for 14,074,250 Hz — a third group is hertz within the kilohertz

Each group after a `.` is padded to three digits, so `14.07` is 14.070 MHz, not 14.007. Type a number with **no `.` at all** and it is taken as plain hertz.

The digit layout toggles between **10 Key** (calculator order, 7-8-9 on the top row) and **Phone** (1-2-3 on the top row). Pick whichever your fingers already know — the choice persists across reboots.

Resize the keypad: Pinch it or swipe up/down on it to toggle between the normal and a compact layout. Your choice is remembered across reboots. The compact layout reveals more of the spectrum behind it.

Reposition the keypad: Drag it by the **"Enter freq"** title label (the only non-button area at the top of the panel) to move it anywhere on screen, clamped to stay fully visible. The position is remembered so it reopens exactly where you left it.

The background behind the keypad is intentionally semi-transparent (40% dim) so the spectrum and waterfall stay visible while you're entering a frequency.

Cancel and Enter are the only exits — tapping outside the keypad does nothing, so an accidental background touch cannot silently discard what you were typing.

2.6. Memory Channels

32 memory channels (4×8 grid), free to hold any frequency/mode — not tied to a band. Each stores frequency, mode, and name. Swipe up from the bottom edge to open the memory picker.

Each slot shows its **label in large text**, with the mode and frequency dimmed beneath. Mode is also colour-coded — CW (green), DiGi (teal), USB (brick red), LSB (purple) — so the grid is scannable at a glance without reading every label.

To recall a channel: Tap it. The QMX retunes immediately.

To create a new channel (empty slot): Tap the empty slot — the frequency/mode picker opens directly, pre-filled with the current VFO so you can adjust both before naming it. No long-press needed.

To edit an existing channel: Long-press it in place. Change the name, frequency, or mode, then tap **Save**.

To move a channel to a different slot: Long-press and drag it to any empty slot. Release to drop. Only the data moves — the grid positions stay fixed.

To save the current QMX frequency/mode to a slot:

1. Swipe ↑ to open the memory picker
2. Long-press the slot you want to overwrite

3. The current frequency/mode pre-fills the editor — adjust the label if you like, then tap **Save**

To delete a channel — drag it onto the wastebin: the bottom-right cell (channel 32) is a **wastebin**. Long-press any filled channel and drag it onto the wastebin to delete it (it fades out). This is the way to clear a slot you no longer want.

Example channels on first use. A brand-new device ships with a few example channels already filled in (rather than 32 blank cells), so the grid is explorable straight away. These only seed empty slots — they never overwrite anything you've saved.

First-run tour. The very first time you open the memory picker, a brief (~10-second) one-time animation shows that channels can be **dragged** to move them and **dropped on the wastebin** to delete them. It plays once and never again.

Out-of-band frequencies are rejected immediately — if you enter a frequency outside a recognised amateur band, the keypad stays open so you can correct it rather than silently saving a wrong value.

2.7. Band Presets

Tap the **band name** on the top bar to switch between configured bands. The band selector shows:

- **160m, 80m, 60m, 40m, 30m, 20m, 17m, 15m, 12m, 10m, 6m** — standard bands (whichever your radio has configured)
- **11m** — the CB segment (26.9–27.5 MHz), shown on QMX+ radios that expose it
- **Custom** — user-added bands (via settings)

On radios with many configured bands (QMX+), the picker lays the bands out in **two side-by-side columns** so every band is visible without scrolling. Selecting a band jumps to the **nearest** band centre, so closely-spaced bands like 10 m and 11 m are no longer confused for one another.

Switching bands **remembers the last frequency you visited on each band**, so you can flip between 20m and 40m without losing your place.

2.8. Zoom & Pan

The QMX sends a 48 kHz-wide slice of I/Q, so **×1 shows 48 kHz** centred on the dial. Zooming narrows that window with a true zoom-FFT — you get finer resolution, not a stretched picture.

The zoom level is displayed on the top bar (e.g. **2.0x**). Tap it to choose a preset:

Zoom	Visible span
×1	48 kHz — the whole slice
×2	24 kHz
×4	12 kHz
×8	6 kHz — CW pile-up detail
×16	3 kHz
×24	2 kHz — individual CW signals well separated

Pinching sets any value in between; double-tap returns to ×1 and re-centres.

Gesture	Effect
One-finger fast horizontal swipe	Pan ("stroll") the view — retunes to the new centre on release
Pinch (two fingers)	Zoom ×1.0 – ×24.0
Two-finger drag	Pan the zoomed window
Double-tap	Reset zoom and pan to ×1.0, centred
Top-bar **Zoom** -> tap	Pick a preset

One-finger pan (stroll). A fast horizontal swipe — more than about 70 px of movement within the first 250 ms of touching down — slides the spectrum and waterfall under your finger in real time, with a live frequency tooltip, and retunes to wherever you release. It works at any zoom level, alongside the two-finger pinch and pan, and it is the quickest way to move along a band without dropping into a deliberate tune-drag.

At zoom levels above ×1 the display **centres on the passband**, not the VFO dial — which matters in USB and LSB, where the passband sits to one side of the carrier. It re-centres whenever you change mode, filter width or zoom, so the active receive area stays on screen, and the passband lines and frequency axis track correctly at every zoom level. What happens as you *tune* is governed by **Still Spectrum**, below.

The zoom level is **persisted**, and comes back at full zoom-FFT resolution on the next boot.

2.9. Still Spectrum

By default the spectrum and waterfall **hold still and the VFO marker moves across them**, so a signal stays where you last saw it while you tune towards it. This is how a Flex behaves, and it is what makes the waterfall readable as history: a signal's past sits directly above its present, under the frequency it belongs to, instead of the whole picture sliding sideways every time you touch the dial.

The view re-frames only when you tune far enough to need it:

Where you are	What the display does
Inside the view	nothing moves at all
Your passband reaches the screen edge	still nothing — it keeps holding, and your passband begins to slide off the edge
Half a passband width past that	a page, carrying part of the old screen across so you can see where you came from

The trigger is **your filter passband reaching the edge of the screen**, not a fixed percentage of the view. That distinction matters because the passband is not centred on the dial — in USB it runs roughly +200 to +2900 Hz — so a percentage rule re-frames too early at one edge and too late at the other, and mirrors itself in LSB. RIT is counted in, since it changes what you are actually listening to.

The display holds completely still right up to the page — it is never dragged along by the tuning. The cost is that your passband slides off the screen edge for a moment before the jump, which is why the allowance is **half** a filter width and not a whole one: long enough that a station right at the edge can still be worked, short enough that you are not tuning blind. How far you can tune between pages therefore depends on your filter — roughly a screen-width less half a filter, so about 4.7 kHz of a 6 kHz view on a 2.7 kHz SSB filter.

Picking a spot brings it into view only if it is not already usable where it is. Tap a spot that is comfortably on screen and the display does not move at all — only the VFO marker jumps to it, which is the whole point of holding still. Tap one that is off screen, or so close to an edge that your passband would not fit, and the display re-frames on it rather than leaving it stuck at the edge.

Changing mode, filter width or zoom re-centres the view on the passband, as before.

Switch it off under **Settings -> Radio & display -> Still spectrum**, and the display returns to re-centring on the dial at every step with the marker in the middle.

> **Warning - ×1 is always dial-centred**
>
> Holding a view still needs somewhere for it to stay while the capture window slides underneath it. At ×1 the view is already the whole 48 kHz the radio sends, so there is no room and the setting has nothing to work with. Zoom to **×2 or beyond** for the still display.

2.10. The Hatched Region

The QMX's local oscillator sits **12 kHz below the dial**. The 48 kHz it delivers therefore covers **dial−36 kHz to dial+12 kHz** — all of it real, but only 12 kHz of it above the frequency you are tuned to.

Above dial+12 kHz there is no data, and there never was. That part of the display is drawn **hatched** and does not respond to tapping, on the Tab5 and in the browser alike, with a caption saying why it is empty.

Before v1.10.5 the display filled that quarter of the ×1 view by wrapping the bottom of the band into it, and the frequency scale labelled it as dial+12 to +24. The signals shown there were real, but they were about 48 kHz from where the scale claimed — so tapping one tuned you to the wrong place entirely.

If a signal you want falls into the hatching, tune down or zoom in. The radio cannot hear above dial+12 kHz, and no display setting can change that.

2.11. Decoded CW

In **CW or CW-R**, a single line along the bottom of the waterfall shows the Morse the radio is decoding — on the Tab5 and in the browser alike.

The decoding is not done by the Tab5. **The QMX has its own CW decoder** and hands the text over the CAT link, so this costs the panadapter nothing: no audio processing, no effect on the spectrum, no effect on FT8. It works on QMX firmware 1.03 and later — there is nothing to enable on the radio for most people, since its own **Decoder -> Enable Rx** setting is on by default *(suggested by Uwe DL8UG)*.

The line reads:

```
CW ~18wpm:  CQ CQ DE OZ1LAV OZ1LAV K
```

- The **green header** carries the estimated sending speed and never moves. The number is zero-padded, so the text after it does not shift as the estimate crosses ten.
- The **cyan text** is what the radio decoded, in a fixed grid of 72 characters.
- When the line fills, it **wraps and writes over the oldest end** rather than scrolling. Nothing slides sideways, so a callsign you are half-way through reading stays where it is. Two blank spaces travel ahead of the writing position — after the first wrap, that gap is what tells new text from old.

It is written to the microSD card too, if one is in the slot: everything decoded goes to `qmx-panadapter/cw-decode.txt` with a timestamp on each line, so a callsign you only half-caught can be resolved afterwards *(Michael K Johnson KZ4LY)*. The line on screen holds one screenful; the card holds the session.

With WiFi on, the card is written every 30 seconds rather than continuously, so the last few characters of a session may still be in memory when you pull the card. Switch the radio off and give it half a minute if you want the very end of an over.

While the web page is open the card writes wait, because writing the card stalls the spectrum stream for several seconds on this hardware. They go through anyway after three minutes, so nothing is lost — but if you want the transcript fully up to date, close the browser tab and give it half a minute.

About the speed. It is a *throughput* figure: characters per unit time, counting the gaps between words and between overs. During a real exchange it therefore reads **lower** than the other operator is actually sending, and only approaches their true speed during continuous sending. The radio hands over finished characters, so the element timing a true speed measurement needs is already gone. That is why it is shown with a `~`, and why it reads `??` rather than guessing when there is not enough to go on.

About the noise. With no signal the decoder chews on noise. Two kinds of rubbish are filtered out before anything reaches the screen:

- `*` — the decoder's marker for a symbol it could not resolve. Measured on a live band it was the single most common thing arriving, and it is never part of real text, so it is dropped outright.
- Long runs of **E** and **T** — the two shortest Morse symbols, so noise lands on them more often than on anything longer, and worst just after switching to CW while the decoder settles. A long run is dropped; a short one is kept, because those are real letters and words like TEST and BETTER must survive.

A dropped run leaves one space behind, so words either side of it are not welded together.

Clearing it, and how many rows (from v1.20.1). A **Clear** button empties the pane, and the row count is **2 or 4**, chosen beside the on/off switch. Four rows of wrapped text can make it hard to find the boundary between fresh decode and old; two rows keeps the recent text where you can see it *(Bryan N0LUF)*.

A click *inside* the pane still tunes the radio rather than clearing it. That is deliberate - the waterfall underneath stays click-to-tune - which is why Clear is its own button.

Turning it off. Settings drawer -> **Radio** -> **Show decoded CW**, beside CW centre and the transmit offset. It is on by default and the setting is shared with the browser.

2.12. S-Meter

The **Signal** field in the top bar is a tick-scale bar labelled S1, S3, S5, S7, S9, +10, +20, with a moving green bar beneath it.

- **S1 to S9** — standard S-units, −130 to −73 dBm
- **+10 / +20** — above S9, in 10 dB steps

The scale is **S9 = −73 dBm**, 6 dB per S-unit below S9 and 1 dB per unit above it.

The reading is the peak level in a ±64-bin window centred on the **IF-shifted VFO bin** — corrected for the QMX's +12 kHz IF offset, so it measures the signal actually under your cursor rather than the DC/local-oscillator spike sitting at the centre of the spectrum. It stays live during FT8 capture.

It is a readout, not a control: there is nothing to tap, and there is no peak-hold mode.

2.13. Settings Drawer

Swipe ← from the right edge to open the settings drawer, or tap the right grip handle.

It is grouped — **Station, Device, Radio, Network, Display, FT8, WSPR, Spectrum** — with a **BASIC / ADVANCED** button at the top. Basic shows what an operating session needs; Advanced holds everything, including the tuning and calibration controls you set once. Which sections sit in which view is yours to change, from the web UI's **Settings -> Tab5 config**.

Every control, group by group, is documented once in Settings — deliberately in one place rather than summarised here as well.

Next: Learn about FT8 Receive or Web UI.

3. Audio

Audio decoding is a BETA feature. These implementations are first-generation and subject to further development. Real-world feedback shapes the roadmap.

Getting audio — start here if you hear nothing

Silence is usually one of five things, and they are listed here in the order they actually catch people out. Work down the list; each step says where the control is.

```
type: flow
title: From the radio to your ears, and where each control acts
node: QMX / QMX+ - receiving, set to DiGi / IQ mode
branch: QMX RF gain - per band, 0-99 dB. Acts before anything else
branch: QMX volume - the radio's own AF gain. Reaches everything downstream, the browse
node: USB - 48 000 I/Q pairs per second, one cable
node: Tab5 DSP - 48 kHz. The same I/Q feeds the spectrum and the demodulator
node: AGC - ceiling, attack and release, or a fixed gain
node: the demodulated audio goes to ONE of these at a time
branch: Tab5 speaker and headphones - I2S codec, 48 kHz stereo. RX Volume acts here, a
branch: a browser - 12 kHz mono mu-law over the WebSocket, 3 s buffer. Your computer's
note: dim | one listener at a time, deliberately: while a browser holds the stream the
```

Four controls act at four different places, which is why "turn it up" is not one question.

3.1. Check the mode

Only **CW, CW-R, USB, LSB and AM** produce audio. **DiGi and FM produce none, by design** — see Which modes produce audio.

This catches more people than anything else, because **FT8, FT4, JS8 and WSPR all run in DiGi**. If you are on a digital mode and expecting to hear the band, you will not. Switch to USB to listen.

3.2. RX audio is off by default, and enabling it needs a restart

Open the **settings drawer** → **Audio Settings** and tick **RX Audio (speaker/headphone)**. Then **restart the Tab5**.

The restart is not optional and it is not a bug: the audio codec is opened at boot, before the USB host claims the DMA memory, so it cannot be opened safely part-way through a session. From **v1.20.1** the Tab5 tells you this with a toast when you tick the box. Before that it said so only in the serial log, which is how three operators arrived at silent audio in one week.

The same switch and volume are on the web API from v1.20.1, in the `rxaudio` group of `/api/settings`.

3.3. In the browser, press the play button

The browser does **not** start playing on its own. Click the ▶ button at the top right of the page, just to the right of **F**. It turns into ■ while it plays.

This is the single most-missed step — two users reported "no web audio" in one week and in both cases the stream was working and the button had never been pressed. See Listen in the browser.

3.4. Know which volume control does what

There are three, and they act in different places — see the chain above for where:

Control	Where	What it changes
QMX volume	Drawer → Radio	The **radio's own AF gain**, over CAT — the same number the QMX shows when you click its volume knob. Affects everything downstream, including the browser stream
RX Volume	Drawer → Audio Settings	The **Tab5's** speaker and headphone level only. Does not affect the browser stream
Your browser's own volume	The browser or the operating system	The browser stream only

A fourth control sets the level before any of these: **QMX RF gain** (Drawer → Radio), per band, 0–99 dB, QMX default 54. If signals look weak on the waterfall as well as sounding weak, this is the one to look at — not the volume.

> **Note - Full volume and still quiet?**
>
> The level is governed by an **automatic gain control**, not by the volume slider. If a signal is quieter than you expect at full volume, the AGC ceiling is holding it back — see The AGC controls.

3.5. The speaker and the browser are exclusive

While a browser is listening, the Tab5 **mutes its own speaker**, and hands the audio back when you stop listening or close the page. One listener at a time is deliberate. So if the Tab5 went quiet, check whether a browser somewhere on your network is holding the stream.

Panoramic CW Split

Listen to two CW stations on different frequencies, panned left and right across the stereo stage. The CW receiver demodulates each sideband independently, allowing separation by frequency.

How It Works

The RF passband is split into two channels:

- **Left**: Lower-frequency station (panned fully left)
- **Right**: Upper-frequency station (panned fully right)
- **Center blend**: Dry/wet mix between the two

Three live controls tune the effect mid-QSO without leaving the exchange, grouped under **Panoramic split — adjust while listening**:

- **Width** — Frequency spread between the two stations. Wider = greater separation.
- **Blend** — Dry/wet audio mix. Left is pure lower channel; right is pure upper channel.
- **Overlap** — Frequency range for each station. Wider ranges allow more signal capture.

Enabling Panoramic CW

1. Open the settings drawer and tap **Audio Settings** (just below SelfSpotter)
2. Tick **RX Audio (speaker/headphone)**, then **Binaural CW (stereo separation)**

> **Warning - Switching RX audio on needs a restart of the Tab5**
>
> The audio hardware claims its memory during start-up, before the USB host takes that pool, so it is started at boot or not at all. If you boot with **RX Audio** off and tick it later, the setting is saved but you will hear nothing until the Tab5 is restarted — it tells you so on screen when you tick it. Your QMX will need a power cycle after that restart, as it does after any Tab5 restart.

> **Note - Which modes produce audio**
>
> **CW, CW-R, USB, LSB and AM. FM is still silent** — there is no FM demodulator, so an FM signal has to be listened to in USB or not at all.
>
> AM is demodulated on the Tab5 itself, from the I/Q the radio sends, so the bandwidth is set by the Tab5 and not by the filter in the radio. Tap **BW** in the top bar to choose it: **2.0, 3.0, 4.0, 6.0, 8.0 or 9.6 kHz**, quoted as the width of the whole signal, the way amateur AM figures always are. 6.0 kHz is the default and the standard — 3 kHz per sideband, what most commercial rigs hard-code. The same list is on the web page's **BW** button.
>
> Go **narrower** when someone is sitting on top of you: 2.0 and 3.0 kHz sound like a telephone line and that is the point, they keep a signal copyable in heavy QRM. 6.0 kHz is the widest on offer and the one to use on a clear channel. Broadcast channels are spaced 5 kHz, so 6.0 already reaches into the neighbour's carrier on a crowded shortwave evening.
>
> **9.6 kHz is the widest the Tab5 offers, from v1.20.2.** Two wider settings were available earlier and have been withdrawn: **10.0 and 20.0 kHz** ran the receive chain at half the decimation the others use, and measured on the bench that leaves the second CPU core with no idle time at all and the audio stutters.
>
> **8.0 and 9.6 kHz cost nothing extra** - matched 60-second windows measured AM at 4.0 and at 6.0 kHz at the same processor load, so width is free up to that point. They were briefly withdrawn in v1.20.1 as unproven and are back in v1.20.2: one operator runs 9.6 kHz daily and another asked for 8.0 kHz, which settles whether they work.
>
> CW and SSB are untouched at any setting.

The three pan controls sit directly under the Binaural CW row in the same window. They stay greyed out until both boxes above are ticked, because they shape a split that is not being produced otherwise.

On the Air

All three controls are **live-tunable** during an exchange:

- Slide them while the QSO is running
- No reboot, no mode change required

- Changes take effect immediately

Binaural CW — First Implementation

Frequency-based station separation in CW. Each station gets a distinct audio tone, allowing the ear to distinguish them without losing either signal.

Bandwidth Requirements

Binaural separation works best with **bandwidths of 300 Hz and above**.

Do not expect good separation below 200 Hz — the audio bandwidth is the limiting factor. At 200 Hz or narrower, the two stations' spectra start to overlap, and separation breaks down. This is a fundamental constraint of the audio bandwidth, not the algorithm.

Recommended Bandwidth Settings

Bandwidth	Clarity	Use Case
300 Hz+	Excellent	General CW operation; two stations easily distinguished
200–300 Hz	Fair	Tight pileup; separation possible but marginal
<200 Hz	Poor	Not recommended; too much spectral overlap

Audio Settings

Audio decoding shares the Tab5's compute and memory budget with the spectrum, waterfall, and FT8/FT4 decode paths. Not all features can run simultaneously.

> **Note - Renamed in v1.16.4**
>
> This window used to be called **Resource Management**. It is **Audio Settings** now, because that is what it is — volume, AGC, binaural and the panoramic split. The old name described how it works rather than what it does.

Opening Audio Settings

Open the **settings drawer** and tap **Audio Settings**, directly below SelfSpotter.

> **Note - Top-bar shortcut removed in v1.16.8**
>
> A long-press on the top bar used to open Audio Settings too. It shared its area with the Band, Mode, BW and Frequency pickers, so it was easy to open the wrong thing by accident. The drawer button is now the only way in.

The window lists RX audio and the background network feeds that compete with it for the same scarce memory, each with a tick box:

- **RX Audio (speaker/headphone)** — with RX Volume and the three AGC controls described below
- **Binaural CW (stereo separation)** — with the three panoramic-split controls
- **SelfSpotter**, **RBN**, **DX cluster**, **PSK Reporter — who's hearing me** — these grey out and are held off while RX audio is on
- **POTA / SOTA spots**, **PSK Reporter — report my decodes** — these keep running

The list scrolls; the **Close** button stays put at the bottom.

Toggle audio on and off as needed — the resource footprint is immediate, and disabling a feature instantly frees that headroom.

Typical Constraints

- **Audio ON, Waterfall ON, Decode ON** → Some margin left; FT8/FT4 decode still runs steadily
- **Audio ON, Waterfall ON, Decode ON, Web streaming ON** → Tight; any spike can cause temporary frame drops
- **Audio ON, full WiFi + web load** → Audio is the first to suffer if the load spikes; consider disabling audio or the waterfall if you need reliable web streaming

Headphones

Plug headphones into the Tab5's 3.5 mm jack and the **internal speaker goes quiet on its own**. Unplug them and it comes back. Nothing to switch.

Before v1.16.4 you heard both at once, which made headphones close to useless in a quiet shack *(Roy KI0ER)*.

> **Tip - The internal speaker is small and trebly**
>
> Several operators have said so, and they are right — it is a tablet speaker. Headphones or a powered external speaker on the jack are a large improvement, and now that the internal one mutes itself either is practical.

Which modes produce audio

Audio and the panoramic split are not the same thing, and it is worth being clear about which you get where.

Mode	Audio	Panoramic / binaural split
CW, CW-R	Yes	**Yes**
USB, LSB	**Yes**	No — single channel, both ears the same
AM	**Yes**, to about 3 kHz	No — AM has no stereo content to split
DiGi	No	No
FM	No	No

So you **do** get audio on SSB and AM; what you do not get there is the left/right split, because that is a CW feature. In DiGi and FM there is no audio at all — nothing is broken if you hear silence in those.

Known Limitations

- **Audio breaks up while a web page is open.** Known and unfixed as of v1.16.3. With a browser connected to the panadapter page the audio task misses some of its deadlines and you hear a regular crackle; closing the page clears it. Measured, but the cause is not yet found — if you need clean audio, close the web page.
- **Stereo rendering requires a headset or speaker with stereo output.** Mono output collapses both channels; panning does not work.
- **The split is CW-only** (see the table above). On SSB you hear the signal, but it is not placed left or right.
- **Limited to the audio passband.** If the audio filter is 300 Hz wide, the maximum usable separation is about 300 Hz. A wider filter in the radio settings increases the available range.
- **Level is set by an automatic gain control, not by the volume slider.** If a signal is quieter than you expect even at full volume, the AGC's ceiling is what is holding it back. All three AGC controls are adjustable live — see below.

The AGC controls

All four audio level controls live in **Audio Settings** (settings drawer → Audio Settings), on their own rows under **RX Audio**:

Control	Range	Default	What it does
RX Volume	0 – 100	60	Output level to the speaker or headphones.
AGC Ceiling	50 – 1500	200	How far the AGC may lift a *weak* signal. Raise it if quiet stations stay quiet.
AGC Attack	1 – 50 ms	3 ms	How fast the gain comes *down* when a signal arrives. Shorter catches sharp CW edges; too short can sound clipped.
AGC Release	10 – 500 ms	150 ms	How fast the gain comes back *up* afterwards. Longer is steadier on CW; shorter follows speech better.

All four are remembered across restarts and travel in a config backup.

AGC presets. Four buttons under the sliders set the timing in one tap:

Preset	Attack	Release	For
Fast	1 ms	30 ms	CW
Med	3 ms	150 ms	general, and the shipped default
Slow	5 ms	400 ms	SSB, where a fast release pumps on speech pauses
Off	—	—	no automatic gain at all

Move either slider and the caption reads **Custom** — the sliders *are* the custom setting, so there is no fifth button.

Off is a real bypass, not a very slow AGC. The gain follows the signal at every attack and release value, so no combination of the sliders switches it off. With **Off** selected the gain is pinned at **AGC Ceiling**, and that slider becomes a plain manual gain control.

⚠ **What to expect when you try them.** The AGC only starts regulating once a signal is strong enough; below that it already sits at the ceiling, so the gain is the same number whether the AGC is on or off. On a quiet band **every preset correctly sounds identical** — the difference is heard in how fast the noise returns between signals when the band is busy. The attack values (1–5 ms) are all far quicker than the ear resolves; **Release** is the audible one.

Volume and Ceiling are not the same thing. Volume sets how loud the output is; Ceiling decides how hard a weak signal is lifted before it gets there. Running Volume to 100 does not mean you have run out of level — if it is still too quiet, raise **AGC Ceiling**.

What changed in v1.16.3

- **AGC Ceiling was called "Gain".** The name was wrong: it never was a plain gain, it is the ceiling on the AGC's own weak-signal boost, and calling it Gain led people to look for an AGC that was already there. Its range also went up from 800 to 1500, because the Tab5's internal speaker wants more headroom than the old ceiling allowed.
- **Attack and Release are new** — the AGC's timing was fixed before, and is now yours to set.
- **Presets arrived in v1.16.5** *(Samuel W7STF)*, along with a genuine **Off**.

- **RX Volume moved here from the settings drawer**, and the drawer's RX Audio section is gone entirely. That includes its on/off box: **Audio Settings is now the only place to switch RX audio on or off.**

What changed in v1.16.4

- **The window is called Audio Settings**, and it has a door of its own in the settings drawer. The top-bar long-press still works.
- **Headphones mute the internal speaker** automatically.

Strong signals

A peak limiter sits after the automatic gain, so a loud station cannot arrive at full scale and startle you in headphones. It holds the peaks down and leaves the average level alone, which means raising **AGC Ceiling** to hear weak stations does not make the strong ones painful.

This is the part of the beta I most want reports on: which mode, roughly how strong the signal was, and whether more ceiling fixed it.

Where the receive signal comes from

Until now the panadapter always listened to the **radio's USB audio**. There is now a **receive source** selector, on the Tab5 and in the browser, with a level meter and a clip indicator beside it.

Source	State
QMX USB	The default, and what every earlier release did.
Line in (3.5 mm jack)	**Declared beta** - see the warning below.
Generic I/Q	Declared, not implemented. Selecting it is refused.
Internal mics	Declared, not implemented. Selecting it is refused.

Line in *(asked for by Tony, @tony1tf, in issue #17)* feeds the panadapter from the Tab5's own 3.5 mm headphone jack instead of the radio's USB. Tony ran a test build on his own Tab5 to find where the jack lands — the mic ring arrives on the audio chip's third channel — which is the measurement the whole feature rests on. The jack is a single real audio channel, not I/Q, so the second channel is synthesised: a 255-tap filter builds the quadrature pair the decoders need. The jack wiring is drawn on screen so you can see which contact carries what.

⚠ **No signal has been decoded through line in yet.** The path runs, samples flow and the levels read correctly with nothing plugged in - that is the whole of what has been verified. Treat it as unfinished. Changing the source needs a restart, and the Tab5 does the restart itself when you press **Save**.

⛔ **In line-in mode the USB radio connection is off by design.** One or the other, not both - so CAT and the S-meter go quiet while line in is selected. The Tab5 stops pretending to have a dial with it: the frequency readout reads **Line-in mode**, the band shows `---` and the band-plan strip is hidden, because an audio spectrum has no band plan and no RF frequency. The frequency axis under the display reads audio Hz.

The left half of the spectrum is empty and there is a step in the middle. That is correct and it is the honest picture: the jack carries one real channel, the second is synthesised, and a synthesised channel cannot recover what the missing one would have told you. The step sits at audio 0 Hz, which is the centre of the span.

Listen in the browser

From **v1.20.0** the panadapter streams its own demodulated audio to any browser on your network. Open the web page and click the ▶ button at the top right, just to the right of **F**. It becomes a ■ while it is playing; click it again to stop.

What you hear is exactly what the Tab5's own speaker gets — the same demodulated audio, tapped after the filter and the AGC — so the two can never disagree about what the radio is doing. It works even if the Tab5 booted with its speaker turned off.

The speaker and the browser are exclusive. While a browser is listening the Tab5 mutes its own speaker, and it hands the audio back when you stop listening or close the page. There is no setting for this: one listener at a time was the point.

Only the modes the Tab5 demodulates produce audio: **CW, CW-R, USB, LSB and AM**. In any other mode the button lights up but nothing arrives, and hovering it says so.

Hover the ■ button for the health of the stream — it reports frame gaps, re-primes and chunks dropped to clock drift. On a quiet network all three stay at zero.

Notes on what this costs and what it does not:

- It runs over the web page's existing connection, so it needs no extra port and no extra setup.
- It costs about **9 percentage points of one CPU core** — measured — and nothing measurable on the core that serves the web page.
- Audio is 12 kHz μ-law, which is a telephone-grade codec chosen because it survives a lossy Wi-Fi link without a jitter buffer deep enough to be annoying.
- **Only one browser gets the page at a time.** Opening the panadapter page in a second browser takes the connection from the first, which takes the audio with it.

Feedback

This feature is **beta and in active development.** Your feedback shapes the next iteration:

- Does panoramic CW work in the field?
- What bandwidth do you typically use?
- What other audio features would you like to see?
- Did you hit the resource limits?

Send feedback to the discussions page or open an issue.

4. Live spots

Live spots put other stations **on your spectrum**, at the frequency they are actually operating on, so you can see who is where without leaving the radio.

Three sources feed the same display:

Source	What it tells you	Default
POTA (Parks On The Air)	Park activations currently spotted, any mode	**On**
RBN (Reverse Beacon Network)	CW stations the skimmer network is hearing right now	**Off** (opt-in)
DX cluster	What other operators are spotting by hand - this is where **phone** activity comes from	**Off** (opt-in)

DX cluster is the only one of the three that shows SSB. No machine recognises a callsign spoken into a microphone, so RBN cannot see phone at all; the cluster sees it because a human typed it in.

All three need WiFi. Nothing here transmits, and nothing here touches the radio until you tap a spot. Each source switches on separately in **Network -> Live spots**.

4.1. What you see

Spots are drawn **over the spectrum**, the way a FlexRadio or similar SDR shows them — the callsign sits around the middle of the spectrum with a thin vertical line dropping from it down to the frequency axis, so the line points at the frequency the station is on. The line is only 2 px wide, so the trace stays readable underneath.

Colour tells you what a spot means:

Colour	Meaning
Amber	A POTA activation
Bright green	An RBN (CW skimmer) spot
Grey	You have **already worked this station on this band**

Grey is the useful one: it answers "do I need this station?" at a glance. It is band-aware, so the same operator on a different band is *not* greyed out — that is a new band-slot.

Spots fade as they age. A spot is a claim about *now*, and an old one pointing at an empty frequency is worse than no spot at all:

- Under 5 minutes old — full brightness
- Around 15 minutes — about half faded
- 30 minutes — gone

The callsign and its line always fade together.

Off-screen counts. When there are spots on your band that fall outside the window you are looking at, a small count appears in the bottom corner of the spectrum — `< spots (3)` on the low side, `spots (5) >` on the high side. It says "spots" in words rather than a bare `<3`, because a lone number against the edge of the spectrum tells you neither what it counts nor that you can tap it. These are **scoped to the band you are on**, so the arrow never points at something on 40 m while you are on 20 m. Each is coloured like the spot it will take you to.

Crowded bands. Only so many callsigns fit legibly side by side, so on a busy segment the closest-spaced spots lose their name — and a spot with no name is not drawn at all, rather than leaving a line pointing at nothing. Stations you have **not** worked are labelled in preference to ones you have, and fresher spots in preference to older ones.

4.2. Tapping a spot

Press, slide, lift. Put a finger on a callsign and it highlights. Keep the finger down and slide along the lane and the highlight follows, so you can correct your aim on a crowded band instead of mistuning. Lift, and the radio tunes to whichever spot is highlighted.

A quick tap is simply the short version of that. And if you change your mind, slide off the lane before lifting: the selection is abandoned and nothing is tuned.

Either way the radio tunes to the station *and* switches mode:

Spot mode	Radio goes to
CW	CW
FT8, FT4, other data	DiGi
SSB	USB above 10 MHz, LSB below
Unknown	Frequency only — the mode is left alone

Bandwidth is deliberately **not** forced. The QMX keeps a filter per mode and loads it when the mode changes, so the right width follows by itself.

Tap an off-screen count (`< spots (3)` / `spots (5) >`) and you jump to the nearest spot on that side, which brings it into view. Those counts have a deliberately large touch area — the visible text is small, but the target around it is not.

Tapping the spectrum anywhere else still tunes normally, and pinch-zoom and the one-finger pan are unaffected. Only the callsigns themselves and the two counts are spot targets.

A callsign's target is the text, and nothing above or below it. That matters because a long callsign is a wide label sitting over the trace, and a click inside it goes to *that station* rather than to the frequency under your pointer. If a signal you want is behind a label, aim at the same column higher or lower in the spectrum and it tunes there normally.

In the browser, hovering a callsign shows you a marker at that station's real frequency with its name, so you can see where the click will take you before you commit. On the Tab5 the two gestures are already distinct: tuning to a point needs a brief hold, while a callsign is a quick tap.

4.3. Turning it on and off

Settings drawer (swipe in from the right edge) — **Live spots** section:

Control	Default	What it does
Live spots (POTA)	On	The whole feature. Off leaves the spectrum completely clean
Add RBN (CW skimmers)	Off	Adds RBN as a second source. Needs your callsign set
DX cluster spots	Off	Human-typed spots — the only source of SSB activity. A second connection, so opt-in
SOTA spots (summits)	Off	Summit activations from spothole.app, a volunteer-run server. Opt-in for that reason
Mode filter the spots	**On**	Shows only spots you can work in the mode you are in

Every one is saved and appears in the configuration file you can download and restore from the web UI, under `[settings]`.

About the mode filter, since it is on by default and changes what you see: tapping a spot sets the **mode** as well as the frequency, so an unfiltered lane can drop a CW operator into FT8 without warning — which is exactly what happened to Michael KZ4LY, and why this exists. With it on, changing mode changes which spots are on screen, and the off-screen counts change with them. A spot whose mode is unknown is always shown, so nothing is hidden on a guess. Switch it **off** and every spot is shown, with a two-letter tag on the label (`CW`, `SB`, `FT`) telling you what tapping it will do — deliberately absent when the filter is on, because then every spot is already your mode.

The spots overlay belongs to the **Panadapter** page. It is not drawn on the FT8 screen.

4.4. About RBN

RBN is **off by default, on purpose.** It is a continuous global feed rather than an occasional fetch, and it arrives over a persistent connection on the part of this board that has historically been the most delicate. Off by default means it can never affect anyone who has not asked for it.

If you do switch it on:

- **Your callsign is required.** The feed asks for one when connecting; it is how RBN attributes load, not a password. Set it under **Callsign & Grid**.
- **Only your current band is kept.** RBN reports the whole world, and the display can only show one band at a time, so spots outside the band you are on are discarded as they arrive. Changing band clears the picture and it refills within a few seconds.
- **Duplicates are merged.** The same CQ is typically reported by ten or more skimmers; you see one entry per station, at the best reported signal.
- **RBN spots** are held for up to **10 minutes** after any skimmer last heard them — shorter than the 30 minutes above, because an active caller is re-spotted every couple of minutes and silence means they have stopped.
- It is the **CW/RTTY** skimmer feed. RBN runs a separate FT8/FT4 feed that the panadapter deliberately does not subscribe to — you are already decoding those yourself, far better, from the antenna.

4.5. About the DX cluster — where phone spots come from

RBN cannot show you an SSB station. It is a network of **automated skimmers**, and a skimmer decodes CW, RTTY, FT8 and FT4 — machine-readable modes. There is no such thing as an SSB skimmer, because no machine can recognise a callsign spoken into a microphone. So every phone station on the band is invisible to it.

A **DX cluster** is people typing. That is the whole difference, and it is why this exists as a third source:

- **Phone spots** — SSB activity you cannot see any other way.
- **Park and summit references** — a human writes `POTA ES-2081` in the comment, so a cluster spot can also tag your chase in the log.
- **Everything else people bother to spot** — DX, contest activity, anything worth telling other operators about.

Switch it on with **DX cluster spots (phone)**, next to the other two sources in the settings drawer, or from the web UI. Like RBN it is **off by default**: it is a second continuous connection, and the same caution applies.

Things worth knowing:

- **Your callsign is required**, for the same reason as RBN — the node asks for one to identify the connection.
- **Only your current band is kept**, again like RBN.

- **Mode is worked out for you.** Most cluster spots carry no mode field at all. If the spotter typed one (`LSB`, `CW`, `FT8`) it is used; otherwise the frequency is looked up in your band plan — 14.020 is CW, 14.285 is phone. The same inference you would make reading the line yourself.
- **Skimmer spots relayed onto the cluster are ignored.** Some nodes forward RBN traffic; those entries would duplicate what RBN already gives you, so they are dropped rather than drawn twice.
- Spots are held for up to **15 minutes** — longer than RBN, because a human spot describes something slower-moving than a skimmer's automatic report.

> **Note - Not every node is equally busy**
>
> Cluster nodes vary enormously. In testing, one node produced two spots in four minutes while another produced seventy-five in fifteen. A quiet node looks exactly like a broken connection, so give it a few minutes before concluding anything.

4.6. About SOTA — summit activations

SOTA spots come from **spothole.app**, an aggregator run by Ian Renton M0TRT, used with his permission. That is worth knowing because it shapes how the Tab5 treats it:

- **Off by default**, so nobody polls somebody else's hobby server without choosing to.
- **Fetched once every two minutes**, which is plenty for activations that last an hour.
- **When it is unreachable, nothing happens.** The spots already on screen stay until they age out normally — no banner, no error, nothing to dismiss. A volunteer server being down for an afternoon is an ordinary state, not a fault in your radio.

Almost every SOTA spot carries a portable suffix — `EA2GM/P` rather than `EA2GM` — because the operator is on a summit rather than at home. That is why the callsign shown is the one that was spotted, exactly as spotted.

> **SOTA is nearly all CW and SSB.** Of a typical sample, none are in a digital mode. So a summit will rarely appear in your FT8 log; the value of the source is seeing where the activity is on the spectrum, and being able to tap it.

4.7. Refresh and timing

- POTA is fetched about **once a minute** — spot rates on that service change on that sort of timescale, and the fetch briefly pauses the web UI's spectrum stream so the two never compete for the link.
- RBN and the DX cluster arrive continuously and are merged into the display every **10 seconds**.
- Ageing is re-evaluated every second, so spots dim and disappear on their own even when nothing new arrives.

If spots never appear at all, check in this order: WiFi connected (bottom bar), **Live spots** switched on, and — for RBN only — a callsign set.

5. Spot map

The **Live Spots** lane answers "who can I work?". The **spot map** answers the other question: **who is hearing me?**

It is a full-screen map of the stations that have reported *your* signal — the CW skimmer that copied your CQ, the FT8 station whose decode of you reached PSK Reporter, the WSPR receiver that logged your beacon — each drawn as a line from your grid square to theirs.

This is Uwe DL8UG's work, contributed to the project.

> **Note - Always on, no setting to find**
>
> The three feeds below connect from the moment the Tab5 boots and stay connected — there is nothing to switch on. Earlier versions asked you to enable it first and only ran the feeds while the map was open; both of those were tried and dropped, in that order, on the operator's own instruction, once he found the list was still empty the first few minutes after opening it either way. See What it costs, running all the time.

5.1. Opening it

Open the settings drawer and tap SelfSpotter — the button right below **Need guidance?**, near the top of the drawer. The map opens over whatever page you were on, and **Exit** (top right) closes it. A Bluetooth keyboard's `Esc` closes it too.

> **Note - Not a swipe any more**
>
> Earlier versions opened this with a swipe down from the top edge of the screen. That gesture collided with the top bar's own Band/Mode/BW/Zoom taps and was easy to trigger by accident, so it was replaced with the drawer button above.

The button works from the panadapter, FT8 and WSPR pages alike — the drawer itself opens the same way from any of them (swipe in from the right edge).

The header shows your **callsign, dial frequency, and the current UTC date and time** — so a screenshot of the map stands on its own without needing anything else to say when or where it was taken.

There is nothing to switch on first — the feeds have been running since boot (see below), so the map is never starting from a cold, empty world.

5.2. What feeds it

Three sources, all of them reports of **your own** transmissions:

Source	Where it comes from	How fresh
CW (RBN)	A Reverse Beacon Network skimmer copying your CQ	as it happens
Digi (PSKR)	PSK Reporter's live feed, over MQTT	as it happens
WSPR	wsprnet.org, polled every 3 minutes	trails a cycle

All three need WiFi. **None of them transmits anything** — they are lookups of what other people have already published about you.

A report stays on the map for **half an hour**, then drops off. The map is a picture of who is hearing you *now*.

The **Source** checkboxes in the left sidebar show or hide each one. **Flush**, in the header beside **Exit**, empties the map immediately so you can start a fresh picture — useful when you change band. It does not stop any of the feeds; they simply begin filling it again. *(Contributed by Uwe DL8UG. It was in the sidebar until v1.14.4.)*

5.3. The three tabs

MAP draws a line from your position to each station that reported you, over a world map — filled land, lighter than the sea, with a coastline outline on top. Line colour matches the source: **blue** for CW, **amber** for digital, **green** for WSPR. A line older than 30 minutes fades rather than disappearing, so a quiet stretch still shows what was heard recently. Your own position is a dot, drawn on top of everything.

- **Drag** with one finger to pan.
- **Pinch** with two fingers to zoom, up to 50×. Zoom is anchored on your own station, so your QTH stays put while the world grows around it.

LIST is the same data as a table — **RECEIVER, GRID, COUNTRY, MODE, BAND, FREQUENCY, SNR, KM** and **AGE** — when you want the numbers rather than the picture. Every column is sortable; tap a header to sort by it.

GRID is the locator the receiving station itself sent, not one worked backwards from a position, so it is blank for CW skimmers: they report a callsign and not a location. **COUNTRY** names the *country*, which can differ from the DXCC entity the map plots — a Hawaiian station's marker sits on Hawaii, while its country reads United States.

CONDITIONS is HF propagation from hamqsl.com: day and night ratings per band group, plus solar flux, A and K index, sunspot number, geomagnetic field and signal noise. It refreshes about hourly, which is as often as the source updates.

5.4. What you must set first

Your grid square. The map draws lines *from* you, so without it there is nothing to draw from — the sidebar says so in red, and the map stays empty. Set it in **Settings -> Station -> Callsign & Grid square**.

Your callsign, for the same reason: it is what the three feeds are asked about.

QRZ.com callbook credentials are optional. RBN reports name the skimmer that heard you, not where it is, so the map looks that up on QRZ to place it. Without credentials an RBN line still gets drawn, using the centre of the skimmer's DXCC entity — right country, wrong town. PSK Reporter and WSPR carry a grid square in the report itself and never need the lookup.

These are set **in the web UI, not on the Tab5** — open `http://<tab5-ip>` in a browser and use **Miscellaneous -> Set QRZ Callbook login**. They are your QRZ.com username and password for the Callsign Lookup service, which is a different thing from the QRZ Logbook API key used for uploading contacts.

5.5. What it costs, running all the time

Worth knowing, since there is no longer a setting to weigh this against:

It holds a connection on your behalf, permanently. From boot, the Tab5 holds a live session to PSK Reporter's broker subscribed to your callsign, and polls RBN and wsprnet. That used to be something you asked for first; now it starts with everything else.

It costs memory that other things need. Measured on the bench, running the feeds costs about 6.6 KB of internal RAM and 2.6 KB of the DMA pool — and that DMA pool is the one that microSD mounts, USB and encrypted uploads draw on when they need it. Every unit pays this now, whether or not the map is ever opened.

The feeds run whether the map is on screen or not — opening it doesn't start them and closing it doesn't stop them — so the picture is already populated with whatever has come in since boot the first time you look, rather than starting from nothing.

5.6. If the map stays empty

Nobody has heard you yet. The commonest answer, and not a fault: the map only fills once you have actually transmitted and somebody has reported it. Call CQ, or let WSPR or FT8 run for a few cycles, then look again.

No grid square set — the sidebar says so in red. See section 4.

WiFi is down. All three feeds need it.

Only some sources are ticked. Check the **Source** boxes in the sidebar.

You are running CW but RBN spots are absent — a skimmer has to actually copy you. RBN coverage is thinner on some bands and at some hours than others.

6. Web UI

Access the panadapter remotely from any browser on your WiFi network.

6.1. Quick Start

1. Enable **WiFi** in the settings drawer
2. Open your browser to `http://qmx.local`
3. You'll see the same spectrum and waterfall as the Tab5

No installation, no configuration, and **no IP address to write down**.

> **Why a name and not an address.** The Tab5 picks its own network - it remembers up to six and moves to whichever is on the air - so its IP address changes without anyone deciding it should, and the only place it is shown is the Tab5's own bottom bar, which is no help when the Tab5 is in the shack and you are not. `qmx.local` does not change.
>
> If `qmx.local` does not resolve, the IP still works exactly as before - read it off the Tab5's bottom bar. Windows and macOS resolve `.local` names out of the box; a few Linux setups need `avahi-daemon` installed, and some guest/hotel networks block the multicast that makes it work.

6.2. On a phone

The layout is built for a landscape screen and reflows fine there. In **portrait** on a phone (v1.9.4), the top and bottom bars now scroll sideways if a control doesn't fit — swipe them like the decode list — and the page itself scrolls vertically if the whole layout is taller than the screen. Before this a portrait phone (Randy N4OPI, iPhone Safari) could lose the top and bottom bar controls entirely with no way to reach them.

6.3. Remote Control

The web UI provides full remote control:

- **Frequency** — tap the VFO to open the numeric keypad (drag it by its title bar to reposition; toggle between a 10-key and a phone-style digit layout), or click the spectrum to tune
- **Mode** — buttons for USB, LSB, CW, DiGi
- **Bandwidth** — selectors for SSB filter width or CW passband

- **Band** — jump between configured bands
- **Memory** — recall, edit and clear your 32 saved channels (see Memory channels below)
- **Zoom** — pinch or scroll to zoom the spectrum
- **RIT** — click the **RIT** pill to arm it, then click a caller to receive off your transmit frequency without moving the dial. See RIT

Everything mirrors the Tab5 display with sub-second latency.

6.4. Spectrum Waterfall

The waterfall is **live-streamed** from the Tab5 every ~100 ms. You see the same real-time signal activity as on the device itself.

Click anywhere on the spectrum to tune to that frequency. The waterfall updates continuously — no refresh needed.

Whole-band plan strip. Along the bottom of the view (above the status bar) is a colour-coded CW/Digi/Phone strip spanning the entire current band, mirroring the one on the Tab5. A draggable "visible window" marks the slice currently on screen — drag it, or tap anywhere on the strip, to retune — and a marker shows the VFO position.

Adjustable split. Drag the divider between the spectrum and the waterfall to give either one more room. Your chosen split is remembered in the browser.

> **In FT8 or FT4 mode the live stream pauses.** The browser stops streaming the spectrum/waterfall and shows a notice plus the log and upload controls instead. This is deliberate — while you're operating digital modes the stream would compete with the on-device decoder and the WiFi link, so pausing it keeps FT8 decoding and WiFi noticeably steadier. Switch the Tab5 back to Panadapter mode and the stream resumes automatically.

6.5. FT8 Control

When the Tab5 is in **FT8/FT4 mode**, the browser pauses the live spectrum stream (see Spectrum Waterfall above) and instead shows a **live TX status banner**, a **Call CQ** button, and the **log and upload controls** — download your ADIF, upload to QRZ/eQSL/LoTW, grab the diagnostic log. Everything else about operating FT8 (watching the decode list, tapping a station to reply) happens **on the Tab5 itself**.

Band presets. A dropdown beside the TX tone lists the standard FT8 or FT4 calling frequencies — the same list the Tab5's own **Preset** button offers, and it switches sub-mode and frequency together, so picking an FT4 frequency puts the Tab5 into FT4. The list follows whichever mode the Tab5 is in.

The status banner mirrors the Tab5's own TX label so you can watch the radio from another room: **red** while transmitting — including the "call 2 of 4" counter when a CQ stop limit is set — **amber** when a transmission is armed or a QSO is waiting, **green** on QSO complete, **orange** on timeout, and the persistent **"CQ stopped after N calls - no answer"** once an auto-stopped CQ run ends. The browser tab's title also shows a red dot while transmitting, so even a background tab signals when the radio is on the air.

A finished QSO waits for you. When an exchange completes, the browser shows `<callsign> QSO complete` in green and **leaves it there** until you do something else — every other message in that line clears itself after twenty seconds, this one does not. The radio does not wait: it returns to idle and carries on. The message is a record that the contact finished, kept for whoever walks back into the room *(Randy N4OPI)*. The Tab5's own label is unchanged; it is only the browser that has to cope with nobody watching.

Live spots on the browser spectrum

The **POTA** and **RBN** spots the Tab5 draws on its own trace are now drawn on the browser's spectrum too, from the same store on the device - so the two screens can never disagree about who is on the air.

- **Amber** is a POTA activation, **green** an RBN (CW skimmer) spot, and **grey** a station you have already worked *on this band*.
- Spots fade with age and are gone at 30 minutes, exactly as on the Tab5.
- **Click a callsign to tune to it** - frequency *and* mode (CW, DiGi, or USB/LSB by band). Bandwidth is deliberately left alone: the QMX keeps a filter per mode.
- Counts at the bottom corners - `< spots (3)` - say how many more are on this band just outside your window; click one to jump to the nearest.
- On a crowded band only so many callsigns fit; unworked and fresher spots keep their labels, and a spot that cannot fit one is not drawn at all rather than left as a line pointing at nothing.

Spots are sent only while the Tab5 is showing the panadapter, and only when they have actually changed, so they cost the WiFi link almost nothing between refreshes.

The decode list, from another room

In FT8/FT4 the browser used to show a transmit banner and nothing else - you could see that your radio was transmitting, but not who was answering. The FT8 panel now carries the **decode list**: callsign, message, **country**, SNR, **DT and DTR**, audio tone, **distance and bearing**, slot (E/O) and age.

DT and DTR are two different quantities, not a repeat. DT is corrected against the band consensus and is character-for-character what the Tab5 prints — the WSJT-X-style number, blank until two separate stations have been heard. **DTR** is the raw boundary-relative offset and carries about 0.5 s of receive audio latency, so it is not comparable with WSJT-X, but it is the one that moves when a transmitter's timing moves. Hover either heading for the same explanation on the page. *(Randy N4OPI, who put the two screens side by side and found them disagreeing under one heading.)*

The country column replaced a GRID column, because the grid was already sitting in the message text next to it. The country is the one thing in that row you cannot read off the message, and unlike the Tab5 — where it has to squeeze into three letters — the browser has room to spell it out. Both screens work it out the same way from the callsign, so they cannot disagree. *(Randy N4OPI)*

The **KM** and **BRG** columns arrived in v1.8.3 (asked for by Tony Abbey). They sit after the tone, in the same order the Tab5 uses, and the header reads **MI** instead of KM if you have *Distance in miles* ticked. The Tab5 works the distance out and sends it, rather than the browser calculating its own, so the two screens cannot disagree — and where a station has not sent a grid the column shows a dash rather than a made-up number.

It is the **same list the Tab5 shows**, not a second opinion: the ordering comes from the device (the station you are working first, then anything addressed to you, then CQ calls, then strongest signal) and your Options-window settings are applied to it. The station being worked is highlighted, and anything carrying your own callsign stands out, as on the Tab5.

Click a station to work it (at the operator's own request). The confirm dialog states what will happen, and then the browser runs the **same intelligent Transmit a Tab5 row-tap runs**: anything that opens an exchange starts the **full automatic QSO** — whether you are answering a CQ or answering someone who called *you* — because you are not in front of the radio to click each step. A closing 73 still goes out as a single message. The outcome comes back in plain words under the TX banner - "Armed: ...", "Working X (auto QSO)", or a refusal if the station has aged out of the list, because transmitting at a station that may have left is worse than asking you to click again.

You can pick your next station before the current one has finished. From the moment your closing 73 is armed the exchange is settled, but it is not *over* for the best part of a minute — long enough for the caller you wanted to drop off the list. Clicking one during that window used to answer "Busy: working X" and discard the choice. It is now **held** and acted on the moment the machine is free, reported as "Queued: X, after this exchange". The callsign is what is remembered, not the prepared message, so the reply is built fresh from the live decode when it is actually sent. **Auto Pounce** on the Tab5 itself behaves the same way.

Who is calling you. Below the table, the **pileup** - stations answering you while you are busy - each clickable with the same confirm. And the **grey-list** (stations the auto pickers skip after repeated no-answers) is shown with a one-click clear, so "why is it ignoring that station?" is answerable from another room.

Switching the Tab5 back to the panadapter

If you left the Tab5 in FT8/FT4 and want the spectrum back, use the **Radio** menu in the bottom bar — it carries the Panadapter <-> FT8 switch.

Next to **Call CQ** is **TXCQ ANY / EVEN / ODD**, which chooses the 15-second slot your CQ transmits in — the same three states as the button on the Tab5, and the same setting underneath, so changing it on either surface is reflected on the other. ANY takes whichever window comes first; EVEN and ODD lock it.

Call CQ from the browser

Asked for by Dennis WN4FLA. A CQ run that has timed out, or that has reached its CQ stop limit, otherwise needs a walk back to the Tab5 to start it again. The **Call CQ** button under the status banner does it from wherever you are.

- **It asks first.** The button **keys the radio**, and a mis-click from another room should not put a carrier on the air, so it confirms ("Start calling CQ on the Tab5?") before anything is sent.
- **It calls exactly what the Tab5 would.** The active CQ preset, the current TX tone (honouring **TX Hold**, or picking the nearest clear slot as usual) and the **TXCQ ANY / EVEN / ODD** parity are all the ones set on the device — the two buttons share one code path, so they cannot drift apart. To change any of those, long-press **Call CQ** on the Tab5.
- **It takes about a second.** The request is handed to the Tab5's own display task rather than acted on inside the web request, so the button greys out and reads "Calling..." briefly. Watch the status banner, not the button, to see the CQ start.

- **Only in FT8/FT4 mode.** The button is part of the FT8 panel, and a request that arrives when the Tab5 is not in FT8 is discarded rather than queued — so it can never fire minutes later, unasked, when you next switch modes.
- If your callsign and grid are not set, the Tab5 shows the reason on its own screen and nothing is transmitted.

> **Not yet confirmed on the air.** The endpoint, the hand-off, the preset/tone/parity reuse and the error path are all verified on hardware; the final key-down is inferred from sharing the Tab5 button's code. Please report how it behaves.

Apart from Call CQ, transmit is still initiated on the Tab5 — replies and pounces need the decode list in front of you, and only one interface should be keying the QMX.

6.6. Mid-QSO controls

While an exchange is running, a **Mid-QSO** row appears under the FT8 controls:

- **Re-send** — send the current message again
- **RR73** / **73** — jump straight to either
- **Pause** — skip exactly one transmission and keep the exchange
- **Cancel** — stop transmitting and end the exchange

Pause is the one to reach for when you want to listen on your own slot: to check the frequency is still clear, or to hear whether the station is really answering *you*. It shows a **pause** symbol normally and a **play** symbol on yellow while a transmission is being skipped, and pressing it again takes the pause back. A slot you skip on purpose never counts against giving up on the contact.

The Tab5's own button does exactly the same thing, so the two screens always agree about whether you are paused.

Randy N4OPI asked for this on the web page specifically — he runs his station from another room and rarely touches the Tab5 itself.

6.7. Choosing your TX tone

TX tone on the FT8 panel opens the same picker the Tab5 has, against the same live occupancy of the 200-2800 Hz window - it is the device's own map, the one its automatic clear-slot picker uses, so a slot shown busy here is one the device would avoid anyway.

- **Green** is free, **red** a station or its guard band, **pink** the station you are working, **white** where you will transmit.
- **Click a slot** to choose it - a mouse can go straight there, where the Tab5 needs a drag. **−50 / +50** nudge, and **Find a clear slot** walks outward to the nearest free one, exactly as the automatic picker does.
- **TX Hold** keeps the exact tone you picked. With it off, each transmission takes the nearest clear slot, and a CQ that gets clashed moves itself.
- **Grey means unknown, not free.** If nothing has been decoded yet the device has no picture of the band, and the strip says so rather than showing reassuring green.
- **The occupancy picture refreshes itself every 3 seconds while the picker is open** (v1.9.4) — it used to be read once, when you opened it, so a slow decision (reading the strip, weighing E vs O) could cross a 15-second slot boundary and land you on a slot that filled in while you were choosing. Your own in-progress pick is never disturbed by the refresh, only the busy/clear colouring under it.

Changing the tone mid-QSO is fine - your partner tracks your time slot, not your audio frequency - but it is refused **mid-burst**, and the reason is shown rather than the change silently half-applying.

Both time windows, always (asked for by Roy KI0ER). The strip is two rows - **EVEN above, ODD below** - because two stations only collide if they transmit in the *same* window, and only you know which window you are about to pounce into. The verdict says where your pick stands: "*Clear in EVEN - busy in ODD*", or judges just your own window once a transmission has fixed it. The Tab5's own mini strip and picker show the same two rows.

6.8. Memory channels

The **Memory** button opens your 32 channels as a grid. **Tune** recalls one - frequency and mode together - and **Edit** changes or clears it. An empty slot offers the frequency you are on now, so storing the current VFO is two clicks.

A frequency outside an amateur band is **refused**, exactly as the Tab5 refuses it: a channel you cannot legally tune is worse than no channel.

6.9. Antenna Tune from the browser (QMX 1.04+)

With a QMX on firmware 1.04+, an **Antenna Tune** button appears in the bottom bar. It **keys the radio with a steady carrier** the moment you click it - there is no confirmation step, because tuning into a load is an ordinary part of operating and the 60-second stop is the real safety net.

While it runs, a **panel appears in the middle of the page** with the live **SWR**, the **power** in watts, and the **seconds left** before the tune stops itself, plus a **STOP Tune** button. The panel stays up whatever else you click, so you can keep both hands on an ATU and still read the SWR; the rest of the page carries on working underneath it. The SWR figure is also colour-coded - green below 1.5, amber below 2.5, red above - and the number itself is always shown, so the colour is never the only thing telling you.

The bottom-bar button doubles as the same readout and the same stop, for when the menu happens to be open.

Both ends carry a **60-second safety stop**: the device's own timer fires even if the browser tab dies, and the radio's prior mode is restored, never left keyed. Stopping Tune from the QMX's own front panel is honoured too.

6.10. Settings, with a real keyboard

The **Settings** button in the bottom bar edits the things you *type* - which is exactly what the Tab5's touchscreen is worst at:

- **Your callsign and grid.** Previously reachable from a browser only by downloading the config file, editing it and uploading it back.
- **The three CQ messages**, which preset is active, and the CQ stop limit.
- **The FT8 include/exclude filter terms** (both pairs) and the filter toggles.
- **Everyday switches**: POTA and RBN spots, PSK Reporter, grey-listing, distance units, I/Q balance, band-plan region, and the QMX volume in dB.
- **WiFi**: add another network from a laptop. The Tab5 remembers up to six, and a **Preferred network** field (under network name and password) decides which one wins — see below.
- **Show RIT button**, matching the Tab5 setting — hide the RIT button on the panadapter if you never use it.
- **The rest of the radio and display settings**, which used to be Tab5-only: CW pitch, IF calibration, SWR protection, the battery charge limit, the 180° screen flip, Fox/Hound mode, simulation mode and the spot mode filter.

Saved straight to the Tab5 - its own settings drawer shows the same values next time you open it.

Preferred network

The Tab5 remembers up to six networks. Before this field existed it always took whichever remembered network was **loudest**, which is the wrong answer on a site with more than one — a DMZ and a house network, say, where the loud one is not the one you want the panadapter on.

Set **Preferred network** to the SSID you want and the Tab5 takes it over any stronger alternative whenever it is reachable. Leave it empty and nothing changes, so nobody who does not need this is affected.

Both this field and **Network name** list the networks your Tab5 actually remembers, under the field itself. Only a name from that list will work.

> **Warning - v1.16.3 — the field was there but it did almost nothing**
>
> In v1.16.3 the preference was only consulted after the Tab5 had **failed** to connect twice. So it worked when your usual network was down, and did nothing in the case it was written for: both networks up, the usual one answers immediately, and you wanted the other. It also did not take effect until the next reboot. If you set it on v1.16.3 and saw no difference, that is why — and it was not something you did wrong.
>
> **Fixed in v1.16.4**: the preference is read when the Tab5 connects, and saving it moves the Tab5 straight away *(Randy N4OPI)*.

If the preferred network does not answer, the Tab5 falls back to the strongest remembered one after a few seconds, so an unreachable preference costs you about ten seconds at start-up and nothing else. It is re-tried on every later fallback too, so it wins again the moment it comes back.

Switching to a network the Tab5 already knows needs only the **Network name** — leave the password blank and the stored one is used. (Up to v1.16.3 a blank password made the Tab5 discard the name silently, which read as the setting not saving. Fixed in v1.16.4.)

The FT8 transmit tone saves properly too, from v1.16.4. Setting `tx_tone_hz` on the settings page used to answer *Saved* and write it down without telling the radio, so bursts kept going out on the old tone until the next restart. The tone control on the FT8 page was never affected.

> Not carried in a config backup. A restored config does not bring the preferred network with it; set it again on each unit.

Also here: the **Display & waterfall** group - brightness, waterfall black level, contrast, FFT window, colour map, display sleep, the 180° flip, the dB scale range, spectrum smoothing and **spur suppression** - the controls you want to tune from a laptop *while watching the Tab5's screen*. Every change applies live and is stored, exactly as the drawer's own sliders do, and the browser's own spectrum and waterfall follow them as well. **Adaptive floor is gone from both the browser and the Tab5 drawer** as of v1.8.3 — it could not change anything (see Spectrum & Waterfall), and a control that cannot change anything is worse than a missing one. The **auto-answer robot** switch is here too, carrying the same permanent warning the Tab5 shows beside it: it transmits unattended - never leave it running unsupervised.

One deliberate omission: **your WiFi password is never sent to the browser**, so the field is always blank; leave it blank and the stored one is kept.

6.11. Help, in the browser

The **Help ?** button in the bottom bar opens the same two doors the Tab5 has, served **by the Tab5 itself** - so it works with no internet at all, and the text always matches the firmware you are running.

- **A list of symptoms and questions in plain words** - "My radio is not showing up", "Nothing appears in the decode list", "How do I change what my CQ says?". Pick the one that fits and the manual opens at the section that answers it.
- **Rows the Tab5 can see are happening right now are highlighted** and moved to the top, exactly as on the device. It ranks; you choose.
- **The whole manual**, with a Contents list, Back, and links between chapters.

See Getting Help for what the device can detect and why it never navigates for you.

6.12. CAT Control (Advanced)

The **CAT** section lets you send raw Kenwood-style commands directly to the QMX:

```
FA;         -> reads current frequency
FA14074000;  -> sets frequency to 14.074 MHz
MD;         -> reads current mode
MD2;        -> sets USB mode
```

This is for advanced troubleshooting — most users don't need it.

6.13. Bottom Bar Menus

The bottom bar groups its actions into four popup menus, plus a battery indicator (e.g. `🔋 87% (8.0V)`):

QSO Logs (n) ▲ — only visible when QSOs exist; *n* is the QSO count:

- **ADIF download** ↓ — QSO log as an ADIF file (import into WSJT-X, EQSL, etc.)
- **Today only, dated file** ↓ — just today's contacts, named `qso-YYYY-MM-DD.adi`. For anyone who files each day's log separately, the date is already in the filename rather than something to add by hand afterwards *(Gyula HA3HZ)*
- **ADIF restore** ↑ — merges a previously downloaded (or any logger's) ADIF file back into the log, skipping any contact already there (matched on callsign, date and time). For after an erase-and-reinstall: **Config upload** never touches the QSO log on purpose, so this is the only way to get worked-station history back *(Randy N4OPI)*. A prompt lets you say whether the restored contacts should be marked as already uploaded to QRZ/eQSL/LoTW (the usual answer is yes, since "restore" almost always means a log that was already sent) or as not-yet-uploaded, so the next upload sends them. **From v1.11.3 the first prompt asks whether the file contains *corrections*.** Say yes and a contact already logged with the same callsign, date and time is **replaced** by the version in the file, instead of being skipped as a duplicate — which is what a merge otherwise has to do, since a correction keeps the very fields it is matched on. This is the way to repair records in another logger and put them back *(Gyula HA3HZ: "the corrected file cannot be installed, the previous incorrect version remains")*. Corrected contacts are sent to QRZ/eQSL/LoTW again afterwards, because the copy held there is the one carrying the error, and all three ignore a repeat. Callsign, date and time still cannot be changed this way — a contact with a wrong date is a different contact

- ↳ **Restore from SD card** — the same merge, but read straight off the microSD card the Tab5 already backs up to, with no file to find and nothing to choose. Contacts already logged are skipped, so pressing it twice does nothing. The card mirrors the *present*, so a deletion that survives a restart is on the card too — which is why the copy from just before the log last got smaller is kept beside it as `qso.prev.adi` (download it from **Files ▲ -> SD Files** and feed it to **ADIF restore** ↑ if you need it) *(Gyula HA3HZ)*
- **QRZ upload** ↑ — upload ADIF to QRZ Logbook (requires API key on first use, saved for future sessions)
- ↳ **Change QRZ API key** — appears once a key is stored, and replaces it. New in v1.8.3: before that the prompt only ever appeared when *nothing* was stored, so a key typed wrongly or later reissued could not be changed from the page at all (reported by Brian WA6JFK)
- **eQSL upload** ↑ — upload ADIF to eQSL (requires username/password on first use, saved)
- ↳ **Change eQSL login** — same, for the eQSL username and password
- **LoTW setup** / **LoTW** ↑ — upload ADIF to ARRL's Logbook of The World (see LoTW Upload below)
- **Cloudlog upload** ↑ — upload to your own Cloudlog or Wavelog. Asks for the server address, your API key and the station profile ID on first use. Plain `http://` is accepted for a server on the same network as the Tab5, given as a numeric address such as `http://192.168.1.20`; anything else needs `https://`. The check is repeated at every upload, so away from home the upload refuses instead of sending your key across someone else's network. If the server is on your own LAN this is the only upload that needs no internet at all *(Mark G4MEM)*
- ↳ **Change Cloudlog server** — appears once a server is stored, and replaces the address, key and station profile
- **Check log** — looks through the log for records that are missing something a programme will want: a callsign, a date that is not in the right form, no station callsign, a reference that does not look like a reference, or — if you say you were activating — no reference of your own. It asks which of those two you mean when you press it. It can only tell you a record is not *obviously* incomplete: the Tab5 cannot see POTA's or SOTA's rules or their databases, so it never tells you a file will be accepted. Born from "*I wish POTA and SOTA had a feature allowing you to test your file for syntax and completeness without submitting it for credit*" — two trips to a park to get contacts to test with *(Don Adams WB0LQW)*. The Tab5's own Activation panel shows the same count under the contact total, so you can read it without a laptop.

- **View / edit log** — opens the QSO log right in the browser (call, mode, band, frequency, date/time, reports, grid — newest first). **Click any column header to sort** by it (click again to reverse) — sorting by Date groups an activation's QSOs together. **Search** filters the list as you type, matching any part of a callsign, band, mode, date, grid or reference; several words must all match, so `ea 20m` narrows to Spanish contacts on 20 m. A search that finds nothing says so — "*so this one has not been worked*" — which is the question worth asking of a log *(Gyula HA3HZ)*. **Tick the box on any row** and **Export selected** saves just those contacts as an ADIF file on your computer; the box in the header ticks everything **currently shown**, so a search and one tick gives you a single day, band or park as its own file. The export carries each record exactly as the Tab5 wrote it, so fields the table does not display are preserved. Each row has a ✕ to delete that one record, and a **Delete all** button clears the whole log (no undo, so it asks you to type `DELETE` to confirm — download the ADIF first if you want a copy). Handy before a POTA activation: start with an empty log and the ADIF at the end is exactly the file you submit. **Every field is editable** *(Gyula HA3HZ)* — the two reports, **Grid**, **Their ref**, and from v1.12.0 also the callsign, band, mode, frequency, date and time. Which record a logbook matches is a decision for the logbook, not one the Tab5 should make on its behalf. **Be aware of what that means**: callsign, band, mode, date and time are exactly what QRZ, eQSL and LoTW match a contact on, so changing one of those turns the record into a different contact as far as they are concerned. If the QSO has already been uploaded, the copy the logbook holds is unchanged. Click one and type the corrected value, or leave it empty (for a report, that records that none was exchanged). **Their ref** is the park or summit the *other* station was activating: while you are operating, that reference is on the POTA spots page on your phone and in nothing the radio sends you, so you write it down and enter it when you get home *(Don Adams WB0LQW)*. It fills itself in when the contact came from a spot, whether or not you were activating yourself — that is what the ADIF `SIG`/`SIG_INFO` fields mean, and a hunter's log should say which park they worked. It is only *Park-to-Park* when you were in a park too, which is why the column is not called that. **An empty cell is not a fault**: the reference comes from the POTA spot feed matched on the exact callsign, and not everyone activating in a digital mode registers the activation — some simply start calling `CQ POTA` — so an unregistered activator produces no reference at all rather than a wrong one *(Don Adams WB0LQW)*. Type it in by hand if you have it. Type the reference alone — `US-1241`, `G/LD-049`, `DLFF-0123` — and the Tab5 works out the programme from its shape and writes both `SIG` and `SIG_INFO`; clear the reference and both go with it. **Grid** is the other station's locator, correctable because it is the field most likely to be wrong through no fault of yours — a partner's grid arrives once, in their first message, and a marginal contact can complete without it ever being heard cleanly *(Gyula*

HA3HZ, who was correcting them in ADIFMaster). It is checked as a real Maidenhead locator, four or six characters, and refused otherwise: it drives distance and bearing on both screens and goes to three logbooks, so a typo would be a wrong measurement presented as a real one. Clearing it removes the field, which is the honest state when no grid was ever exchanged. Malformed values are refused rather than stored: a date must be eight digits, a time four or six, and a grid a real Maidenhead locator — those fields drive distance, bearing and three logbook uploads, so a typo would become a wrong measurement presented as a real one.

Files ▲:

- **Config download** ↓ — all settings as a text file (backup or transfer to another Tab5). It is also the way to change a stored credential by hand: the `qrz_key`, `eqsl_user` and `eqsl_pass` lines are in there in plain text, and an edited file uploaded back takes effect on the next upload with no restart. Note it holds your WiFi password in plain text too, so treat the file accordingly
- **Config upload** ↑ — restore settings from a backup file
- **SD Files** — opens the **microSD file browser** (`http://<tab5-ip>/files`): browse the card from your computer without pulling it — download logs and config backups, upload files, delete
- **Diagnostic download** ↓ — downloads **both** diagnostic logs: the live session log (always on, nothing to enable) and the flash-persisted copy from before the last reboot/power-off

Miscellaneous ▲ also carries **Prepare for flashing** — closes the USB host cleanly before you unplug the radio or reflash the Tab5, so the QMX is not left half-open.

Radio ▲ — the things that reach the radio itself:

- **Switch to FT8/FT4** / **Switch to Panadapter** — changes the Tab5's own screen from the browser. The label always says where you are going, and it follows the Tab5 if you switch there instead
- **Memory channels** — your 32 channels: click one to tune to it, drag one onto another slot to move it, or onto the bin to clear it
- **Let me use the QMX menus** — stops the Tab5 talking to the QMX so you can use the radio's own menus or its Band Configuration terminal. A bar appears until you hand it back. See Settings
- **Antenna Tune** — QMX 1.04+ only; keys a steady carrier with live power and SWR, and stops itself after 60 seconds

- **Activation** — start or stop a POTA/SOTA session from the browser. While one is running a badge appears in the top bar with the reference and the contact count, so it cannot be forgotten on the drive home
- **Radio menus** — the QMX's own 80×24 menu system, in the browser. Arrow keys, Enter and Back on screen, and your real keyboard works too. Needs the radio's second USB serial port switched on first. See Radio Menus

Miscellaneous ▲:

- **Tab5 screenshot** — current display as PNG, including any open pop-up (band/mode dropdown), not just the base screen

- **Reboot the Tab5** — restarts it there and then; receiving stops for about ten seconds. Until now a restart was only offered after a firmware update, which is no use when you are in another room and something is stuck *(Randy N4OPI)*

- **Power off the Tab5** (from v1.20.1) — shuts the unit down from the browser. It does **not** come back on its own; somebody has to press the button, or cut and restore its power. It asks for confirmation. The radio is put back into receive first, so a transmission in progress cannot be left keyed. Measured to stay off with 5 V still present on the USB-C input. The same control is in the Tab5's own drawer, under Restart, in the Advanced section *(Randy N4OPI, whose headless QMX+ runs from 18650s the Tab5 was draining after the radio shut itself off)*

- **Port mode** — chooses what owns PORT.A, the same connector the relay below uses: **Relay** or **Unit GPS v1.1**. It sits directly above the relay row because it decides whether that row can do anything. Switching is applied live through the same sequencer the Tab5 drawer uses, so both screens always agree; a refusal comes back as `refused: PORT.A mode not changed: a relay pulse or power cycle is running, or the Unit GPS UART failed to start`, and the row snaps back to what the device is actually in. When **Unit GPS** owns the port the whole relay row greys out with the reason in its tooltip. See Settings ▸ Port mode and Unit GPS Time Source.

- **Power-cycle relay** — pulses one of two Tab5 GPIO pins (GPIO53 or GPIO54) for a chosen level and duration. Wire an external relay's trigger input to the pin and its contacts to your QMX's PWR_ON/GND **signals**, and this lets you power-cycle the radio remotely — the piece a remote firmware upgrade otherwise needs someone at the bench for, since the QMX always needs a manual power cycle after a Tab5 flash.

  > **Warning - Experimenter feature — you fit the connector yourself**
  >
  > **The QMX does not have a PWR_ON/GND jack.** Those signals have to be extended from the main board out to a connector you add yourself. Do not go looking for an existing socket, and do not connect them to some other connector that looks plausible *(Randy N4OPI)*. Randy has documented the whole modification, including a schematic — see Power-cycle relay: building the interface below.
  >
  > Both pins are held as **driven outputs** from boot — they never float. From v1.11.3 they rest on the **inactive side of the active level you have chosen**, so a relay wired to them cannot be closed by the Tab5 booting, crashing or being reflashed. Before v1.11.3 they always rested LOW, which for an active-Low setting meant the relay was held **closed** from power-on until the first pulse released it *(Randy N4OPI)*. Changing the active level re-applies the resting level immediately.
  >
  > !!! warning "Not while Port mode is Unit GPS" The relay and the Unit GPS v1.1 are **the same two pins**, so only one of them can be plugged in. With Port mode on **Unit GPS** the row above is greyed, `gpio_pulse` and `gpio_power_cycle` answer `PORT.A is in Unit GPS mode`, and nothing is driven — including the pins resting. Switch back to **Relay** first (and unplug the GPS). See Port mode.

- **Keyboard shortcuts** — assign what the Tab5's snap-on keyboard does (see below)
- **Reset settings** — clear stored settings back to defaults (see Troubleshooting)
- **Reset WiFi** — clear just the WiFi/network state

Power-cycle relay: building the interface

*Designed, built and documented by **Randy N4OPI**, and published here with his permission. The Tab5 side of this — the pin, the level and the pulse length — is all the firmware does; everything below is the hardware that turns a pulse into a QMX that switches itself on.*

Why bother. A Tab5 firmware upgrade is a warm reset, and a warm reset with the radio attached always leaves the QMX needing a manual power cycle. That one step is what otherwise keeps a remote station from being upgradeable from another room — or another country.

 Schematic: M5Stack Tab5 power switch for QRP Labs QMX(+) — a 4-pin Grove plug on the Tab5's EXT socket feeds a 100 ohm resistor into a PC817C optoisolator, whose output drives a 3.5 mm stereo jack on the module, reaching a 2.5 mm TRS jack wired to PWR_ON and GND inside the radio through a patch lead

Parts

Ref	Part	Notes
P1	4-pin Grove plug (HY2.0-4P)	Into the Tab5's side EXT socket. Its four pins are `EXT 5V`, `G53`, `G54`, `GND`
R1	100 Ω, ¼ W, 5 %	In series with the optoisolator's LED
U1	PC817C optoisolator	Keeps the Tab5 and the radio electrically apart — the reason this is safe
J1	3.5 mm stereo audio jack	The output side, mounted on the module — **not** the jack that goes in the radio
—	3.5 mm to 2.5 mm TRS patch cable	Joins the module's 3.5 mm jack to the 2.5 mm jack in the radio. Tip to tip, sleeve to sleeve

Wiring

1. **In the radio.** Fit a **2.5 mm TRS jack** somewhere accessible on the case and wire it internally: **tip to `PWR_ON`, sleeve to `GND`**. This is the only irreversible part of the job, and the only part that involves opening the QMX. Randy brought his out on the rear panel beside the ATU RF input.

2. **The interface.** The Tab5 pin you choose — **G53 or G54** — goes through **R1** into the PC817C's LED, with the Grove plug's `GND` completing that side. The PC817C's phototransistor goes across **J1**, the module's 3.5 mm jack, tip and sleeve, so a pulse from the Tab5 briefly connects `PWR_ON` to `GND`. **The phototransistor side is polarity-sensitive — follow the schematic**, and note that the `EXT 5V` pin is not used at all.
3. **Assembly.** In Randy's words: "*I built it onto a small chunk of breadboard, but with just 2 components it could just as easily be connected point to point and sealed up inside a piece of heatshrink tubing. Make sure you insulate your wires so the heatshrink doesn't squeeze them together and create a short.*"

Settings that work

In **Miscellaneous ▸ Power-cycle relay**: the pin you actually wired (**G53** or **G54**), active level **High**, and **2000 ms** — "*I've found 2000ms to be sufficient*". The firmware's own default is 1000 ms, so this is one to set and save; it is remembered from v1.10.9 on.

> **Note - Why the bottom-edge Dupont pins are not offered**
>
> A neater cable run would leave the Tab5's bottom edge rather than the side, and Randy asked about exposing `G0`, `G1`, `G49` and `G50` for it — then talked himself out of it in the same message: "*The Dupont connector isn't keyed, so a person could make a disastrous mistake if they are also trying to power the Tab5 using the 12V IN pin.*"
>
> There is a second reason he could not have known: **`G0` and `G1` are the I2C bus for the snap-on keyboard**, so a relay pulse there would fight a keyboard on any Tab5 that has one. `G53` and `G54` are on a keyed connector and are held as driven outputs from boot, which is what makes them safe to hand to people — so those two are the only pins the firmware will pulse.

Keyboard shortcuts

Miscellaneous ▸ Keyboard shortcuts assigns what the Tab5's snap-on keyboard does. Each row is a modifier, a key and an action; **Add shortcut** makes a new one, **Remove** deletes it, and **Restore defaults** puts back the nine the firmware ships with.

There are **25 actions** — opening any of the setup windows, the manual, the log, the radio menus, plus zoom in and out, brighter and dimmer, flat spectrum, and releasing the radio to its own front panel. **Alt is left completely unused** by the defaults, so it is free for your own bindings.

The editor is here rather than on the Tab5 for a simple reason: assigning shortcuts is a setup job done once, and doing it on the Tab5 would mean typing on the very keyboard you are configuring.

> **Note - Nothing that transmits can be bound**
>
> A button you deliberately press is one thing; a two-key combination that a slipped finger can produce is another.

6.14. Tab5Shot - what the screen looks like right now

Tab5Shot in the bottom bar opens a live `/ss.bmp` capture of the Tab5's screen in a new tab. It is the quickest way to put what you are seeing into a bug report or a message to the group, without a camera and without reaching for the device.

> **Note - It stops animations while it reads**
>
> The capture comes from the live framebuffer, so the waterfall pauses for the moment it takes. Nothing else is affected.

6.15. Config backup, restore and edit

The **Config down / Config up** buttons in the bottom bar save every setting to a plain text file, so you can edit it on a PC, put it back, or share part of it.

Config down downloads `qmx-config.txt`, an INI-style file grouped into sections:

Section	What is in it
`[settings]`	Callsign, grid, WiFi SSID and password, CW pitch, IF trim, I/Q balance, flat spectrum, zoom, colour map, brightness, dB range, EMA, GPS toggle, keypad layout, QRZ key, eQSL user and password
`[cq]`	The three FT8 CQ-message presets, and which one is active
`[ft8_filters]`	The CQ-run include/exclude filter terms and toggles
`[memories]`	The 32 memory channels, one per line: `slot = freq_hz, mode, label`

Config up picks an edited file and **merges** it back. Only the keys and sections actually present in the file change; everything else is left alone, and unknown keys are ignored, so a file from a newer firmware still loads. Memory channels apply at once, other settings after a restart.

> **Danger - The file contains your passwords in clear text**
>
> It holds your **WiFi password** and your **QRZ and eQSL credentials** unencrypted, because it is meant to work as a complete backup. Keep it private, and strip those lines before sending the file to anyone.

Three things it is good for:

1. **Back up before a clean flash.** Config down, erase and flash, Config up - seconds instead of re-typing everything on glass.
2. **Edit with a real keyboard.** Faster for callsign, grid, CQ messages, or entering a batch of memory channels.
3. **Share a band plan.** Copy just the `[memories]` section into a message; another operator pastes it into their own file and uploads it, and their other settings are untouched.

6.16. LoTW Upload

Uploading to ARRL's **Logbook of The World** requires an existing LoTW account and a callsign certificate (made with ARRL's TQSL program).

One-time setup: open the **QSO Logs** menu -> **LoTW setup**. A guided two-page window walks you through it:

1. **Page 1** explains how to export your callsign certificate from the TQSL program on your PC (**Callsign Certificate -> Save the Callsign Certificate**, which produces a `.p12` file), with a button that opens ARRL's own instructions.
2. **Page 2** imports the `.p12` file, its passphrase, and your DXCC entity (plus optional CQ/ITU zones, and **US state and county** — see below). The `.p12` is parsed **in the browser** — the passphrase never reaches the device.

After setup the button reads **LoTW** ↑. Each click signs all not-yet-uploaded QSOs on the device with your certificate and uploads them to lotw.arrl.org.

US stations: fill in state and county. These were not being sent at all before, which meant US operators' uploaded QSOs earned **no Worked All States and no county credit** — for them or for the stations they worked. Fill them in if your TQSL station location has them. The county is the **name on its own** (`Arlington`), not `VA,Arlington`. Operators outside the US can leave both blank.

Certificate renewal — LoTW certificates expire roughly every 3 years. Use the ↳ **Change LoTW certificate** row beneath the upload button to re-run setup (**Ctrl-click** on **LoTW** ↑ still works, and was the only way in before v1.10.5). From v1.3.3, re-submitting the *same* certificate no longer re-uploads your whole log — only an actually-changed certificate resets the upload position, because a new key means every QSO has to be re-signed. So you can go back in to add a state and county without resending everything.

What the result tells you. From v1.10.5 the message shown after an upload is **LoTW's own reply**, not just our count. That matters: LoTW accepts a *file* and processes the contacts in it afterwards, so a figure like "22 uploaded" only ever described what was sent. If contacts are being rejected, the server's own words are what say why.

If an upload does not appear on LoTW, check ARRL's queue status before assuming a fault — the queue has run hours behind at busy times. LoTW rejects a malformed file at upload time, so anything that reached the queue was signed correctly.

6.17. Upload Behaviour

Uploads work **while FT8 or FT4 is actively running** — the panadapter briefly pauses the FFT and SD-archive activity during the HTTPS transfer, then resumes automatically. A result is shown once the upload completes, reporting how many QSOs were sent.

Each upload remembers where it left off — re-uploading skips QSOs that were already sent in a previous session.

6.18. Network Requirements

- **WiFi must be on** (settings drawer)
- **IP address shown** in settings (or static IP if you prefer)
- **Both Tab5 and browser on the same LAN** (no internet needed)
- `qmx.local` resolves on most networks; guest/hotel WiFi often blocks the multicast it needs, in which case use the IP
- **5 GHz WiFi works** but 2.4 GHz is recommended (longer range)

6.19. Limitations

- **One browser at a time** — the Tab5 streams the spectrum to a single browser. Open a second one (or a phone, or another tab) and it takes over; the first says "*another browser took the live view — click to take it back*" and stops, rather than the two fighting over it. Click that message to reclaim it.
- **Clock sync** — the time-sync window is still on the Tab5 only
- **Real-time chat** — no operator messaging
- **Export formats** — ADIF only (import to EQSL, WSJT-X, etc. yourself)
- **Latency** — ~200 ms typical (WiFi dependent)
- **Stale connection recovery** — if the browser's network drops without a clean disconnect (e.g., putting a laptop to sleep), the web view may freeze briefly before recovering. Recovery is automatic and capped at ~5 seconds.

Next: Set up Time Sync or explore Settings.

7. Web API

The Tab5 runs an HTTP server that exposes a REST API for remote control and diagnostics. All endpoints are on port 80.

7.1. Base URL

```
http://<tab5-ip-address>/api/
```

Example: `http://192.168.1.50/api/status`

7.2. Status Endpoints

GET /api/status

Returns the current state of the panadapter.

Response (JSON):


```
{
  "battery":  { "level": 85, "mv": 8000, "charging": false },
  "wifi":     { "ssid": "MyNet", "rssi": -65, "ip": "192.168.1.50" },
  "freq_hz": 14074000,
  "qmx_fw": "1_03_002",
  "mode": "DIGI",
  "band": "20m",
  "screen": "ft8",
  "ft8":      { "st": "armed", "text": "TX armed: CQ OZ1LAV JO65 - call 2 of 4
       (~7s)" },
  "passband_hz": 2700,
  "signal_dbm": -83.5,
  "zoom": 1.0,
  "pan_bins": 0,
  "cw_pitch_hz": 700,
  "if_cal_hz": 0,
  "flat_mode": false,
  "utc_epoch": 1785340050,
  "qso_count": 42,
  "qrz_key_set": true,
  "eqsl_creds_set": false,
  "lotw_ready": true,
  "bandplan": { "lo": 14000000, "hi": 14350000, "segs": [ { "lo": 14000000, "hi":
       14070000, "c": "20B0FF", "l": "CW" } ] },
  "bt":       { "en": true, "conn": false },
  "tab5_fw": "v1.7.2",
  "bands":    [ { "name": "20", "center_hz": 14100000 } ]
}
```


Notes:

- `screen` is `"panadapter"` or `"ft8"`.

- `ft8` **is present only while the FT8/FT4 screen is live** — the engine does not run otherwise, so its text would be stale. `st` is one of `active` / `armed` / `done` / `timeout` / `wait` / `idle`, and `text` is the same line the Tab5 shows on its own TX status label. This is what drives the browser's TX banner and the red dot in the tab title.
- `bt` is the Bluetooth mouse: `en` is whether it is switched on, `conn` whether a mouse is connected right now. `en: true, conn: false` means it is scanning or waiting for a paired mouse to come back.
- `signal_dbm` and `bandplan` are `null` when unavailable (no spectrum yet; VFO outside a known band).
- `passband_hz` falls back to a per-mode default until CAT reports a real width.

7.3. Control Endpoints

POST /api/cmd

Send a command to the panadapter. The command name is the **`action`** key; each action names its own parameter key (there is no generic `value`).

Request (JSON):

```
{
  "action": "set_freq",
  "hz": 14074000
}
```

Supported actions:

Action	Parameter	Effect
`set_freq`	`hz`	Set VFO frequency
`set_band`	`hz` (band centre)	Switch band — recalls the frequency last used on it
`set_mode`	`mode: "USB"/"LSB"/"CW"/"DIGI"/"AM"`	Change mode (AM needs QMX firmware 1.04+)
`set_bw`	`hz`	Filter width — SSB at 1000 Hz and above, CW passband below
`set_zoom`	`zoom: 1.0/2.0/4.0/8.0`	Zoom level
`set_rit`	`hz`	RIT offset. Retuning clears it
`set_screen`	`screen: "panadapter"/"ft8"`	Switch the Tab5's own view. Applied within about a second (handed to the display task)
`set_ft8_mode`	`mode: "ft8"/"ft4"`, optional `freq_hz`	**Switch between FT8 and FT4**, optionally retuning. Omit `freq_hz` to keep the current frequency. This is an action rather than a setting because it also retunes the radio and clears stale decodes — read `ft8_op_mode` back from `/api/settings` to confirm
`set_activation`	`ref`	Set the POTA/SOTA activation reference
`cq_start`	*(none)*	Start a CQ run, as the Tab5's own **Call CQ** button does. **Keys the radio.** Only acted on while the Tab5 is in FT8/FT4 mode; otherwise discarded, not queued
`reply`	`call`	Work a decoded station — the same intelligent Transmit a Tab5 row-tap runs. **Keys the radio.** Outcome in `/api/status` `ft8.web_r`

Action	Parameter	Effect
`qso_override`	`what: "resend"/"rr73"/"73"/"cancel"`	Mid-QSO override, as the Tab5's own three buttons do. **Keys the radio** except for `cancel`
`clear_swr`	*(none)*	Clear a latched SWR-protection trip, same as tapping the Tab5's own fault prompt. TX stays refused until this is called
`tune_start` / `tune_stop`	*(none)*	Antenna Tune (QMX 1.04+ only). **Keys the radio continuously**; 60 s safety stop on the device. While running, `/api/status` carries a `tune` object with `watts`, `swr` and `secs` (seconds left before the safety stop); the object is absent when no tune is running
`pause` / `resume`	*(none)*	Release the radio so its own menus can be used, and take it back. While paused the CAT poll, the dead-stream watchdog and the stuck-decode watchdog all stand down
`greylist_clear`	*(none)*	Un-skip every grey-listed station
`ota_install`	`url`	Download and install firmware. Refused while transmitting or mid-QSO, with the reason in the reply body. Does **not** restart on its own
`ota_reboot`	*(none)*	Restart into firmware already installed
`usb_shutdown`	*(none)*	Orderly USB teardown before a reflash: radio to RX, CAT and audio closed, VBUS dropped
`usb_replug`	*(none)*	Power-cycle the USB port to recover a wedged device

Action	Parameter	Effect
`power_off`	*(none)*	Power the Tab5 down
`reset_settings`	*(none)*	Clear stored settings back to defaults
`reset_network`	*(none)*	Clear the stored WiFi/network state

Radio menus (the QMX's own terminal, on its second serial port): `open`, `close`, `key`, `text`, `term_view`, `qmx_ports`, `qmx_term_probe` — see Radio menus and `GET /api/term`.

Developer actions. Not referenced by any web-UI element, and easy to regret:

Action	Effect
`cat_raw`	Send one raw CAT string to the radio
`drawer`	Open the settings drawer, optionally at `scroll_y`, so a section below the fold can be screenshotted
`resmon`	Toggle the resource-monitor overlay
`time_redetect`	Re-arm the once-per-boot QMX GPS detection without rebooting
`help`	Open the on-device manual at a topic
`panic_test`	**Crash the device deliberately**, to prove the crash recorder works. ⚠ Radio **off** only — a panic reboot is a warm reset, which leaves the QMX needing a power cycle

⚠ **Check the response BODY, not the status code.** The reply is `{"ok":true}` on success — but an unrecognised action returns **HTTP 200** with the body `unknown action`, and a refused one (for example `ota_install` while transmitting) returns `{"ok":false,"error":"..."}`, also with HTTP 200. A caller that only tests the status code cannot tell success from a typo, and a mistyped action is then a silent no-op.

Effects are applied asynchronously — mode and filter writes are queued onto the CAT poll task, and `cq_start`, `reply`, `qso_override` and `set_ft8_mode` onto the display task — so read `/api/status` or `/api/settings` to see the result rather than assuming it from the response.

GET /api/tone

The TX audio tone and the live occupancy of the 200-2800 Hz window.

```
{ "hz": 1650, "hold": false, "min": 200, "max": 2800, "step": 50,
  "stations": 14, "busy": "0000000000111110...", "partner_hz": 1750, "busy_tx":
        false }
```

`busy` is one character per 50 Hz slot (`1` = a station or its guard band), sent as a **string** because 52 bits do not survive JavaScript's number type. `busy_e` and `busy_o` carry the **EVEN and ODD windows separately** — two stations only collide in the same window, and only the operator knows which one they are about to transmit into; `busy` remains the our-window/union view the automatic verdict uses. `stations` is how many decoded stations fed the mask - **zero means the band picture is unknown, not empty**, and a caller must say so rather than showing the all-clear.

POST /api/tone

`{"hz":1650}` and/or `{"hold":true}`. Out-of-range frequencies are clamped.

A tone change while a QSO is running is applied to the running exchange first (your partner tracks your slot, not your tone). Mid-burst it is **refused with 409** and the body carries the device's own reason - the preference is not stored either, so the two can never end up half-committed.

GET /api/memory

All 32 memory channels: `{"slots":[{"idx":0,"occupied":true,"freq_hz":14074000,"mode":"DIGI","label":"FT8"}, ...]}`. Empty slots carry only `idx` and `occupied:false`.

POST /api/memory

Write one channel - `{"idx":3,"freq_hz":14074000,"mode":"DIGI","label":"FT8"}` - or clear one with `{"idx":3,"clear":true}`. Fields you omit keep their stored value, so renaming a channel needs only `idx` and `label`.

A frequency outside a recognised amateur band returns **400** - the same refusal the Tab5 makes, so the browser cannot be a way around it.

There is no recall endpoint: recalling a channel is `set_freq` plus `set_mode`, which `/api/cmd` already does.

GET /api/settings

The settings a browser can usefully edit: callsign, grid, the CQ presets, the FT8 filters, the everyday toggles, QMX volume, band-plan region and the WiFi SSID. The spot sources (`spots_en` overall, then `rbn_en` and `cluster_en`), propagation feedback (`psk_rx_en`) and the Bluetooth mouse (`bt_mouse_en`) are here too — `bt_mouse_en` is stored immediately but only takes effect after a restart. `swr_limit_x10` (the SWR limit × 10, so `30` is 3.0:1) is **reported but not writable** — a transmit safety limit is set at the device.

The WiFi password is never returned - only `wifi_pass_set` says whether one is stored.

POST /api/settings

Applies a **partial** update: only the keys present are touched, everything else is left alone. So a single toggle is a one-key body, and a stale browser form can never reset fields it did not show.

```
{ "my_grid": "JO65ab", "rbn_en": true, "cq": { "max_calls": 10 } }
```

`wifi_ssid` and `wifi_pass` take effect only when both are supplied - an empty password means "keep the stored one", never "erase it". `qmx_vol_db` is in dB (0-50) and is sent to the radio as well as stored.

Everything the settings drawer can change is reachable here. Besides the keys above that includes `flat_mode`, `distance_in_miles`, `greylist_en`, `ft8_early_decode`, `field_day_en` (with `fd_class` / `fd_section`), `spots_en`, `sota_en`, `rbn_en`, `pskreporter_en`, `resmon_en`, `tx_tone_hold`, `tx_tone_hz`, `cw_pitch_hz` and `cw_cal_hz` — all readable back from `GET`.

`sim_mode_en` is the one to know about if you are automating tests. With it on, the firmware runs complete FT8 exchanges against phantom stations and `ft8_tx.c` hard-refuses to key the radio, so a QSO can be driven end to end with no antenna and no QMX attached. ⚠ Simulated contacts land in the **real** ADIF log, so turn it off afterwards and delete them — they have `FREQ=0`, and the ADIF viewer's "Del N test" button finds exactly those.

Read-only reporters. `ft8_op_mode` (`"ft8"` / `"ft4"`) and `ft8_freq_hz` are returned but not writable here — change them with the `set_ft8_mode` action, which also retunes the radio and clears stale decodes, then read these back to confirm. `swr_limit_x10` is likewise reported only.

GET /api/decodes

The FT8/FT4 decode list as the Tab5 shows it - same ordering (the station being worked, then messages addressed to you, then CQ calls, then strongest signal) and the same Filter-window hides applied. Read-only.

```
{
  "mode": "FT8",
  "miles": false,
  "working": "DK7CVD",
  "rows": [
    { "call": "DK7CVD", "text": "OZ1LAV DK7CVD -07", "grid": "JO31",
      "snr": -7, "hz": 1503, "dt": 210, "age": 6, "heard": 3,
      "km": 612, "brg": 173,
      "sl": "E", "me": true, "cq": false, "pin": true }
  ]
}
```

`dt` is milliseconds, `hz` the station's audio tone, `age` seconds since it was last decoded, `sl` the transmit window it was heard in. `me` marks a message containing your callsign, `pin` the station you are working.

`km` and `brg` (v1.8.3) are distance and bearing from your grid to theirs, worked out on the device with the same maths the Tab5's own list uses. **Both are omitted entirely when either grid is missing** — there is no null and no zero, so a client should render an absent field as "unknown" rather than as a distance. `km` is in miles when the top-level `miles` flag is true, which mirrors the *Distance in miles* setting.

GET /api/help

The guidance rows, ranked by the device from its own live state. `flagged` marks a row the firmware can see is happening right now.

```
{
  "context": { "page": "guide/panadapter.md", "anchor": "Layout", "label":
        "Panadapter" },
  "rows": [
    { "symptom": "My radio is not showing up", "flagged": true,
      "page": "reference/troubleshooting.md", "anchor": "won't reconnect",
      "label": "Radio not connecting" }
  ]
}
```

`context` is the chapter the Tab5's own **User Manual** button would open right now. `anchor` is a case-insensitive heading substring, not a URL fragment.

GET /api/manual?page=guide/ft8-tx.md

One page of the built-in manual, as the raw markdown that built the documentation site. `page=toc.json` returns the contents list. Served from the firmware, so it needs no internet and always matches the running version. 404 if the page is not in this firmware's manual.

7.4. CAT Endpoints

There is no raw-CAT HTTP endpoint. Send CAT-level changes through `POST /api/cmd` (`set_freq`, `set_mode`, `set_bw`, `set_band`), which routes them through the CAT poll task - the only writer to the radio's serial port. The web UI's own **CAT** box uses those actions.

7.5. FT8 Endpoints

The decode list is `GET /api/decodes` (above). Transmit from the browser is limited to `POST /api/cmd {"action":"cq_start"}`; there is no endpoint that sends an arbitrary FT8 message, deliberately - replies are chosen from the decode list in front of you, at the Tab5.

7.6. ADIF Endpoints

GET /api/adif

Download ADIF log.

Response: ADIF file (text/plain)

`?activation=<REF>` limits the download to contacts made during an activation of that reference (matched on `MY_SIG_INFO`), so a park's log can be uploaded on its own rather than the whole file. An unknown reference returns a valid ADIF file with no records, not an error.

`?date=today` or `?date=YYYYMMDD` limits it to one UTC day, and names the download `qso-YYYY-MM-DD.adi` so a daily file is self-identifying once it is saved. A day with no contacts returns a valid empty ADIF, not an error.

POST /api/adif/import

Restore records from a raw ADIF file — the browser's own "ADIF download", or any other logger's export. Body is the ADIF text itself (`Content-Type: text/plain`).

`?mark_uploaded=0` imports the contacts as **not yet uploaded**, so the next QRZ/eQSL/LoTW upload pass sends them. Default (`1`, or the parameter omitted) marks them as already uploaded instead — advancing all three upload cursors to the new record count — since the ordinary reason to restore a log is that these contacts were already sent before whatever wiped the device.

A pretty-printed, multi-line record (as some loggers export) is normalised to this device's own one-line-per-record form before being written. A record whose `CALL`+`QSO_DATE`+`TIME_ON` already matches one in the log is skipped, so re-importing the same file twice — or a file that overlaps an earlier partial restore — adds nothing extra.

Response:

```
{ "ok": true, "added": 12, "total": 47 }
```

`added: 0` with `ok: true` means every record in the file was already logged. `ok: false` means the log could not be written at all (storage full).

POST /api/adif/edit

Correct **one field of one record**: `?idx=<n>&call=<CALL>&field=<F>&value=<V>`.

`idx` is the record's position in the file and `call` must match the callsign already stored there — both, or the device answers `409` and changes nothing, so a browser showing a stale list cannot edit the wrong QSO.

Only three fields are accepted: `RST_SENT`, `RST_RCVD` and `SIG_INFO`. An empty `value` removes the field, which for a report is the honest record of one that was never exchanged. `SIG_INFO` is the reference the *other* station was activating (`US-1241`, `G/LD-049`, `DLFF-0123`); the device derives `SIG` from its shape and writes it alongside, and clears it again when the reference is cleared. Everything else is refused with `400`: callsign, band, mode, date and time are what QRZ, eQSL and LoTW match a contact on, and a LoTW record is signed over exactly those.

Response: `{"ok":true}`, or `{"ok":false,"error":"…"}`

POST /api/adif/clear

Clear all QSOs from ADIF log.

Response:

```
{
  "status": "cleared",
  "count": 42
}
```

7.7. Configuration Endpoints

GET /api/config

Download current configuration (all settings + memory channels).

Response: INI text file

POST /api/config

Upload a configuration file to restore settings.

Request: the config file as the raw request body (not multipart)

Response:

```
{
  "status": "imported",
  "merged": true
}
```

7.8. Logging Endpoints

GET /api/log

Download diagnostic log.

Response: Text file (qmx-log.txt)

GET /api/log/saved

The flash-persisted copy of the diagnostic log from before the last reboot or power-off - the one that survives a crash in the field.

Response: Text file

GET /api/signal

The signal level alone, for a faster S-meter than the 1 Hz `/api/status` poll can give. Same DSP call and parameters as `/api/status` and the Tab5's own S-meter - only the cadence differs.

```
{ "dbm": -83.5 }
```

GET /api/psk_rx

Propagation feedback — which receivers have copied **your** call, from PSK Reporter.

```
{ "enabled": true, "age_s": 96, "receivers": 14, "max_km": 7412,
  "reports": [
    { "call": "W1ABC", "grid": "FN42", "dxcc": "United States", "mode": "FT8",
      "f": 14074812, "t": 1785340050, "snr": -14, "km": 6031, "brg": 291 }
  ] }
```

Requesting this asks for a refresh, but the collector's own **five-minute floor** is enforced on the device, so polling faster than that returns the same data with a larger `age_s` rather than fetching again. `age_s` is `-1` before the first successful query.

`snr` is omitted when the receiver gave none — 0 dB is a real report, so it cannot double as "unknown". `km` and `brg` are omitted together when the receiver's grid is unknown.

Off by default. The endpoint always answers: with the feature disabled nothing is fetched, so `enabled` is `false` and `reports` holds whatever the last query returned before it was switched off — read `age_s` rather than assuming the list is current. It is also empty while offline or before a callsign is set.

POST /api/adif/delete?idx=&call=

Delete one QSO. **Both** the index and the callsign must match the record, or the request is refused with `409` - so a stale browser view can never delete the wrong contact.

GET /api/upload_status

Progress of a running QRZ/eQSL/LoTW upload (`busy`, `kind`, `uploaded`, `failed`, `error`).

7.9. Upload Endpoints

POST /api/qrz_key

Set QRZ API key (stored server-side).

Request:

```
{
  "api_key": "1a2b3c4d5e6f..."
}
```

POST /api/qrz_upload

Upload ADIF to QRZ Logbook.

Response:

```
{
  "uploaded": 5,
  "failed": 0,
  "error": null
}
```

POST /api/eqsl_creds

Set eQSL username and password.

Request:

```
{
  "username": "oz1lav",
  "password": "secretpass"
}
```

POST /api/eqsl_upload

Upload ADIF to eQSL.

Response:

```
{
  "uploaded": 5,
  "failed": 0,
  "error": null
}
```

POST /api/lotw_cert

Store the LoTW callsign certificate and key (extracted from the TQSL `.p12` in the browser during LoTW setup).

POST /api/lotw_upload

Sign all pending QSOs with the stored certificate and upload them to lotw.arrl.org.

Response:

```
{
  "uploaded": 5,
  "failed": 0,
  "error": null
}
```

GET /api/lotw_tq8

Download the whole log as a signed `.tq8` file, for manual upload to LoTW or offline verification.

Response: `.tq8` file (gzipped, signed)

7.10. WebSocket (Streaming)

Single-client endpoint for live spectrum:

```
ws://<tab5-ip>:80/ws
```

Each binary frame is **1026 bytes**:

Bytes	Meaning
`0`	Frame type: `0x01` = spectrum
`1`	Zoom decimation factor (1 = base 1024-bin spectrum, >1 = zoom-FFT with that decimation)
`2–1025`	1024 unsigned bytes, one per bin, quantised −130 dBm (0) to −30 dBm (255)

Bins are **pre-shifted server-side**, so byte 2 is the leftmost screen pixel and a client does not have to know where the VFO sits in the FFT. Push rate is about **10 fps**.

> **Warning - One client at a time**
>
> A second connection takes the stream over from the first. Opening the web UI in two tabs is enough to do it.

7.11. Screenshot

GET /ss.bmp

Returns what is on the Tab5's screen right now, as a BMP.

```
Invoke-WebRequest -Uri "http://192.168.1.213/ss.bmp" -OutFile screenshot.bmp
```

> **Note - It freezes animations while it reads**
>
> The screen is captured from the live framebuffer, so a waterfall or a blinking indicator stops for the duration of the read.

7.12. Error Responses

All endpoints return errors in a standard format:

```
{
  "error": "CAT timeout",
  "status": 500
}
```

Common HTTP status codes:

- **200** — Success
- **400** — Bad request (invalid command)
- **500** — Server error (QMX not responding, etc.)
- **503** — Service unavailable (WiFi off, etc.)

7.13. Example: Tune to 7.074 MHz in Python

```python
import requests

ip = "192.168.1.50"
api = f"http://{ip}/api"

requests.post(f"{api}/cmd", json={"cmd": "set_frequency", "value": 7074000})

requests.post(f"{api}/cmd", json={"cmd": "set_mode", "value": "USB"})

status = requests.get(f"{api}/status").json()
print(f"Now on {status['frequency_hz']} Hz, {status['mode']} mode")
```

Next: See Troubleshooting for API debugging.

8. FT8 Receive

The panadapter includes **on-device FT8 and FT4 decoders** with real-time spectrum waterfall and a live list of heard stations.

8.1. FT8 & FT4 View

Swipe → from the left edge to reach the FT8/FT4 view (the edge cycles Panadapter -> FT8/FT4/JS8 -> WSPR). The same decode list and waterfall serve all three modes — FT8, FT4 and **JS8** — switched with the **Preset** dropdown in the left pane (top). JS8 adds a Conversation tab and free-text sending; see JS8.

FT4 notes:

- Decodes refresh roughly twice as fast as FT8 because slots are 7.5 seconds
- Slot countdown shows 7.5 s instead of 15 s
- All filtering, priority, and auto-reply features work identically in FT4
- Time-sync from decoded signal timing works in both FT4 and FT8 (the bottom-bar clock shows `UTC(FT4)` when synced from an FT4 decode)
- **v0.19.4 made FT4 reliable**: a memory-placement fault previously killed decoding on every other slot (FT8's longer slots mostly hid it), the EVEN/ODD slot markers were on the wrong 15-second grid, and the slot clock could visibly jump between slots — all fixed; if FT4 seemed deaf or erratic on an earlier version, update the firmware

You'll see:

- **Decode list** (left pane) — every station heard, ordered so the ones that matter to you are at the top (see Decode List below)
- **Waterfall** (right pane) — same real-time spectrum as panadapter mode
- **Call CQ button** — start a CQ (requires QMX on the air)
- **Options button** — include/exclude stations, prioritise by SNR/distance, show only CQ callers

8.2. Decode List

The list shows every decoded FT8 message:

```
type: timeline
title: One FT8 slot - 15 s, and what DT and DTR measure
span: 15
seg: 0-12.64 steel a transmission - 12.64 s, 79 symbols
seg: 12.64-15 dim dead air - 2.36 s
mark: 0-0.014 red our TX starts at +14 ms
tick: 0, 5, 10, 15
note: steel | DTR is the RAW offset of a received signal from the slot boundary. It ca
note: trace | DT is that same offset corrected against the band consensus, so the shar
note: dim | the decode runs after the capture, in the dead air and into the next slot.
```

DT and DTR are two different quantities, which the table below spells out and the picture above shows: DTR is the raw offset from the slot boundary and carries about half a second of receive latency, while DT is that offset corrected against the band consensus, so the shared latency cancels and an on-time station reads about zero.

Column	Meaning
SL	Slot parity: **E** (blue) or **O** (amber)
CALL	Their callsign
MESSAGE	The full decoded message text
COUNTRY	Country, spelled out where it fits and shortened where it does not - never a 3-letter code (v1.15.0)
SNR	Signal-to-noise estimate, colour-banded by strength
TONE	The station's audio tone within the FT8 passband (v1.3.1)
DT	Slot-timing offset in seconds, corrected against the band consensus — an on-time station reads ~0.0. This is the WSJT-X-style number. It reads `--` until at least **two separate stations** have been heard, because with one station the correction is that station's own offset minus itself, which is zero by construction and not a measurement (v1.3.1, corrected in v1.17.0)
DTR	*Browser only.* The **raw** slot-boundary-relative offset, uncorrected. It includes about 0.5 s of one-way receive audio latency (the QMX SDR plus USB buffering), so it is **not** comparable with WSJT-X's DT — but it is the number that moves when a transmitter's timing moves, which is what you want when you are checking your own TX. Both columns sort on the raw value (v1.17.0)
KM / MI	Great-circle distance from your grid. A `~` means the station never sent a grid, so the distance is worked out from its country and is approximate (v1.14.4)
HRD	Times decoded since last appearance

The **BRG** column was removed in v1.14.4 to make room for spelled-out country names; a bearing is derivable from the distance and the map.

All three lists — this one, the SelfSpotter LIST tab and WSPR — use the same column order and capitalised headings since v1.15.1, so the same kind of information sits in the same place whichever screen you are on.

Own call highlight — your callsign is shown in **inverted colours** (red fill, white text) so you spot replies to you instantly.

How the list is ordered. Top to bottom:

1. **The station you are working** (asked for by Don WB0LQW) — during an exchange there are several transmissions each way, and the partner's messages used to sort down-screen where you had to hunt for them. Now *everything* from them stays at the top: their reply to you, their CQ while you are mid-exchange, and a message they send to a third station — which is exactly when you most want to see it. It covers a hand-typed reply as well as an automatic exchange, and matches on their **callsign**, so a third station merely mentioning them is not promoted. It releases the moment the QSO completes (their closing `73` contains your callsign, so rule 2 keeps it up there anyway).
2. **Messages addressed to you** — anything containing your callsign.
3. **CQ calls.**
4. **Everything else, strongest signal first.**

Stale entries — the list is a live picture of who's on frequency now; stations not heard again within **2 minutes** drop off automatically.

During your own CQ run — stations transmitting on *your* transmit slot (which you can't hear while transmitting) are hidden from the list and return when the run ends; other stations' CQ rows are hidden during a run as before.

Nonstandard callsigns — special-event and compound calls now resolve to the full callsign instead of showing `<...>`, once the station has been heard in full.

8.3. FT8 Options

Tap the **Options** button to open the Options modal:

- **Include callsigns** — only show these calls (space or comma-separated)
- **Include callsigns containing** — show any call with these substrings (e.g., `/P` for portable)
- **Exclude callsigns** — hide these calls
- **Exclude if contains** — hide any call matching these substrings

- **Never auto-call worked-before** — the one entry in this list that does *not* change the list. Every other filter here hides rows; this one only stops the **machine** calling a station already in your log on this band — CQ answers, the pile-up, Auto-work and the robot all pass over them. Worked stations stay in the decode list either way, greyed, and you can still tap one to work it. Per band: the same call on another band counts as new. (It was called "Exclude stations worked before", which read as a promise about the list.)
- **Hide worked-before** — new in v1.20.2, and the companion to the entry above rather than part of it. This one *does* change the list: tick it and a station already in your log on this band disappears from the decode list altogether, on the Tab5 and in the browser. Off by default, so the grey behaviour above is what you keep unless you ask for this. The two are separate on purpose - working duplicates by hand while the machine leaves them alone is a perfectly normal way to operate. The station you are mid-exchange with is never hidden.
- **Show only CQ callers** — hide replies, show only CQ messages
- **Show CQs in my run** — off by default. While you are running your own CQ, other stations' plain CQ calls are hidden so that replies addressed to you are not buried under unrelated traffic. Turn this on to keep seeing them, so you can spot a station worth breaking off for during your listening slot. It changes the list only; it does not change who the robot or Auto-work-pileup will answer
- **Skip TX1** — when you pounce a station, open with your signal report straight away instead of the grid exchange, for a quicker QSO (falls back to the normal grid exchange if the station has dropped out of the decode list)
- **Allow grey-listing** — off by default. A station that times out two of your pounces in a row is set aside: the robot and Auto-work-pileup skip it, its decode row turns **violet**, and tapping it offers to clear it from the grey-list instead of opening the TX dialog. Handy on a busy band where one station simply never comes back. The list is held in memory only and forgotten at power-off
- **Auto-reply priority** — Strongest SNR, Weakest SNR, Most distant grid
- **Robot mode** — auto-answer CQ (disabled by default; see FT8 Transmit)

All filters persist across sessions. Toggle any filter on/off to enable or disable it without erasing the criteria.

8.4. Decoding Performance

On a reasonably busy band, expect roughly **15–20 distinct messages per slot** in either mode. That is messages, not contacts — one QSO produces several.

Decoding uses **both processor cores**: the decode task works one half of the candidate list on core 1 while a helper takes the other half on core 0, and both run against a waterfall that was built during the capture rather than after it. That is what makes the results appear a second or two into the following slot instead of most of the way through it.

What limits it:

- **LDPC error correction is heavy**, and most candidates the search finds are false alarms that use the full iteration budget before being discarded
- **The spectrum and waterfall** are being drawn from the same audio at the same time
- **USB audio must be serviced continuously** — a gap there costs decodes far more than a slow decoder does

FT4 advantage: because FT4 slots are half as long (7.5 s vs 15 s), you get refresh feedback roughly twice as fast, which feels snappier on a crowded band even though the per-slot decode count is the same.

The panadapter prioritises **receive latency** (decode every 15 s, not accumulate) over volume.

8.5. Real-Time Waterfall

Unlike WSJT-X (which buffers a whole 15-second capture), the Tab5 **builds the waterfall on-the-fly** as audio arrives. You see CW and other traffic in real time, and FT8 stations appear instantly as their symbol blocks are decoded.

The waterfall also shows **non-FT8 activity**:

- CW (carrier lines)
- SSB (broader vertical smears)
- RTTY (FSK sidebands)
- Interference (wideband noise)

8.6. Tap to Reply

Tap any FT8 row in the decode list to **prepare a reply**:

1. A confirmation modal pops up showing your message
2. You can review the target callsign, grid, and SNR
3. Tap **Transmit** to send, or **Cancel** to abort

The reply always follows FT8 protocol (correct parity, proper message sequence) — you can't accidentally send a garbled or out-of-order message.

Pile-up list. If a station answers you while you're busy with another contact, they used to disappear from the list once they stopped transmitting. They're now remembered in a **Pileup** list — the **ADIF-log** button on the FT8 screen turns into a **Pileup** button (a different colour) whenever callers are waiting. Tap it to see who's called you, tap a station to work them, or tap the ✕ to dismiss one. See FT8 Transmit for the full workflow.

8.7. Auto-Reply (Robot Mode)

⚠️ **Experimental** — enabled via a checkbox in the Options modal.

When robot mode is on, the Tab5 **automatically replies to CQ** without waiting for you to tap. It scans each FT8 slot for CQ callers matching your filters, picks the highest-priority station, and sends a full QSO exchange (TX1 -> wait for report -> TX2 -> wait for RR73 -> TX3). Everything is logged to ADIF.

Important: Robot mode **keys the QMX for real** — your signal goes on the air. Never leave it unattended unless you're confident in your filters and your station is in a safe state.

8.8. Search & Window

The decoder uses an **FFT-based search window** to find candidate tone blocks, at whichever tone spacing the active mode uses — **6.25 Hz** for FT8's 8 tones, **20.83 Hz** for FT4's 4.

Frequency stability: The QMX's USB audio clock isn't bit-exact 48 kHz, so decodes slide slowly over time. The panadapter **re-locks to UTC boundaries every 15 seconds** (the slot boundary), preventing long-term drift.

8.9. Time Sync for FT8

FT8 requires **UTC time accurate to ±1 second** for slot synchronisation. The panadapter gets time from:

1. **WiFi + SNTP** (if available) — most accurate, syncs every ~hour
2. **Tab5 RTC** (if set) — accurate to ±1 min, no internet needed (POTA)
3. **QMX GPS** (if QMX has GPS) — re-syncs the RTC every 5 minutes
4. **Manual** — you can set time via the settings drawer

See Time Sync for details.

Next: Learn about FT8 Transmit or go back to Panadapter.

9. FT8 Transmit

The panadapter **keys the QMX and transmits full FT8/FT4 QSOs** — reply to CQ, run your own CQ, auto-answer (robot mode), or conduct a full exchange.

9.1. FT8 vs FT4 — Which to Use?

Both modes transmit via the same CAT interface; the difference is **slot length and symbol rate**:

Feature	FT8	FT4
Slot length	15 seconds	7.5 seconds
Symbols transmitted	79	105
Symbol duration	160 ms	48 ms
QSO time (full exchange)	~90 sec (6 slots)	~45 sec (6 slots)
Use case	Weak signal, SOTA/POTA	Busy bands, contests
Preset picker	Panadapter mode -> Freq dropdown	FT8 mode -> Preset picker

Choose FT4 when:

- The band is crowded (faster exchanges, less QRM)
- You're in a contest (time is critical)
- You want quicker pileup resolution

Choose FT8 when:

- Signal is marginal (longer decode window tolerates more QRN)
- You're portable/POTA (less pressure to transmit fast)
- You prefer the relaxed pace

FT4 has full feature parity with FT8, including time-sync from decoded signals (fixed in v0.19.3 — the time-sync modal shows "FT4" when appropriate and the bottom-bar clock correctly reads `UTC(FT4)` while in FT4 mode).

9.2. Switching Between FT8 and FT4

In FT8 screen (left pane):

- Tap the **Preset** dropdown (shows current frequency/mode)
- Select an FT4 frequency to switch to FT4
- Select an FT8 frequency to switch back

In Panadapter screen:

- Tap the **Band preset dropdown** (top left)
- Some bands have both FT8 and FT4 frequencies listed
- Tap the FT4 variant to switch

Your **settings are sticky** — when you switch back to FT8 or FT4, the panadapter remembers your last frequency, bandwidth, and filter settings for each mode.

The **decode list and pileup are cleared** on the switch (v1.3.3). The two modes use different slot lengths, so stations decoded under the old timing describe a band the new mode cannot hear — and leaving them on screen meant you could tap one and try to work a station that was no longer there.

9.3. Modes of Transmission

9.0.1. Reply to a CQ

In FT8 view, tap a CQ row in the decode list. A confirmation modal appears:

```
type: panel
title: Confirm FT8 Transmission
row: K9ZZ EN52   -07   2138 Hz
row: W1AW FN31   -12   1406 Hz   <- the row you picked
row: SV1ENG JN37 -09   1869 Hz
row: TX1: W1AW OZ1LAV JO45
buttons: Cancel!, Auto Pounce, Transmit+
note: dim | drag up or down the list to move the highlight before you lift your finger
note: amber | Transmit sends this message once; Auto Pounce runs the whole exchange fo
```

- **Transmit** (green) — send this one message on the next correct slot

- **Auto Pounce** (blue) — handle the entire exchange automatically (TX1 -> wait for report -> TX2 -> wait for RR73 -> TX3 73)
- **▲ / ▼ Nudge** — move the target to the row above or below, without closing the modal and redoing the selection gesture
- **Cancel** — abort

Transmit is intelligent (v1.3.0) — it builds the correct *next* message from what that station last sent, exactly like a WSJT-X double-click: their CQ -> your grid (or your report, with Skip TX1 on), their grid -> your report, their report -> `R`+your report, their `R`-report -> `RR73`, their `RR73`/`73` -> `73`. You can run a whole QSO one Transmit tap at a time — and sending the closing `RR73`/`73` **logs the QSO to ADIF** just like an automatic contact. Auto Pounce is offered on any first reply; once you're mid-exchange with a station, its rows offer Transmit only.

A quick tap is enough — you do not need to hold a row before releasing. Hold-and-drag still works for selecting across rows in a busy list.

The reply follows **correct FT8/FT4 parity** — if you're replying to an even slot, you transmit on the odd slot, and vice versa. For FT4 the countdown timer correctly counts down in 7.5-second slot increments, not 15-second FT8 ones.

Skip TX1 (faster pounce). With **Skip TX1** enabled in the Options modal, pouncing a station opens with your signal report straight away instead of the grid exchange — saving one round trip. If the station has already dropped out of the decode list, it falls back to the normal grid exchange automatically.

If your target is working someone else, the panadapter holds instead of keying up over their exchange — the status shows "working *call* - waiting" with a **TAP TO CANCEL** line. Waiting costs nothing (a hold doesn't count toward the timeout), but if you'd rather move on, tap the status to cancel the pounce and pick a different station; the abandoned exchange stays resumable for a few minutes via the resume prompt.

Working a pile-up

When you call CQ (or work a busy run), more than one station may answer at once — and a caller who replies while you're mid-QSO with someone else used to vanish from the live decode list as soon as they stopped transmitting. They are now kept in a **Pileup** list so you don't lose them:

- Whenever one or more stations are waiting, the **ADIF-log** button on the FT8 screen becomes a **Pileup** button in a distinct colour, and reverts once the list empties.

- Tap **Pileup** to see everyone who has called you and isn't worked yet.
- **Hold the button to open the ADIF log** — the log stays reachable even while the button reads "Pileup" (a one-time hint appears the first time it flips).
- Tap a station in the list to work them (the same confirmation modal as a decode-list tap), or tap the small ✕ to dismiss them without working them. The reply is the correct **next** message for that station, built from whatever they actually last transmitted — usually a signal report for someone who has just called you, but an R+report if they came back later with a report of their own (v1.3.2; the same laddering as tapping a decode-list row).
- A station is removed from the list automatically once you start a QSO with them, and a just-completed contact's trailing 73 can't put them back.
- Stations you've worked before still appear in the pileup unless **Never auto-call worked-before** is checked — the pileup follows the same rule as the auto-answer.

The pile-up tracker never transmits on its own — it only remembers callers; you choose who to work. If you'd rather it *did*, check **Auto-work pileup** in the Options modal: the strongest waiting caller is pounced automatically when your current QSO completes — or immediately, if you enable it with callers already waiting and nothing else in progress. It carries the same unattended-TX warning as the robot.

Pick callers myself

Normally, while you are calling CQ, the first station to answer starts the exchange straight away. Check **Pick callers myself** in the Options modal and that no longer happens: an answering station waits in the pile-up until *you* tap them, and from that point the exchange runs automatically as usual.

Only the *choice* becomes manual. You are not signing up to type anything, or to send each message by hand.

Two things it deliberately does:

- **It keeps calling CQ while you decide.** The pile-up carries on building, which is usually what you want when activating — and a radio that fell silent while you thought about it would look like it had stopped working.
- **It overrides Auto-work pileup.** Both settings decide who to work next, so if you have asked to choose, nothing else picks for you a moment later.

Off unless you turn it on, so nothing changes for anyone who does not want it.

9.2. Call CQ

Tap the **Call CQ** button. A modal opens:

```
type: panel
title: CQ Messages (check the active)
row: [x] CQ OZ1LAV JO65
row: [ ] CQ DX OZ1LAV JO65
row: [ ] CQ POTA OZ1LAV JO65
buttons: Cancel!, + call grid, Save+
note: dim | CQ stop and Listen sit top-right and apply on the tap - no Save needed
```

Choose a preset (or edit/save a new one), then tap **Transmit**. The QMX starts calling CQ and listening for replies.

CQ tone selection — the panadapter automatically picks a quiet frequency (6.25 Hz tone spacing, 200–2800 Hz audio range, no other stations nearby). If your chosen tone gets busy during a QSO, the panadapter **auto-relocates to the nearest clear frequency** to avoid stepping on other stations.

Where your transmission sits in the slot

```
type: timeline
title: One FT8 slot - 15 s, and what DT and DTR measure
span: 15
seg: 0-12.64 steel a transmission - 12.64 s, 79 symbols
seg: 12.64-15 dim dead air - 2.36 s
mark: 0-0.014 red our TX starts at +14 ms
tick: 0, 5, 10, 15
note: steel | DTR is the RAW offset of a received signal from the slot boundary. It ca
note: trace | DT is that same offset corrected against the band consensus, so the shar
note: dim | the decode runs after the capture, in the dead air and into the next slot.
```

Your burst starts **14 ms after the slot boundary** and runs 12.64 s, so the last 2.36 s of every slot is dead air. That gap is the whole reason a reply has to be decided before the next boundary rather than after it — the decode of the slot you just heard is still running in it.

What the TX status line is telling you

The left pane carries a persistent status line under the slot countdown. The colour is the quickest read; the text says which station and which tone.

Colour	Line	Meaning
Red	`TRANSMITTING: <message>`	A burst is going out, with your audio tone on its own line. Tap to abort.
Amber	`TX armed: <message>` then `nnnn Hz -> EVEN/ODD, ~Ns`	Waiting for the slot. Tap to cancel.
Red-orange	`FREQ BUSY`	Your tone is occupied (+/-50 Hz guard), and the warning names the frequency. During a CQ run this self-heals - the next cycle hops to the nearest clear tone. Mid-exchange, move it yourself with the TX tone control.
Amber	`working <call> - waiting` + `TAP TO CANCEL`	The station you are calling is mid-exchange with somebody else. Nothing transmits until they sign off or call CQ again.
Green	QSO complete	
Orange	QSO timed out	Tap to clear.
Dim white	capturing / decoding / symbol count	The engine's own status, passed through. Nothing is armed.

> **Note - Amber means armed, not transmitting**
>
> An armed burst has not keyed the radio yet. Cancelling an amber line costs nothing; cancelling a red one aborts a transmission in progress.

CQ auto-stop — by default the panadapter keeps calling until someone answers. If you'd rather call a few times and then pause (a common courtesy on a quiet band), long-press **Call CQ** and tap the **CQ stop** button at the top-right of the preset editor: it cycles through never / 1 / 2 / 3 / 4 / 5 / 10 calls and applies immediately, no Save needed. While calling, the TX status shows the progress ("call 2 of 4"); after the last unanswered call the panadapter listens through one more receive slot (an answer to your final call still starts the QSO normally), then stops and goes idle. Applies to every CQ run, including the automatic resume after a completed or timed-out QSO — each fresh CQ sequence starts the count over.

Listening slot - while you are transmitting you are deaf to your own time window, so the occupancy picture for the window you transmit in is the one that goes stale. The **Listen** button in the preset editor (under **CQ stop**) can spend one slot receiving after every 3, 5 or 10 calls. Off by default, because it changes your on-air cadence, and like CQ stop it applies on the tap with no Save needed.

9.3. Auto-Reply (Robot Mode)

⚠️ **Requires explicit checkbox in the Options modal** to enable.

When robot mode is on and you're idle, the panadapter scans every FT8 decode for CQ callers matching your filters, picks the highest-priority match (Strongest SNR, Weakest SNR, or Most distant grid — your choice), and starts a full QSO:

1. **TX1** — send your call, grid
2. **RX** — wait for their report
3. **TX2** — send their report + your call back
4. **RX** — wait for RR73 or 73
5. **TX3** — send 73 or RR73 (exchange complete)
6. **Log** — ADIF entry created, move to next CQ

Pause — skip one transmission

Sometimes you want to *listen* on your own slot: to check the frequency is still clear, or to hear whether the station you are working is actually answering you rather than someone else.

Pause sits with Re-send, RR73 and 73 in the mid-QSO row, and it is on the browser page as well as the Tab5 (v1.16.2). It skips exactly one transmission and leaves the exchange intact — unlike **Cancel**, which ends it.

The row now stays up until your final is acknowledged (v1.20.2). Until this release it vanished the instant the closing RR73 or 73 was sent - so the one window in which a missed final could still be put right was the one window with no buttons in it. The machine has always kept watching for five minutes and re-sending on its own; now you can see that it is doing so, and act inside it:

- Only **Re-send** and **Cancel** are offered, because RR73 and 73 are built from an exchange that has already closed.
- **Re-send** puts that same final back on the air - the RR73 if that is what you sent, not a bare 73.
- **Cancel**, or a tap on the Tab5's TX indicator, says the contact is over and stops the machine re-sending too.
- Pressing Re-send does not use up the machine's own six tries.

The QSO is logged when the final leaves the air, as it has been since v1.20.0 - none of this delays the log entry.

- It shows a **pause** symbol normally and a **play** symbol on yellow while a transmission is being skipped, so you can see the state at a glance.
- Press it again and the pause is taken straight back, re-arming the transmission if the slot still has room.
- It works while **calling CQ** as well as mid-exchange.
- A slot you skip on purpose **does not count** against the six cycles before the Tab5 gives up on a contact. Pausing can never cost you the QSO.

Asked for by Randy N4OPI, who works his station from another room; the countdown behaviour was caught by Gyula HA3HZ.

An experienced caller who skips a step is followed (v1.8.5): if someone answers your CQ with a **report** instead of a grid — common from operators who already know they have you — the reply acknowledges it with `R` plus your report and waits for `RR73`, rather than sending another bare report and losing a cycle. Reported by Gyula HA3HZ, who saw it twice. Tapping **Transmit** by hand always did the right thing here; it was only the automatic run that was a rung behind.

A busy pick is abandoned, not waited for (Roy KI0ER's reasoning adopted verbatim): when *you* pounce on a station that turns out to be mid-QSO with someone else, the Tab5 politely holds — a deliberate pounce means you want *that* station. But the robot picked its target off a list, so when its pick engages a third station the robot **moves on to a different CQ caller** instead of idling through someone else's QSO. No grey-list strike: busy is not unresponsive.

Each QSO takes **~90 seconds** (6 FT8 slots × 15 s/slot). If a station doesn't respond in 4 slots, the panadapter times out and resumes scanning CQ.

Important: Robot mode **transmits on the air unattended**. Never enable it unless:

- Your station is physically attended
- Your filters are tested and correct
- Your antenna/SWR is known good
- Your QMX power is set appropriately

A permanent on-screen warning appears whenever robot mode is available.

Four things now switch it off for you, all of them because the alternative is transmitting when you did not expect it (Roy KI0ER):

- **It waits before its first call.** Straight after a band change or a restart nothing has been heard yet, so both transmit windows are unmapped and a tone would be chosen from no information at all. It holds, showing *"Auto-answer: listening before first call"*, until it has heard both windows.
- **Cancelling a transmission switches it off.** Tapping the TX indicator to stop a transmission disarms it, abandons the QSO **and** turns auto-answer off — so halting a transmission to go and check your antenna does what you expect, rather than the radio starting again a cycle later.
- **A band change switches it off**, whichever way you changed band — the band buttons, the web page, a spot, a memory recall, or the radio's own knob. Few stations have an automatic ATU, so the antenna is usually not tuned for the new band yet.
- **It is off at every startup**, however the last session ended. Unattended transmission is never the state the panadapter powers up in; switching it on is a deliberate act, once per session.

Each of these says on screen why it stood down.

9.4. FT8 Simulation Mode (Practice QSOs)

⚠️ **For practice only** — no real stations involved, radio never keys up. Since v1.3.0 the simulator needs **no QMX connected at all** — no radio, no antenna, zero RF.

Enable **FT8 Simulation Mode** in the settings drawer to practice everything: manual step-by-step Transmit, Auto Pounce, CQ-runs, pileups, and Field Day exchanges. Useful for:

- Learning the full FT8 exchange sequence

- Testing your setup without risking interference
- Verifying ADIF logging works correctly
- Practicing the pileup tools on a genuine pileup

How it works:

- **Six phantom stations** (three US, three DX) call CQ on their own tones — every message is real FT8, synthesized to actual GFSK audio and decoded through the same receive pipeline real RF uses
- Tap a CQ to pounce (auto or fully manual — they answer either), or **Call CQ** yourself and **four phantoms answer at once**, building a real pileup
- Phantoms are **patient like real operators**: each repeats its message every cycle, up to four times, before giving up and going back to CQing; one you're working stops CQing; one you've worked stops answering your CQs (toggle sim off/on to reset)
- Their replies match **what you actually transmitted** — grid gets a report, a report gets a roger, `RR73` gets a courtesy `73`
- The **Fast pounce** toggle is honoured, so you can watch what it changes: decodes surface just before the slot boundary with it on, just after with it off
- Swiping to the Panadapter and back clears the phantom rows and pileup for a fresh session, and **turning simulation mode off also clears them** (v1.3.3) — phantom stations no longer linger in a list that is supposed to be real, where they were still tappable

> **Note - Fixed in v1.3.3**
>
> With **Fast pounce** turned *off* — which is the default — the phantoms never answered at all, so every practice pounce simply timed out. If you tried the simulator before v1.3.3 and concluded the phantom stations were ignoring you, that was this, not you.

Visual indicator: When simulation mode is active, a **10 px red border pulses around the entire screen** (breathing red frame). This is your visual reminder that you're in practice mode — if the red frame is gone, simulation is off and real stations are in play.

The interlock is in the firmware, not just the UI. While simulation is on, every CAT command that would key the radio is skipped and logged instead — so a QMX that happens to be connected is never keyed, regardless of what the screen is showing.

Important:

- The QMX is **never keyed** in simulation mode, no matter what
- QSOs are logged as real ADIF entries — deliberately, so the logging/upload paths get exercised too. When you're done, the ADIF viewer shows a **"Del N test"** button (only while practice contacts exist — they're recognized by their missing frequency): two taps deletes them all
- All other features (panadapter, web UI, settings) work normally

9.5. ARRL Field Day Mode

During ARRL Field Day, tick **Field Day mode** in the Options modal and fill in the two fields beside it:

- **Class** — your number of transmitters plus the category letter, written together: `16A`, `5B`, `1D`. It is not a power rating.
- **Section** — your ARRL/RAC section, e.g. `EMA`, `WCF`, `NNJ`.

This switches the FT8 exchange from grid-and-signal-report to Field Day's class+section format, using the standard FT8 message type WSJT-X uses for FD (`WA9XYZ KA1ABC R 16A EMA`). Both pounce and CQ-run follow the convention automatically: the opening message is unchanged — FT8's CQ format has no room for class and section — and the *report-equivalent* step carries them instead, with the receiving side echoing back `R`-prefixed.

While the mode is on, **Call CQ** tags your CQ as `CQ FD <call> <grid>`, using the same modifier mechanism as `CQ POTA` or `CQ DX`, so other Field Day stations know to expect this exchange. It **replaces** any other modifier for as long as the mode is enabled — FT8 allows exactly one. The CQ preset editor reflects that: the three presets are dimmed and locked while FD mode is on, since their own modifier cannot apply, and a live preview line shows exactly what will go out.

Completed Field Day contacts log the standard ADIF contest fields — `CONTEST_ID=ARRL-FD`, your section and theirs — alongside the usual call, frequency and time, so they import cleanly into contest-logging software.

9.6. Fox/Hound (DXpedition) Mode

Big DXpeditions run FT8 in **Fox/Hound** mode, where one station (the *Fox*) works five callers at a time from below 1000 Hz while everybody chasing it (the *Hounds*) calls from above 1000 Hz. It is not the ordinary FT8 exchange: a hound that answers the Fox has to **move onto the Fox's frequency** to do it, because the Fox listens only to its own narrow slice.

Set **Fox/Hound (DXpedition)** in the Options window to one of three positions:

- **Off** — normal FT8. Nothing changes.
- **Guided** — the Tab5 tells you when it can see a Fox, and you tap it to call. Every transmission stays your decision.
- **Automatic** — the Tab5 calls a Fox it recognises and runs the whole exchange.

Whichever you choose, the awkward part is handled for you: your first call goes out **above 1000 Hz** on a clear slot, and the moment the Fox answers you with a report, your reply is **moved down onto the Fox's own frequency** — which is the step that makes the contact possible at all. The Fox's `RR73` ends it, and the Tab5 then returns to its calling tone ready for the next one.

The Tab5 does not send `73` to a Fox. The Fox's frequency is the scarcest thing on the band during a DXpedition and a courtesy `73` there is just clutter. The contact is complete, and logged, on the Fox's `RR73`.

How a Fox is recognised — and where it trusts you instead. For anything it does *by itself*, the Tab5 is strict: not frequency alone (plenty of ordinary stations work below 1000 Hz) but a station down there visibly working a **queue** — several different callsigns inside a few minutes, a pattern nothing else on the band produces. Until it has seen that, automatic mode will not call anything a Fox.

A tap from you needs no such evidence. While Fox/Hound is enabled, tapping any station in the Fox region runs the Fox/Hound exchange, because enabling the mode is your declaration that you are chasing a DXpedition — the same way WSJT-X's Hound tick works. The machine is deliberately more cautious than you are.

The catch is leaving the mode on after the DXpedition has finished: a tap at an ordinary station low in the passband would then QSY onto their frequency and skip your `73`. So the transmit-confirmation window **tells you before you commit** — it adds a line reading "*as HOUND — will QSY onto 500 Hz, no 73 sent*" whenever Fox/Hound rules are about to apply. If you see that on a station you did not think was a DXpedition, set Fox/Hound back to **Off**.

Three of the Tab5's usual courtesies stand down while a Fox contact runs, and all three would otherwise work against you:

- it **keeps calling** while the Fox is busy with other stations — a Fox is always busy, and waiting for a free frequency means never calling at all;
- it **does not re-send** a closing message, because the Fox never asks for one;
- it **does not grey-list** the Fox for ignoring you, which in a pileup of hundreds is entirely normal.

Automatic mode will not call a Fox already in your log on that band, so it works one contact and then leaves the DXpedition alone.

> **Note - Hound only — the Tab5 cannot be a Fox**
>
> A Fox transmits up to five signals simultaneously. The Tab5 keys the QMX one tone at a time over CAT, so a multi-signal transmission is not something this radio can be asked for. This is a hardware limit, not a missing feature.

9.7. Message Status

The FT8 left pane shows a **status label** indicating what's happening:

Status	Meaning
ACTIVE (red)	Transmitting right now
ARMED (amber)	Ready to transmit next slot
QSO Complete (green)	Exchange finished, logged to ADIF
QSO Timeout (orange)	No response after 4 slots, exchange aborted
(dim white)	Idle, no active QSO

Tap the status label to **abort** the current QSO (only works if ARMED or ACTIVE; once COMPLETE, it's logged).

Resume after timeout — if a QSO times out because the partner faded, and they come back within ~5 minutes calling you, the exchange **resumes where it left off** automatically (or tap their row to resume manually) instead of restarting from scratch.

If they never heard your final (v1.3.4, extended in v1.16.8) — a partner who does not decode your closing `73`/`RR73` keeps sending you their report, waiting for it. The Tab5 notices: if a station just worked comes back with a report rather than `RR73`/`73`/`RRR`, the final is sent again, up to six times within five minutes, *before* anything else can start a new contact. This now tracks **up to four partners at once** rather than only the most recent one — finishing a QSO and immediately working the next caller in a pileup no longer drops the first station's re-sends. The QSO is not logged a second time. Taking over by hand no longer produces a duplicate entry either — the same callsign on the same band inside ten minutes is recognised as the same contact.

9.8. Power & SWR Readout

After each transmit burst, the status bar briefly shows:

```
Last TX: 5.2W SWRx1.25 [N=79]
```

- **5.2W** — average output power measured via QMX `PC;` CAT command
- **SWRx1.25** — SWR measured via QMX `SW;` CAT command (only valid during TX)
- **[N=79]** — number of symbols transmitted
  - FT8: always 79 symbols (~12.7 s at 160 ms/symbol)
  - FT4: always 105 symbols (~5.0 s at 48 ms/symbol)

These readings are **informational only** — the panadapter does not enforce limits. Monitor your antenna system independently.

9.9. CQ Presets

You can save up to 3 custom CQ messages:

1. Tap **Call CQ**
2. Tap **Edit Presets** (or long-press the active preset)
3. Type your message: callsign, modifier (optional), grid
4. Tap **Save**

Presets persist across power cycles. Common modifiers:

- **DX** — calling DX only
- **POTA** — Parks on the Air activation
- **SOTA** — Summits on the Air activation
- *(blank)* — standard CQ

The preset editor's top-right **CQ stop** button sets the CQ auto-stop limit (see Call CQ above) — it cycles never / 1–5 / 10 and applies on tap, independent of Save/Cancel.

9.10. Frequency & Tone Control

Your transmit tone is chosen for you by default — the nearest clear 50 Hz slot, scanned against the stations currently decoded. You can see it, move it, and from v1.3.4 pin it.

The TX tone button is in the FT8 left pane, to the right of the `TXCQ` parity button, and always shows a number (1500 Hz until you change it). Tap it to open the picker.

The mini occupancy strip sits directly under the slot countdown in the same pane: the same 50 Hz grid and the same colours as the picker's full-size strip, so where the band is busy — and where you are in it — is answerable at a glance without opening anything.

Both time windows, always (asked for by Roy KI0ER). The strip is **two rows — EVEN above, ODD below**, marked `E` and `O` in the same blue/orange the slot countdown uses. Two stations only collide if they transmit in the *same* window, and only you know which window you are about to pounce into — so both pictures are on screen **before** you choose, not after a transmission has fixed your window. Your white marker sits on your own window's row once one is locked (both rows until then), and your partner's pink sits in *their* window. The picker's full-size strip is split the same way, and its verdict says where your pick stands: "*Clear in EVEN — busy in ODD*".

The picker **repaints every second while it is open**, so the occupancy you are choosing against is the live picture rather than a snapshot taken when you tapped the button. If it has just told you something in words, that message stays on screen for a few seconds rather than being overwritten by the next repaint.

In the picker:

Control	What it does
Occupancy strip	The whole 200-2800 Hz window as 52 slots. **Green** free, **red** occupied, **white** you, **pink** your QSO partner
Touch and drag	Drag along the strip to pick a slot. The bar turns grey and follows your finger, the readout tracks it live, and it commits when you lift off
-50 / +50	Nudge one slot at a time
Find clear slot	Jumps to the nearest free slot, scanning outward from where you are
TX Hold	Pins the tone: every CQ and every reply goes out on it. See below
Apply nnnn Hz	Commits the tone and the hold setting

The readout states in words whether your chosen slot is clear or occupied, and lists the nearest free slots as numbers.

Apply whenever you like, including in the middle of a transmission. If a burst is on the air the change is accepted and applied the moment that burst ends — the transmission in progress finishes on the old tone, because stopping it half way would send a corrupted frame, and the very next message carries your new one.

That used to be a refusal, and it mattered more than it sounds (Roy KI0ER). A burst covers about 12.6 s of a 15 s slot and you transmit every other slot, so roughly four attempts in ten landed mid-burst and were rejected — leaving the QSO running on the offset it started with. Your choice depended on your timing, which is not a choice.

Slot parity is untouched, so moving your tone mid-QSO does not disturb the exchange — your partner tracks the slot, not the frequency. Nothing in FT8 requires either station to stay on the offset it began on, only to stay in its time window. This is the same freedom WSJT-X gives you, and it is the way out from under a station that has landed on top of you.

TX Hold is WSJT-X's "Hold Tx Freq". With it **off** — the default — each transmission takes the nearest clear slot, and a CQ that gets clashed relocates itself on the next cycle. With it **on**, the tone you picked is the tone used for everything, a clash is reported but never acted on, and nothing moves you off the slot you chose. The line under the checkbox says which of the two you are getting. Both the tone and the hold setting survive a power cycle.

> **Note - The strip shows decoded stations, not raw spectrum**
>
> A station too weak to decode will not appear as occupied, and an all-grey strip means nothing has been heard yet rather than "the band is clear".

> **Note - Occupancy is per time window**
>
> Two stations only collide if they transmit in the *same* slot, so a tone busy in the odd window may be completely free for you in the even one. The strip accounts for this whenever your own transmit window is known — that is, once a reply or a CQ is armed, since a reply inherits the opposite of your partner's slot and a CQ carries your `TXCQ EVEN/ODD` choice. With `TXCQ ANY` and nothing armed there is nothing to work from, so both windows are shown combined. When the window *is* known, the free-slot line names it.

Waiting for a busy station

On a crowded band several stations answer the same CQ and the caller works one of them. If that is not you, the Tab5 no longer keeps calling: while their last decoded message is addressed to somebody else, **nothing is transmitted**, and the status line shows `working <call> - waiting`. As soon as they send `73` or `RR73`, or call CQ again, it picks up where it left off.

You are not committed to the wait (Roy KI0ER's report). The status line carries a **TAP TO CANCEL** line while the hold is on; tapping it drops the pounce so you are free to work somebody else. The abandoned exchange stays resumable for a few minutes, in case the station frees up and you want back in.

This also protects the grey-list. Giving up on a station used to count against it, so a popular station could end up permanently skipped by the robot and Auto-work-pileup for no reason other than being busy. A wait is not a failed attempt. The wait is capped at about six minutes so a station that vanishes mid-exchange still times out normally.

> **Warning - Fixed in v1.16.4 — the six-minute cap did not work**
>
> Before v1.16.4 that cap never actually took effect: the counter reset itself instead of stopping, so the Tab5 could wait **more than half an hour** for a station working somebody else. Worse, each held slot cancelled the call it had just queued, so it never got back to calling them either — which is why it could look like the radio had simply given up and gone quiet *(Gyula HA3HZ)*. Both are the same fault, and both are fixed.

To tune to a different frequency while CQ is running:

1. Tap the **Freq** label on the top bar
2. Enter a new frequency
3. The CQ continues on the new frequency next slot

9.11. ADIF Logging

Every completed QSO is **automatically logged to ADIF** on the Tab5's internal storage:

- **Callsign**, grid, report (SNR), time (UTC)
- **Frequency**, mode (`FT8`)
- **Duration** (QSO start to finish)
- **Operator** (your callsign)

Download the log via the web UI (**QSO Logs** menu -> **ADIF download** ↓) or Settings -> ADIF Log.

9.12. Upload to QRZ, eQSL, LoTW & Cloudlog

Via the web UI:

1. Open the **QSO Logs** menu in the bottom bar and click **QRZ upload** ↑, **eQSL upload** ↑, **LoTW setup** (reads **LoTW** ↑ once configured), or **Cloudlog upload** ↑
2. Enter your API key (QRZ), username/password (eQSL), the guided certificate setup (LoTW), or your server address and API key (Cloudlog) when first prompted — credentials are saved for future sessions
3. Click again to upload

See Web UI -> LoTW Upload for the LoTW certificate setup details.

Cloudlog and Wavelog

Cloudlog and Wavelog are the only logbooks here that you host yourself, so you supply the address as well as the API key. Both use the same interface, so either works.

You are asked for three things: the server address, your API key, and the station profile ID (the number in the Cloudlog URL — leave it as `1` if you only have one).

Plain `http://` is allowed, but only to a server on the same network as the Tab5, and only to a numeric address such as `http://192.168.1.20`. That covers the usual case of a Cloudlog running on a machine at home with no certificate. Anything else needs `https://`.

Two things follow from that, both deliberate:

- The check is made **every time you upload**, not once when you save the address. So if you set it up at home and then operate from a park, the upload refuses rather than sending your API key across a network you do not control. Upload when you get home and it goes through.
- For plain `http://` the address must be numeric, not a name. A name has to be looked up, and what answers the lookup away from home need not be your server. Use `https://` if you want to use a hostname.

If your server is on your own network this is the only upload that needs no internet connection at all, which suits POTA and SOTA better than the other three. Contacts are sent in batches, and Cloudlog does its own duplicate checking, so uploading again after a failure is harmless.

Logs are batched — each upload session records which QSOs have been sent, so re-running skips already-submitted entries.

Uploads work **while FT8 or FT4 is actively running** — the panadapter briefly steps the FFT and SD-archive activity aside for the duration of the HTTPS transfer, then resumes automatically. You typically miss one FT8 slot and won't notice. A progress result is shown once the upload completes.

9.13. Troubleshooting

"QMX not responding to TX command"

- Check USB cable (must be data cable, not charge-only)
- Verify QMX is in the correct mode (USB/LSB/CW/DiGi, not AM)
- Try a manual `TX;` command via the web UI's CAT tab

"⚠ FREQ BUSY warning, but no other station is there"

- The clash detector reserves a ±50 Hz guard band around each decoded station, so it flags a neighbour that is close rather than exactly on you
- It counts only stations **in your own transmit window** (it used to count both windows, so it could warn about a station that could never collide with you; Roy KI0ER caught it)
- The warning names the frequency. Tap **TX nnnn Hz** and either **Find clear slot** or drag to a green slot on the occupancy strip
- Remember the strip only knows about stations it has *decoded* — a very weak one will not show

"QSO timeout — no response"

- The called station didn't hear you (too weak, bad timing, etc.)
- Try a different station, or switch to manual mode and check levels first
- See Troubleshooting for more

Next: Set up Time Sync for accurate FT8 timing, or explore the Web UI.

10. JS8

JS8 Normal runs on the Tab5 itself — the decoder, the error correction and the transmitter, with no PC in the loop. JS8 is a keyboard mode built on FT8's modulation: it carries free text rather than the fixed exchange FT8 uses, so stations hold short conversations with each other.

JS8 arrived in v1.17.0 and is **declared a beta**. What is proven and what is not is spelled out in section 6. Read that before relying on it.

10.1. Selecting JS8

Swipe → from the left edge to reach the FT8/FT4/JS8 view, then pick **JS8** from the **Preset** dropdown in the left pane. The dropdown is three wide now — FT8, FT4 and JS8.

All four JS8 speeds are selectable from v1.19.0 — Slow, Normal, Fast and Turbo — from the **JS8 speed** row in the settings drawer, and from the JS8 tab in the browser. Normal is the default and the one to use: 15-second slots, the same slot length as FT8.

On the Tab5 you can also **tap the big MODE label** in the left pane while JS8 is selected, and pick the speed there. That is where to look if JS8 suddenly decodes nothing.

⚠ **The speed is not a quality setting.** Each one is a different physical layer — slot length, symbol rate and tone spacing all change — so a station on Turbo cannot hear a station on Normal at all, and almost everybody runs Normal. The setting is remembered across a reboot and a power cycle, so if you leave it on Fast, Turbo or Slow, JS8 will stay silent until you put it back. The MODE label always names the speed in force — **JS8 NORM**, **JS8 FAST**, **JS8 TURBO** or **JS8 SLOW** — so you can confirm it is right, not only notice when it is wrong. Tap that label to change it. FT8 and FT4 are unaffected either way.

⚠ **Only Normal is proven on air.** The other three are built and they transmit, but no contact has been made on any of them, and the Tab5 has not decoded another station using one. Treat them as untested. Turbo was long thought to be out of reach over the control link; a measurement in v1.19.0 showed the tone command takes about 250 microseconds, so the Tab5's own side of it is comfortably fast enough. **What has not been measured is the radio's side** — whether the QMX can retune that quickly in practice. Until somebody completes a contact, that is an open question and not a promise.

⚠ **JS8 and FT4 share two of their calling frequencies.** If you were expecting FT4 activity on a frequency and see JS8 instead, that is why — the modes overlap on the band plan, not a fault.

10.2. The Stations and Conversation tabs

With JS8 selected, the right pane carries two tappable tabs:

- **Stations** — who is on the band. Each station that has identified itself, with its grid and how long ago it was heard.
- **Conversation** — what has been said, newest first, with four columns: **AGE**, **Hz**, **CALL** and **MESSAGE**.

Rows are tappable in both. The same two views appear in the browser, so you can watch a conversation from a laptop while the Tab5 sits with the radio.

10.3. What the decoder reads

Three kinds of JS8 traffic arrive:

- **Directed messages** — a message aimed at a particular station, shown with the sender's callsign.
- **Heartbeats** — the periodic `HB <call> <grid>` that JS8 stations send to say they are listening.
- **Free text** — a plain message, which JS8 splits across as many 15-second slots as it needs. The Tab5 joins the pieces back together and shows the message once it is whole, with a partial line while the rest is still arriving.

A free-text frame carries **no callsign of its own** — that is how JS8 works, not a limitation here. The Tab5 names a free-text message after a station that identified itself on the same audio frequency, and if two stations are close enough together to be equally plausible it shows the **frequency instead of a name**. A wrong callsign on somebody's message is worse than no callsign, so it declines to guess.

10.4. Sending free text

On the Tab5: tap **Free text**, the button under Call CQ. It is shown in JS8 only. Type the message, and the window tells you how many frames it takes and how many seconds it holds the frequency *before* anything is keyed, along with the text exactly as it will go out. **Send** starts it; **Stop TX** aborts a run that is already going out, and stays reachable while transmitting.

In the browser: the same composer sits under the conversation pane on the FT8/JS8 page, with the same plan line and the same Send and Stop.

The text is upper-cased, runs of spaces are collapsed and anything JS8 cannot carry is dropped. That is the encoder's own rule, and what the plan line shows you is the encoder's answer rather than a second copy of the rule — so what you read is what goes on the air.

⚠ **A long message is a long transmission.** One frame goes out per 15-second slot, so a twelve-frame message holds the frequency for three minutes. Read the plan line before you send.

The same three actions are scriptable, which is how the feature worked before the composer existed:

```
POST /api/cmd   {"action": "js8_text", "text": "HELLO FROM THE PANADAPTER"}
```

- `{"action": "js8_text_plan", "text": "..."}` reports the frame count and the seconds **without transmitting**.
- `{"action": "js8_text_cancel"}` stops a run that is already going out.

Progress is in `/api/status` under `ft8.ftx_sent` and `ft8.ftx_total`.

10.5. Calling CQ

The **Call CQ** control works in JS8 as it does in FT8, and sends `CQ <your call> <your grid>`.

Your grid goes on the air as **four characters**, always. If you have entered a six-character grid it is truncated for transmission — a six-character grid in a JS8 CQ is read by other stations as a callsign, which makes the call unintelligible.

10.6. What is proven, and what is not

Stated plainly, because "beta" on its own tells you nothing:

Proven on air:

- Receiving. Real stations decoded off air, including other people's free text arriving intact.
- Sending free text. A real JS8Call at the other end reassembled a three-frame message and displayed it complete, with the end-of-message marker it draws for a finished transmission.
- Calling CQ. A CQ from the Tab5 was decoded by a second station.

- **Compound callsigns.** `G0ABC/P`, `W1ABC/4` and special-event calls used to be discarded without trace — the frame that carries them needs a different codec, and before v1.18.0 the Tab5 refused that frame rather than risk putting somebody else's callsign on your screen. `SV2BBO/A` was decoded twice on 20 m and appeared in the station list under its full callsign, slash and all.

Not proven:

- **A full automatic two-way JS8 QSO has not been completed** — meaning no complete exchange has been watched from both ends. ⚠ Do not read that as "the automatic exchange is not implemented". **It is, and it has been since v1.17.0**: with automatic replies switched on your Tab5 answers JS8 callers with a signal report by itself. Gyula HA3HZ found four stations answered unattended in his own log after I had told him it was not possible. The automatic-reply setting is where you turn it off.
- **The Slow, Fast and Turbo speeds** (section 1). Built, selectable and they transmit; no contact made on any of them, and nothing decoded from another station using one.
- The fix that stops a message being shown under the wrong callsign (section 3) was built from a real on-air case but has **not** itself been re-tested on air — reproducing it needs a third station on a nearby frequency, which is not something that can be arranged to order.

10.7. Signal reports

JS8 signal reports come from the same measurement as FT8's and are calibrated against `jt9`, the reference decoder from the WSJT-X family, using 154 paired decodes of the same recordings. A report here should be within a decibel or two of what WSJT-X or JS8Call would say about the same signal.

If somebody asks you for one — from v1.20.2. JS8 has an explicit command for it, `SNR?`, and a station sending it to you by name used to get silence back. The selection chip now reads **? THEIRCALL** on the Tab5, and the browser says "*THEIRCALL asks for a report*", for ten minutes after the request. Pressing **SNR** with nothing else selected answers whoever asked.

⛔ **It does not answer by itself**, and that is deliberate: an automatic reply is the radio transmitting with nobody deciding to. If you have already tapped a station, that selection wins - a request from somebody else cannot redirect your report as the button goes down. A request addressed to `@ALLCALL` or another group is ignored, because it is not a question to you.

11. Time sync

FT8 requires accurate UTC time — within ±1 second of the real thing. The panadapter syncs time from multiple sources in priority order.

11.1. Time Sources (Priority Order)

```
type: flow
title: Which clock source wins - highest authority first
node: Unit GPS on PORT.A - a satellite clock, and it carries its own DATE
node: QMX+ GPS second-tick - only when the QMX GPS flag is set, and then it OUTRANKS SI
branch: phase-locked to the second boundary, about +/-25 ms, re-locked every 5 minutes
branch: drift-free and independent of WiFi, so better than our own SNTP
node: SNTP over WiFi - authoritative whenever WiFi is up
node: Tab5 RTC - holds the clock until something better arrives
node: QMX time-of-day - offline fallback ONLY
branch: undisciplined, the QMX's own clock drifts freely; trusting it above SNTP would
node: manual entry - last resort, no network and no GPS
note: amber | the BOOT order is not the authority order. The Tab5 RTC is applied first
note: steel | an FT8-derived nudge sits below all of these and is not really a source:
note: dim | WSPR and FT8 both search only a narrow slice of time either side of the slo
```

⚠ **The boot order is not the authority order.** The Tab5 RTC is applied first of all, at boot, before SNTP or the QMX can answer — that makes it the first source in *time*, not the highest in authority. Anything above it replaces it as soon as it arrives.

1. **Unit GPS v1.1 on PORT.A** (if selected in Settings -> Device -> Port mode) — satellite UTC **including the full date**, phase-locked to the second, works offline
2. **GPS-disciplined QMX** (auto-detected) — phase-locked to the GPS second, ~10 ms, works offline; on radio firmware 1.04_004 and later it brings the **date** as well
3. **WiFi + SNTP** (if available) — ~10 ms, re-syncs every ~1 hour
4. **Tab5 RTC** (if set) — persists across power cycles (±1 min accuracy)
5. **The QMX's own clock** (`TM;` over CAT) — offline fallback only, used when SNTP has gone stale. This is a *plain* QMX, with no GPS discipline: its clock drifts freely, which is exactly why it sits below the RTC rather than above SNTP
6. **FT8/FT4-derived** — a *nudge*, not a source. It is blocked outright whenever a real absolute reference exists (SNTP, a live Unit GPS, or a GPS-disciplined QMX), so it only ever disciplines a unit that has none
7. **Manual set** — enter time manually via the settings drawer

GPS and SNTP are the accurate sources and are used whenever present; if you're offline with neither, the panadapter falls back to the RTC (or FT8-derived / manual). The Unit GPS outranks the others while it has a fresh lock — see Unit GPS Time Source.

> **Note - Both GPS sources show `UTC(GPS)`**
>
> A Unit GPS and a GPS-disciplined QMX are both GPS, so the clock suffix is the same word for both. Which one you have is told apart by the **GPS indicator** on the bottom bar (see Bottom Bar Time Display) and by the apply lines in the log, not by a different suffix.

Unit GPS Time Source (PORT.A)

With **Settings -> Device -> Port mode** set to *Unit GPS v1.1*, the M5Stack Unit GPS on PORT.A disciplines the clock. It is the only offline source that brings the **date** with it, so it answers the "Is today's date right?" question by itself.

What it needs	Port mode = *Unit GPS v1.1*, the Unit GPS plugged into PORT.A, sky view for the antenna
What it gives	UTC time and the **full date**, from the satellite's RMC sentence
Priority	Highest, while it has a fresh lock (see the table above)
Date	Verified automatically on every accepted fix — the modal does not appear
Accuracy	Phase-locked to the second; no measured figure is documented yet, and none will be until it has been measured against a reference on hardware

The bottom-bar **GPS indicator** — one GPS symbol, coloured by state, the same construction as the Bluetooth glyph beside it — shows what the *receiver* is doing, which is a different question from the clock suffix (which shows who is *disciplining* the clock). The colour is the whole message:

Symbol	Meaning
amber	Running, no fresh fix yet — antenna unplugged, indoors, or still acquiring
green	Locked: a fresh fix, and this is the clock's authority
red	Had a lock, it has gone stale (cable out, sky lost) — the clock falls back to whatever else is live
(hidden)	Port mode is **Relay** — there is no receiver to describe

Locking takes 30 s to a few minutes from cold with a clear sky. **Switch the mode before plugging the GPS in** (enable-then-plug); see Settings for why.

Module GPS v2.1 on the 30-pin bus

> **Warning - Works, but there is no control for it on screen**
>
> A **Module GPS v2.1 (M5Stack M003-V21)** stacked on the 30-pin bus is handled by the same driver as the Unit GPS — it is the same AT6668 receiver, and the only difference is which pin the data arrives on (GPIO38). The receiver has now been run on hardware and delivers NMEA, but there is still no control for it on screen: it is switched on over the API only:
>
> ```
> POST /api/cmd   {"action":"gnss_mbus","on":true}
> ```
>
> **Set DIP switch 6 on the module — the first switch of the group the silkscreen labels "RXD" — and leave every other switch off.** The groups are named from the host's side, not the module's: the module's data output drives the host's *receive* line, so it is the RXD group that selects it. The TXD group produces nothing at all.
>
> Being on the 30-pin bus, it does not conflict with the PORT.A relay, so both can be fitted at once.
>
> Measured on the bench with the antenna at a window: a lock, a fix 64 ms old, about 530 bytes per second of well-formed NMEA, and the clock taking it every second. Indoors it sat without a fix for three minutes, which is normal for this receiver.

11.2. Offline (POTA / Portable)

If you're operating **without WiFi** (POTA, portable, SOTA):

0. **Best: take a Unit GPS.** With Port mode set to *Unit GPS v1.1*, the satellite gives you time **and the date** with no internet, no RTC remembered from home and no question to answer — see Unit GPS Time Source. Everything below is what to do when you have no Unit GPS.
1. **Set the Tab5 RTC before you leave home** (settings -> Time)
2. The RTC is powered by a **supercap battery** and holds time for **30–40 hours** without power
3. When you turn the Tab5 on in the field, it reads the RTC immediately
4. **Turn the QMX on whenever you like — you do not have to do anything about its clock.** A QMX without GPS comes up at 00:00 and its clock free-runs, so the panadapter does not take the time from it while the Tab5's own clock is good. It sets the radio's clock instead, and only when the radio is more than a few seconds out.
5. Optional: if your QMX has GPS, the panadapter syncs the RTC from QMX GPS every 5 minutes

No internet needed — FT8 timing works offline.

> **Note - Why step 4 says that**
>
> Earlier versions took the time from a GPS-less QMX when there was no WiFi, which meant switching the radio on in the field replaced your accurate RTC time with the radio's 00:00 — and FT8 stopped decoding. Reported by Don WB0LQW. The Tab5's supercap RTC is the better clock of the two offline, so it now wins, and the radio gets set from it.
>
> If you have no accurate time at all — the RTC was never set, or it has been unpowered for more than a day or two — then a QMX reading *is* used, because something is better than nothing. Failing that, set the clock by hand (**FT8 -> Options -> Sync Time**), which also accepts seconds.

> **Warning - The date is checked separately from the time**
>
> The QMX, its GPS and the hours/minutes/seconds boxes all give only a **time of day**. If the Tab5 has been off longer than its clock lasts and there is no internet, the date it has is simply the last day it was used — so a perfectly set time can still be logged under the wrong day. Reported by Don WB0LQW, whose POTA log came out two days behind.
>
> So when the date cannot be checked, the Tab5 asks about a minute after it starts: **"Is today's date right?"** Step it with **- day / + day** if needed and tap **This date is right**. If the Tab5 asked by itself and WiFi then comes up, the question closes by itself; if **you** opened the window to change the date, it stays open until you are done with it. QSOs are still logged if you do not answer — nothing is ever held back.
>
> The Tab5's own clock surviving a power-off is **not** taken as proof that the date is right. That clock keeps running as long as its small battery holds, but it says nothing about who last wrote it — another firmware flashed onto the same Tab5 can leave a date decades out with the battery perfectly healthy, which is what happened to royord. The clock is trusted only when this firmware's own record agrees with it, the clock has not gone backwards since that record, and the gap is short enough that somebody has plausibly been watching.
>
> The **Unit GPS on PORT.A is the exception**: its sentences carry the satellite date, so the date is verified by the fix itself and the question never has to be asked. See Unit GPS Time Source.
>
> The **Set and Sync the Clock** window also shows the date on its second line, marked *(unverified)* when it has not been checked. Tap that line to change it.

11.3. WiFi + SNTP

When WiFi is active:

1. The panadapter connects to an SNTP server (default pool.ntp.org)
2. Gets the current UTC time
3. Sets the system clock and writes it to the Tab5 RTC
4. Checks periodically (~hourly) and re-syncs if time has drifted

SNTP sync is **automatic** — you don't need to do anything. The bottom bar shows the current time (updates every second).

11.4. QMX GPS Time Sync (auto-detected)

If your QMX has **internal GPS** (QMX+ models often do), there is **nothing to enable** — the Tab5 detects it automatically:

1. **On radio firmware 1.04_004 and later it asks the GPS receiver directly** (`GP;`). A reply settles two things at once: this radio *has* a GPS — the receiver's own answer, not a guess — and **today's date**, which the radio's clock has never carried. Both work offline and on the first try.
2. It then catches the QMX's `TM;` seconds *flip* (the true GPS second boundary) and **phase-locks the clock to that edge** (~10 ms, drift-free), not just to the whole second. Re-locks every 5 minutes.
3. On older radio firmware, or a radio with no GPS, there is no `GP;` answer. Then the `TM;` flip is compared against SNTP: if they agree tightly — **and the Tab5 has not itself set that radio's clock** — the QMX is flagged GPS-disciplined. If they don't agree (a non-GPS QMX with a free-running RTC), it's used only as a low-priority offline fallback, and the check is **repeated every 5 minutes** until it does agree.

> **Note - Why the time still comes from `TM;` when `GP;` is the receiver**
>
> `GP;` is asked for the **date** and for the GPS answer, not for the clock phase. Measured on the bench against a reference: the second boundary `GP;` reports is roughly **a second behind** UTC, because a GPS receiver emits the sentence for one second during the next one. The radio's own clock, set from the satellite at power-on, is within about **100 ms** — an order of magnitude better. So the date comes from the receiver and the phase comes from the radio, each from whichever is actually good at it.

> **Note - A GPS that locks late is no longer missed (v1.16.3)**
>
> Before v1.16.3 that check ran **once**, about 45 seconds after boot, and its verdict stuck for the whole session. A receiver still hunting for satellites at that moment was written off as "no GPS" and never looked at again, short of a reboot. That is not a rare case: a cold start, or an antenna that has just been moved, can easily take longer than 45 seconds.
>
> Measured on the bench: a QMX+ with a genuine fix failed the one-shot three times, then locked — and the panadapter picked it up **25 minutes after boot**, which the old code could never have reached. If your GPS takes its time, you no longer have to do anything.

A GPS-disciplined QMX shows **UTC(GPS)** in the bottom bar and is a top-tier source (as good as SNTP); a non-GPS QMX shows **UTC(QMX)** and only helps when offline.

> **Note - Why step 2 asks who set the clock**
>
> The Tab5 pushes its own time into a non-GPS QMX to set that radio's clock — which leaves the radio agreeing with the Tab5 *because the Tab5 put it there*. Until v1.8.5 a close-enough agreement was taken as proof of GPS, so a radio with no GPS at all could be labelled `UTC(GPS)`, and worse, the Tab5 then stopped maintaining the clock of the one radio that had no other source. A clock the Tab5 has set is no longer accepted as evidence about itself. The flag clears when the radio's clock is plainly its own again — a QMX's clock is not kept across a power cycle, so switching it off and on is what re-arms detection.

> **Note - The label needs the radio to still be there**
>
> A remembered verdict says *this radio is GPS-disciplined*, not *GPS is keeping time right now*. With the QMX unplugged the Tab5 has no GPS of its own, so the label falls back to whatever is actually in charge — usually `UTC(NTP)`. Reported by Don N2VGU, whose Tab5 read `UTC(GPS)` with the radio disconnected; the time was correct throughout, only the label was wrong.

11.5. Setting the time by hand

There is no "Set Manual Time" control in the settings drawer. Earlier versions of this guide said there was, and there never has been on recent firmware — Don WB0LQW went looking for it during a POTA activation and could not find it. Apologies. The clock is set from the FT8 screen:

1. Switch to **FT8**
2. Tap **Options** (left pane)
3. Tap **Sync Time** (bottom right of the modal)
4. A panel appears with three boxes: **[HH] : [MM] : [SS]**
5. Tap **HH** or **MM** to type them on the numpad (0–23, 0–59)
6. **Hold the SS box and let go exactly on the minute** — the seconds are set to 00 the instant you release
7. Tap **Apply**

Step 6 is the one that matters off-grid. FT8 needs the clock right to about a second, and until v1.8.1 the seconds could not be set at all — hours and minutes were editable and the seconds were not, which left no way to get inside a second with no WiFi and no GPS.

The clock is set **when you release**, not when you tap Apply. The release is the measurement, so an Apply a few seconds later would be a few seconds late. Use a watch with a second hand, or listen for the gap between FT8 transmissions. While you hold the box it turns amber and tells you what letting go will do.

If you have typed hours and minutes first, those are used. If you have not, the Tab5 takes the nearest minute to its own clock — so this corrects the seconds without disturbing an hour and minute that are already right.

The time is written to the RTC as well as the system clock, so it survives a power-off.

11.6. FT8-Derived Sync (Offline Fallback)

The panadapter can also **estimate the UTC offset** from decoded FT8/FT4 signal timing and nudge the clock:

1. Decode several messages (requires on-air activity)
2. Measure their timing relative to the slot boundary
3. Estimate the error and apply a damped correction

This runs only as an offline fallback. While SNTP or a GPS-disciplined QMX is available they are authoritative and FT8-derived sync is **ignored** — deliberately. The FT8 slot offset the device measures is dominated by ~½ second of one-way *receive audio latency* (QMX SDR + USB buffering), which is **not** a clock error; letting it pull the clock would actually drag your transmit timing late. So it only touches the clock when you're off-grid with no SNTP/GPS.

Most useful when: WiFi is unavailable, no GPS QMX, and your RTC has drifted.

Fine-Tune Time with "Sync Time" (FT8 Only)

In **FT8 mode**, the Options modal includes a **"Sync Time" button** that opens an interactive time-setting panel:

1. Tap the **Options** button (FT8 screen, left pane)
2. Tap the **Sync Time** button (bottom right of the modal)
3. A panel appears with three fields: **[HH] : [MM] : [SS]**
    - **HH** / **MM** (hours/minutes) — tap to edit via numpad (0–23, 0–59)
    - **SS** (seconds) — auto-syncs from FT8 signals (blue frame = actively syncing, grey = locked). **Hold it and release on the minute** to set the seconds to 00 yourself, which is what you want when there are no FT8 signals to sync from
4. Tap **Apply** to write the time to the RTC and system clock

The **SS field** updates automatically from decoded FT8 messages — each decode gives a sub-second correction estimate. Tap **SS** to toggle between auto-syncing (blue) and locked (grey). While locked, seconds still count at the captured offset, so you don't lose precision after locking.

Use case: You're operating portable without WiFi, your RTC is ~5 seconds off, and there's on-air FT8 activity. Set HH/MM manually, let SS auto-sync to the decoded signal timing for a few seconds, lock it, and you're done — FT8 timing is now precise.

FT8-derived sync shows **SS (sub-second)** in the bottom bar when active.

11.7. Bottom Bar Time Display

The center of the bottom bar shows the current UTC time and which source is active:

Indicator	Meaning	Updates
UTC(GPS)	GPS: either the Unit GPS on PORT.A, or a GPS-disciplined QMX (auto-detected, phase-locked to the GPS second, ~10 ms)	every fix / every 5 min
UTC(NTP)	WiFi + SNTP	~1 hour
UTC(FT4) / **UTC(FT8)**	FT4/FT8-derived offline sync	Continuous (when decoding, offline only)
UTC(RTC)	Tab5 supercap RTC	At boot
UTC(MAN)	Manual time set	On-demand
UTC(QMX)	Non-GPS QMX RTC fallback (offline only)	Every 5 min
UTC	Fallback (no sync source)	—

The suffix tells you at a glance which source is **currently in charge** (the active authority, not merely the last one that wrote the clock). A GPS source (`UTC(GPS)`) and WiFi/SNTP (`UTC(NTP)`) are both accurate — if you see either, you're set. If you're off-grid you'll see `UTC(FT8)`/`UTC(FT4)` (on-air digital activity), `UTC(QMX)` (non-GPS QMX clock), or `UTC(RTC)` (the supercap RTC you set before leaving home).

At the right-hand end of the bar, just left of the Bluetooth glyph, sits the **GPS indicator** when Port mode is *Unit GPS v1.1*: one GPS symbol, coloured by state, with no words beside it — the Bluetooth glyph's construction. It reports the **receiver**, not the clock: amber while it is acquiring, green when locked, red when a lock has gone stale. The suffix only says `UTC(GPS)` when the symbol is green — the two never disagree, because the suffix is derived from the same lock the symbol shows.

11.8. Slot Timing

FT8 operates on **15-second slot boundaries** aligned to UTC. The panadapter:

1. Reads the current system time
2. Waits until the next 15-second boundary (hh:mm:00, hh:mm:15, hh:mm:30, hh:mm:45)

3. Starts a 15-second capture window
4. Decodes the received FT8
5. Transmits (if armed) at the start of the next boundary

This alignment is **automatic** — you don't configure slots. But **time accuracy is critical**:

- ±500 ms error -> can miss decodes or transmit off-slot
- ±1 s error -> very few decodes, transmit often off-time
- ±2 s error or worse -> FT8 doesn't work

8b. GPS Stats - the sky view

Tap the UTC clock on the bottom bar, or open **Radio -> GPS Stats** in the drawer, and the Tab5 draws what the GPS receiver can see:

- A **sky view** - the satellites in use plotted by bearing and elevation, with elevation rings and the compass axes drawn behind them
- The fix itself, the satellite count, and **which receiver answered**
- The same page is in the browser, composed by the device so the two cannot drift

The bottom-bar clock now distinguishes its source: **U-GPS** for a GPS attached to the Tab5, **GPS** for a GPS-equipped QMX+ answering over CAT.

Two receivers work, and both are confirmed on hardware:

Receiver	Where	How to enable
Unit GPS v1.1	PORT.A	Drawer -> **Port mode (relay / Unit GPS)** -> *Unit GPS v1.1*
Module GPS v2.1	30-pin M-Bus	**DIP switch 6**

⚠ On the Module GPS v2.1 the silkscreen DIP labels read backwards until you realise the groups are named from the **host's** side, not the module's. v1.16.12 shipped with this receiver supported in code but never run on hardware; it runs now.

Your grid fills itself from the fix. If the GPS has a position, the station grid is filled in for you - and a grid you have typed by hand is never overwritten.

⛔ The Unit GPS and the remote power-cycle relay share PORT.A. One cable, two uses: the relay cannot fire while the GPS owns the port, so unplug the GPS before switching back.

11.9. Time Sources Summary

Source	Accuracy	Updates	Works Offline?
Unit GPS (PORT.A)	phase-locked to the second — no measured figure documented yet	every fix (1 Hz)	Yes (plus the **date**)
QMX GPS	~10 ms (auto-detected, tick phase-lock)	every 5 min	Yes (if QMX has GPS)
SNTP	±10 ms	~1 hour	No
RTC	±1 min	at boot	Yes (30–40 h)
Manual	±1 sec	on-demand	Yes (until power cycle)
FT8-derived	offline fallback only	continuous	Yes

Next: Configure Settings or troubleshoot Time Sync Problems.

12. Settings

Swipe in from the right edge to open the settings drawer. All settings are saved automatically.

The top two buttons are not settings but the two ways into help: **User Manual**, which opens this guide at the chapter for the screen you came from, and **Need guidance?**, which lists symptoms and questions in plain words. See Getting Help.

12.1. Basic and Advanced

The drawer is grouped under headings — **Station**, **Device**, **Radio**, **Network**, **Display**, **FT8**, **WSPR** and **Spectrum** — and the button beside the **Settings** title chooses how much of it you see. It reads **BASIC** or **ADVANCED**; tapping it swaps between them. (It said EXPERT until v1.10.0, which described the reader rather than the contents.)

Advanced holds everything. Basic holds the sections an operating session actually reaches — Activation, QMX volume, RF gain, SWR protection, Antenna Tune, WiFi setup, Display brightness, and the FT8 group's distance and reporting options, plus the four WSPR sections when you are on the WSPR page. That is what Basic is *for*: it is the operating set, not a beginner mode. Nothing is lost in Basic; it is only hidden.

Which sections appear in which view is yours to set (v1.10.0). In the web UI, open **Settings** and click **Tab5 config** next to Save: every section in the Tab5's drawer is listed in drawer order with a **Basic** and an **Advanced** tick. A section can be in both, in one, or in neither — neither being a legitimate answer meaning you never want to see it. It is deliberately edited from the browser rather than from the Tab5, so it stays out of the way of anyone who just wants to operate.

A firmware update that adds a new setting **shows** it rather than hiding it behind a layout you saved before it existed — the device records which sections were known when you saved, so anything newer than that takes the shipped default instead of reading as "unticked".

This layout is stored on the device and is deliberately **not** part of a config backup: it is a preference about the menu, not a station or radio setting.

12.2. Radio

These reach the QMX itself over the CAT link.

QMX volume — the radio's own AF gain, in decibels, the same number the QMX shows on its LCD when you click its volume knob. Read back from the radio when you open the drawer, so it agrees with the rig even if you last changed it there.

QMX RF gain — the per-band RF gain from the radio's Band Configuration, 0–99 dB (the QMX default is 54). Because it is **per band**, the Tab5 reads it from the radio rather than remembering a number that would belong to whichever band you were on last; it shows "reading…" until the radio answers, and **"radio not connected"** if the radio is not there at all. Until v1.8.3 it could stick on "reading…" for a whole session: the value was only ever repainted when you *next* opened the drawer, so a single unanswered query left it stuck (reported by Samuel W7STF). It now fills itself in as soon as the radio replies. It writes when you let go of the slider, not while you drag, because this is stored configuration rather than a session setting. Changing it moves the noise floor, so the flat-spectrum reference is re-learned.

Show decoded CW — the line along the bottom of the panadapter carrying the Morse the radio is decoding, in CW/CW-R. On by default. The QMX does the decoding itself, so switching this off frees no processing — only the screen space; see the Decoded CW section of the Panadapter guide for what the line shows and how the noise is filtered. The same setting is in the browser under **Radio & display**.

CW profiles — a CW centre frequency together with the filter widths to offer with it, so a setup you use often is one tap rather than two menus on the radio (suggested by Uwe DL8UG: "*setting this on the QMX is always a bit of a chore, and half the time you do not have the right values in your head anyway*").

There are four slots. Edit them in the **browser's Settings window**, under **CW profiles**: a name, a centre frequency, and a tick for each width you want the bandwidth list to offer. Apply one from the button beside it, or from **CW profiles** in the Tab5's settings drawer, where the four slots appear as four buttons.

- The centre must be **500–950 Hz in steps of 25**, which is what the radio accepts. Anything else is corrected as you leave the box, so what you see is what will be stored.
- Names are cut to nine characters — that is what fits on the Tab5's buttons.
- Leave the centre empty to clear a slot.

> **Warning - Applying a profile changes your radio's stored configuration**
>
> It writes the CW centre and all eight filter rows, then asks the radio to reload — which takes about a second, and briefly interrupts reception. It is a deliberate, once-a-session action rather than something to tap while operating. Every write is read back and retried, and the bandwidth list is re-read from the radio afterwards, so what the Tab5 offers is what the radio really has.
>
> Note that setting the CW centre also moves the radio's CW offset and sidetone with it, which is the QMX's own behaviour and not something the panadapter chooses.

CW transmit offset — transmit a little away from the station you are listening to, so a QRP call is not buried in the pile of everyone zero-beating the DX (suggested by Roy KI0ER). The centre of the slider is off, and the sign chooses whether you transmit above or below.

Keep it small: around 100 Hz or less is typical, and the slider stops at ±300. The point is to land inside the other station's filter, not outside it — most CW operators run a 500 Hz filter or narrower, 200 Hz is not unusual, and the passband is not equally usable right to its edge: the tone gets muddy and a QRP signal has least energy exactly where the filter is already attenuating (Michael KZ4LY). Roy KI0ER uses +60 Hz on his own rig. Earlier versions of this page suggested 400–600 Hz, which is outside many operators' filters altogether.

> **Note - Your QMX may keep showing two frequencies afterwards**
>
> The offset works by putting the radio in split — receive on VFO A, transmit on VFO B. When you switch the offset off, the panadapter returns VFO B to VFO A and drops split, and it confirms both by reading the radio back.
>
> The QMX's own LCD may nevertheless keep showing a second frequency until you change band or power-cycle it. That is a display artifact: nothing repaints the second row of the LCD when the VFO mode returns to A over CAT (confirmed by Stan KC7XE, and known on the QRP Labs list). **You are not transmitting off frequency** — VFO B equals VFO A and split is off.
>
> Do not use the `MU;` command to tidy it up. It does clear the display, but it also silently drops the radio's I/Q mode, and the panadapter's spectrum goes flat while the radio keeps streaming audio as though nothing had happened.

The QMX has no XIT, so this is done with **split**: you receive on VFO A and transmit on VFO B, which the Tab5 holds at your receive frequency plus the offset. Set it once and it follows you — a tap on the panadapter, a spot, a memory recall, a band change, the web page, and the radio's own tuning knob. It applies in **CW and CW-R only**, and leaving CW clears it. If you are running your own split, the Tab5 leaves it alone.

Let me use the QMX menus — hands the radio back. The QMX's own menu, and its Band Configuration terminal application, talk over the same USB serial port the Tab5 polls several times a second, so the two fight over it. Tap this before using the radio's own menus: the Tab5 stops sending anything, the spectrum freezes, and a blue bar across the top says so and gives the radio back when you tap it. Resuming re-checks IQ mode, which a trip through the radio's menu can switch off.

While the radio is released the Tab5 will not transmit — an armed FT8 burst and Antenna Tune are both refused rather than keying a radio you are holding.

Radio menus — the QMX's own menu system on the Tab5's screen, so you do not have to reach the radio at all. Unlike the button above it, this does **not** stop the panadapter: it uses the radio's *second* USB serial port, so CAT keeps running while you are in the menus. You have to switch that port on once, on the radio — System config -> GPS & Ser. ports -> USB serial ports -> 2. For a QMX+ with no control panel this is the only way into its menus. See Radio Menus.

12.3. Operator Info

Callsign — Your amateur radio callsign (required for FT8/FT4 logging).

Grid Square — Your Maidenhead grid square (e.g., JO45; required for FT8/FT4 exchanges).

12.4. WiFi

Tap **WiFi setup** in the settings drawer to open the WiFi window.

WiFi On/Off — the WiFi icon button in the WiFi window (shown with a diagonal red slash when off). Toggling it takes effect immediately — off disconnects and stays off; on reconnects right away. Turning WiFi off is useful for field operation (no WiFi overhead, extended battery life).

SSID — Your WiFi network name. Tap **Scan** to list nearby networks and pick one (as of v1.3.6 this also works when your stored network is out of range — hotels, POTA sites — where it previously always reported "No networks found").

Password — Your WiFi password (pre-filled with your saved password; tap the eye icon to show/hide it). Tap **Save** to store it and connect.

Once connected, the settings show your **IP address** — use this to access the web UI from a browser.

Static IP address / Subnet mask / Gateway / DNS server — set in the web UI's Settings window under **WiFi** (there is no Tab5 equivalent for typing them — four addresses on glass is no kindness). **Leave the address empty for DHCP**, which is what every unit does by default, so nothing changes unless you fill it in. It takes effect the next time the Tab5 connects.

Leave the mask, gateway and DNS blank unless you have a reason not to. They are filled in from the address the Tab5 currently holds, so on an ordinary network you only have to type the address itself — and on anything that is not a plain 255.255.255.0 network, that is also the part most easily got wrong by hand.

An address on the wrong network is refused. The Tab5 compares what you type against the network it is on and will not accept an address that would put it somewhere else, telling you both. If you are deliberately setting a unit up for a different network, confirm when it asks and it will be stored as typed.

Use DHCP — a button in the Tab5's settings drawer, under **WiFi setup**. It appears only when a static address is configured. One tap arms it, a second clears the address and restarts on DHCP. This is the way back if a static address ever leaves the web page unreachable, and it needs nothing but the Tab5 itself. The bottom bar always shows the address the unit actually has.

12.5. Time Sync

SNTP Server — NTP pool (usually `pool.ntp.org`). Change only if you have a local NTP server.

Setting the time by hand — there is no control for this in the drawer, despite what earlier versions of this guide said. The clock is set from the FT8 screen: **Options -> Sync Time**. See Setting the time by hand, which also covers setting the seconds — the part that matters with no WiFi and no GPS.

FT8-Derived Sync — Estimates the UTC offset from decoded FT8 signal timing. **Offline fallback only**: it is automatically ignored while SNTP or GPS is available (those are authoritative), and only nudges the clock when you're off-grid with no better source. See Time Sync.

QMX GPS — Detected **automatically**, no setting to toggle. If your QMX (typically a QMX+) is GPS-disciplined, the Tab5 recognises it at connect by comparing the QMX's own second-tick against SNTP, and then phase-locks to the GPS second boundary (~10 ms) as an offline time source. On a non-GPS QMX nothing happens. The bottom-bar clock shows **UTC(GPS)** when a GPS-disciplined QMX is the active source. A clock the Tab5 has itself pushed into the radio is never accepted as evidence of GPS — see Time Sync.

Unit GPS (PORT.A) — not set here. Choose it under **Device -> Port mode**, since the first question is what is plugged into the connector. When it is on, it beats the QMX's GPS and SNTP while it has a fresh lock, and it is the only offline source that also gives the date. See Unit GPS Time Source.

12.6. Display

Flip 180° — Invert the display for upside-down mounting or cable routing.

QMX volume — The radio's own AF gain, **in decibels: the same number the QMX shows on its own LCD**, not a percentage. Available in both Panadapter and FT8 modes. It reads the radio back each time you open the drawer, so if you turn the volume with the rig's own knob the slider follows instead of disagreeing with it. This exists mainly for QMX+ builds with no control panel, where there is no volume knob at all (Randy N4OPI's request). Nothing is sent to the radio until you move the slider, so switching on can never change your volume unexpectedly.

The slider runs 0–50 dB rather than the QMX's full 0–199 dB protocol range, so the useful part of the travel is not crammed into a couple of centimetres. 50 dB is Randy N4OPI's figure from a real radio with the antenna connected — an earlier "not loud enough" reading turned out to have been taken with the antenna off, where the missing loudness was band noise, not gain. If you turn the radio past 50 dB with its own knob the slider knob sits at the end of its travel, but the number still shows the radio's true dB — it has to agree with the LCD.

Two things worth knowing, both from Randy N4OPI's side-by-side check against a real QMX (the numbers do read identically): the QMX only shows the figure on its own LCD when you give its knob a click, and the Tab5 slider reads the radio when the drawer *opens* — so if you turn the rig's knob while the drawer is on screen, the slider catches up the next time you open it rather than tracking live.

Display sleep — Dropdown (Off / 1 / 2 / 5 / 10 / 30 min). After the chosen idle time the backlight turns off; FT8, the radio link, and the web UI keep running. Tap the screen to wake — the wake tap is swallowed, so it can't tune or press anything. A **two-finger double-tap** blanks the display immediately.

Brightness — Screen brightness (0–100%).

Frequency format — How a frequency is punctuated, everywhere it is shown: the readout in the top bar, the FT8/FT4 preset button and its band lists, the WSPR band picker, the frequency keypad and the scale under the spectrum.

The default, `14.074.000`, is what the Tab5 has always shown and what the QMX shows on its own LCD. It is not a national convention — Icom, Yaesu and Kenwood group a frequency the same way, using the periods as visual anchors between MHz, kHz and Hz rather than as a decimal point. Matching the radio sitting next to the Tab5 is why it is the default.

The alternative, `14,074,000`, is US written grammar: a comma every three digits *(Don N2VGU, who reads and writes frequencies that way and whose instruments offer the choice)*. It changes as soon as you pick it — there is nothing to restart.

Spectrum Mode —

- **Normal** — absolute dBm scale
- **Flat** — relative to per-bin noise floor (signals pop above baseline)

12.7. Port mode (PORT.A)

Advanced -> Device. One connector, two possible owners, so there is one setting with two values:

- **Relay (default)** — the remote power-cycle relay drives GPIO53/54, exactly as it always has. A board that has never seen this setting is on **Relay**, so nothing changes when you update.
- **Unit GPS v1.1** — the M5Stack Unit GPS owns those pins as a UART instead, and disciplines the clock from the satellite: time **and the full date**, with no internet. See Unit GPS Time Source.

Switching is live — the pins are handed straight over, and a toast confirms which side has PORT.A. It is refused while a relay pulse or power cycle is running (finish it first), and refused if the GPS UART cannot be brought up (the mode then stays as it was, rather than changing to something that is not working).

> **Warning - Never leave the Unit GPS plugged in while Port mode is Relay**
>
> Relay mode **drives both pins**. The Unit GPS transmits on one of them. They are mutually exclusive by hardware: either plug the GPS in and select *Unit GPS v1.1*, or unplug it and stay on *Relay*. The warning is shown right under the control, and the bottom-bar GPS indicator is hidden entirely while Relay owns the port.

Tip: switch the mode first, then plug the GPS in. The reverse order works but puts the GPS's transmitter against a driven pin for the moment between the two.

Sharing PORT.A between the two — "Relay: drive only the selected pin" (v1.20.2). Relay mode normally holds **both** GPIO53 and GPIO54 as driven outputs, whichever one is set to pulse, so that changing the pin cannot leave the other one resting on its active side with a relay wired to it. If you have a Unit GPS on one pin and a relay trigger on the other, that is the wrong behaviour: entering Relay mode then drives a pin the GPS is transmitting on, and it keeps driving it for as long as you stay in Relay mode - no pulse required. Tick this setting and only the pin you selected is driven; the other is left floating. A pulse on the floating pin is then refused rather than silently doing nothing.

⚠ Only use it when the unused pin is unconnected or biased by its own circuit. A floating pin's level is decided by whatever is wired to it, not by the Tab5, so this is the wrong choice if you have relays on both pins.

The same control is in the web UI's **Miscellaneous** menu, above the remote relay row — the relay row greys out and says why whenever Unit GPS owns PORT.A. See Web UI.

12.8. Battery Care

Charge Limit — Optionally stop charging once the battery reaches a set percentage (default **80%**), to reduce long-term wear on the pack. Enable it and choose the target in the settings drawer; charging restarts automatically if the level later falls well below the target (5% hysteresis). Leave it off to always charge to 100%.

Accurate charge reading while charging — the displayed battery percentage (and the charge-limit trigger) now compensate for the voltage rise that occurs while charging current is flowing. Previously this made the reading jump around while plugged in, and made the charge limit either stick just short of the target or oscillate; both are fixed.

"(limit)" on the battery reading (v1.9.4) — once the cap trips, the battery text on both the Tab5 and the web UI grows a `(limit)` suffix. Without it, "capped on purpose at 80%" and "not charging for some other reason" looked identical.

12.9. Waterfall Controls

Black Level (dB) — How far above noise floor to show as black (default 9 dB). Lower = more colour detail.

Contrast (dB) — Span of the colour ramp (default 45 dB). Lower = more contrast, higher = more gradation.

FFT Window —

- **Blackman-Harris** (default) — best frequency resolution

- **Hann** — smoother peaks
- **Nuttall** — sharpest edges

Waterfall rate — in **rows per second**, default 10. The steps are 1 to 10, then 20, 30 and 40. Before v1.16.12 this was a 1× to 4× multiplier over a fixed 10 rows/s, which could only go *faster*; slower is what was actually wanted, both to put more history on one screen and to cut the waterfall's share of the drawing work. The spectrum and S-meter update rate is unchanged either way.

The same control is in the browser, and the two lists match.

Spur suppression — withdrawn in v1.8.9. The control is no longer in the drawer.

At some dial frequencies the QMX puts a comb of evenly spaced artifacts into the spectrum, and this setting was meant to remove them. It only ever worked at zoom ×1: above that the display is drawn from a second, finer analysis that the suppression never reached, so turning it on or off made no difference at all — and zooming in is exactly what you do when you are looking at a spur.

It also matters less than the original measurements suggested. Those were taken with the antenna disconnected, where the strongest tooth stood nearly 40 dB above the noise. With a real antenna on the same radio it measures about 23 dB, because band noise rises and buries the weaker teeth.

Nothing is lost by its absence, and the work is kept for a future release.

Wherever something is being removed, the thin line under the frequency labels turns teal. You can always see what is being touched.

> **Note - The 25 Hz nudge is real**
>
> Learning a frequency briefly moves the radio 25 Hz and puts it back. You may hear it on CW. It happens once per frequency, about two seconds after you stop tuning, and not at all on a frequency already learned. This is why the feature is off by default.

12.10. Panadapter & Zoom

Still spectrum *(Radio & display, on by default)* — the spectrum and waterfall hold still and the VFO marker moves across them, so a signal stays where you last saw it while you tune towards it. Switch it off and the display re-centres on the dial at every step with the marker in the middle. Applies from **×2 zoom up**; at ×1 the view is already the whole 48 kHz the radio sends, so there is no room to hold it still. The full behaviour, including when the view re-frames, is in Panadapter -> Still Spectrum.

Tune snap *(Advanced)* — the grid a tap on the spectrum lands on, in SSB and the digital modes: **Off**, 250 Hz, 500 Hz or 1 kHz. The default is 500 Hz, which is what earlier versions always did.

Choose **Off** to tune exactly where you tap (asked for by Samuel W7STF). The grid exists because SSB stations sit on whole kilohertz and the ones that stray are usually at the half, so landing on a round number is normally what you want — but it is a preference, and now it is one you can set. CW keeps its own 10 Hz grid, AM its 1 kHz, and tap-to-RIT overrides all of them, because an offset onto one caller's tone needs resolution rather than tidiness.

Distance in Miles — show distances in miles instead of kilometres, in the FT8 decode list **and on the WSPR page** (off by default). Until v1.12.0 the WSPR list and its Best DX panel always printed kilometres regardless of this setting, and the control was not reachable from the WSPR page at all.

Band-plan region — Sets which region's band plan drives the coloured CW/Digi/Phone strip along the bottom of the screen. **Auto** derives it from your grid square; you can also force Region 1/2/3.

Band Presets — Add or remove custom bands. Standard bands (160–10 m) are always available.

12.11. Bluetooth mouse and keyboard

A Bluetooth mouse is the one pointer that works **while the QMX is plugged in**. A USB mouse cannot: the radio occupies the Tab5's only USB host, and sharing it through a hub does not work on this hardware — both devices end up disabled. A Bluetooth mouse never touches that port, so the radio keeps its connection.

For anyone operating with cold or unsteady hands, that is the point: every menu, button and drawer control becomes a click instead of a precise tap on glass.

> **Warning - The mouse must be Bluetooth 4.0 or later**
>
> Only **Bluetooth Low Energy** mice work. The Tab5 gets its Bluetooth from a co-processor that has no Bluetooth Classic radio in it at all, so an older Classic mouse cannot be made to work by any firmware change.
>
> Most mice sold since about 2014 are Low Energy. Look for **Bluetooth 4.0** or later on the box. A dual-mode mouse works on its Low Energy channel.
>
> **How to tell which problem you have.** A Classic mouse never appears to the Tab5 at all — the Bluetooth symbol keeps looking and never turns blue, because the Tab5 listens only for mice that announce themselves the Low Energy way. A mouse that *does* connect but moves the pointer oddly is a different matter, and that one is worth reporting.

Setting it up

1. Tick **Bluetooth mouse -> Enable** in the settings drawer.
2. **Restart the Tab5.** Bluetooth can only start once the radio link to the wireless co-processor is up, so the switch takes effect on the next boot — the toast says so at the time.
3. Put the mouse into pairing mode and wait a few seconds. It connects on its own.

You only do this once. The pairing is stored, survives a reboot and a firmware update, and the mouse reconnects by itself from then on.

The Bluetooth symbol in the bottom bar sits just left of the WiFi indicator and has three states:

Symbol	Meaning
Dim grey	Bluetooth is switched off
Pale	On and looking for a device
Blue	A mouse or keyboard is ready to use

What works: moving the pointer, left click, and the scroll wheel — the wheel scrolls whatever is under the pointer, so the decode list, the settings drawer and the manual all scroll.

> **Note - The pointer disappears when the mouse sleeps**
>
> A Bluetooth mouse switches itself off after about half a minute of not being moved, to save its battery. The pointer vanishes with it and comes back the instant you touch the mouse — reconnecting takes under a third of a second. This is the mouse looking after its own battery, not a fault.

A Bluetooth keyboard

A Bluetooth keyboard works the same way and needs no separate setting — the same **Bluetooth -> Enable** switch covers both. Pair it and it types into every text field, exactly as the snap-on keyboard does: **Enter** presses Save in any window, **Esc** presses Cancel, **Tab** moves to the next field, and the arrow keys move the cursor.

A keyboard and a mouse can be connected at the same time, in either order.

The on-screen keyboard steps aside whenever a Bluetooth keyboard is connected. That is most of the reason to have one: the on-screen keyboard covers a large part of the display, and with a real keyboard you get that space back.

> **Warning - US keyboard layout only for now**
>
> A Bluetooth keyboard sends key *positions* rather than characters, and the Tab5 currently reads them through a US layout. Letters, digits, **Enter**, **Tab**, **Esc** and the arrow keys are the same on every layout, so typing a callsign, a grid, a frequency or a password works on any keyboard. Some punctuation on a non-US layout will produce a different character. National layouts are planned.

> **Note - A sleeping keyboard costs you nothing**
>
> Like the mouse, a Bluetooth keyboard switches itself off when idle. Pressing any key wakes it, and reconnecting takes a few seconds — but anything you type while it is waking up is held and delivered once the link is back, so you do not lose the first characters.

12.12. Firmware updates

> **Warning - The next update after v1.14.4 needs a USB-C cable, once**
>
> v1.14.4 is the last release that installs over the air for a while. The one after it claims 2.81 MB of flash that no partition has ever used, which means rewriting the partition table — and an over-the-air update can only ever write the app, never the map of the flash. That is deliberate: it is what stops a failed download taking the layout with it.
>
> So the next firmware arrives as a **flasher download** — the same USB-C data cable and the same `flash.bat` / `flash.command` you used the first time. One time, and everything after it is over the air again.
>
> **Your settings and your log are kept.** Press **Enter** at the flash-type prompt. Do *not* press **E** — that erases the whole chip, including your QSO log and your LoTW private key, and it is not needed here.
>
> The Tab5 will tell you when the time comes: the update window says so instead of offering a download, and it refuses to fetch an image it cannot install.

Updating is one route, and you are always asked *(v1.14.4)* — when a newer release appears, the version at the bottom of the screen changes to show it. Tap it, and a window opens with **Download now**. The download runs in the background, and when it finishes the same window offers **Restart now**.

Nothing is fetched and nothing is installed until you press something. The earlier "download updates in the background" option is gone: it had been suppressed by a memory guard for months and never actually ran, so removing it made the behaviour match what it had always been in practice.

⚠ Applying an update Keeping It Up To Date.

12.13. Live Spots

Draws other stations on the spectrum at the frequency they are operating on. Full details, including what the colours mean, in Live Spots.

Live spots (POTA) — On by default. Fetches Parks On The Air activations about once a minute and draws them over the spectrum. Needs WiFi. Switching it off leaves the spectrum completely clean.

RBN spots (CW skimmers) — **Off** by default, and deliberately so: RBN is a continuous global feed over a persistent connection, unlike POTA's occasional fetch. Adds stations the skimmer network is hearing right now, filtered to the band you are on. **Requires your callsign** to be set (the feed asks for one on connect — it identifies you, it is not a password).

DX cluster spots (phone) — **Off** by default, same reasoning: a second persistent connection. This is the source that carries **SSB**. Skimmers are machines, and no machine recognises a callsign spoken into a microphone, so phone activity is structurally invisible to RBN however long you leave it on. A cluster is people typing, so it carries voice contacts, and park and summit references written into the comment. Mode is worked out from the spotter's comment, or from your band plan when the comment does not say. Relayed skimmer spots are dropped, so the cluster cannot double what RBN is already showing.

Whatever you switch on, **one station is one entry**: the same callsign within 2 kHz is merged, an activation spot outranks a plain sighting, and a dropped duplicate folds back in as corroboration — drawn brighter, meaning a receiver copied them just now rather than someone having typed it an hour ago.

12.14. Spot Map

There is no setting here any more — the full-screen map of stations that have reported *your* signal runs its feeds from boot, always, and opens from the **SelfSpotter** button in the drawer (below **Need guidance?**), not a setting or a switch. Full details in Spot Map, including what running the feeds all the time costs.

Wants **your grid square** (the map draws lines from it) and **your callsign** (it is what the feeds are asked about). QRZ callbook credentials are optional and only place RBN skimmers more precisely.

12.15. FT8 Settings

FT8 On/Off — Globally enable/disable FT8 mode.

Field Day Mode — ARRL Field Day mode (on/off, class, section).

Simulation Mode — Practice QSOs with six phantom stations — no QMX needed at all, radio never keyed (red breathing border on screen). See FT8 Transmit.

Fast pounce (early decode) — On by default. Decodes surface ~1.8 s *before* the slot boundary (WSJT-X style), so a fresh CQ can be answered in the very next slot and mid-QSO replies land on the beat. Trade-off: the capture window closes early, so a station transmitting late in the slot can occasionally be missed. ⚠️ *Not yet A/B-verified on a live band — if your decodes-per-slot drop with it on, turn it off and please report your numbers.*

> **Tip - v1.16.4 made this actually deliver**
>
> Starting the decode early only helps if the *decision* to transmit is taken early too, and until v1.16.4 only one of the Tab5's two processors was allowed to take it. Which one happened to find the station calling you was chance — so the same situation answered immediately about half the time and a full cycle (30 s) late the rest, which is exactly how it looked from the operator's chair *(Gyula HA3HZ: "sometimes it works well, other times the transmission is delayed")*. Both halves may now decide. Verified on the air over three contacts, every message going out in the immediate slot.

Distance in miles — Show the decode list's distance column in miles instead of kilometres.

Report to PSK Reporter — **On by default.** Uploads the stations you decode to PSK Reporter, the same as WSJT-X, so you appear on the map as a monitoring station and other operators can see where they were heard.

What is sent, over the internet only and **never on the air**: your callsign and grid square, and for each station you decode their callsign, their grid (if their message contained one), the frequency, the signal report and the mode. Reports are batched and sent at most once every five minutes.

It does nothing at all until both your **callsign and grid** are set, and it is disabled outright in **Simulation Mode**, so practice contacts can never reach the public map. Uncheck this box to switch it off entirely.

> The first report goes out 5–5½ minutes after switching on, so nothing appears immediately. To check it is working, look for **QMX Panadapter** under "Software in use" at pskreporter.info/cgi-bin/pskstats.pl.

FT8 Filters — Include/exclude stations, set auto-reply priority, enable robot mode, grey-listing (see FT8 Receive for details).

Keyboard — M5Stack Tab5 snap-on keyboard support (if connected). See The snap-on keyboard below for everything it can do.

12.16. WSPR

The **WSPR** group appears only while the WSPR page is up, and holds **Allow transmitting**, **Declared power**, **Transmit schedule**, **Band hopping** and **Publish spots to wsprnet**. They are described where they make sense — see WSPR.

12.17. The snap-on keyboard

If the M5Stack Tab5 70-key snap-on keyboard is attached it is detected and needs no configuration. It does considerably more than fill in text fields.

Snap it on any time (v1.9.4) — attaching it after boot works too, found and ready to type within a couple of seconds; the same is true of taking it off and reattaching mid-session. Its two LEDs are deliberately dark once it's found - the keyboard works identically regardless of when it was attached, so there's nothing left for a light to usefully distinguish.

Typing. Any window with text fields takes the keyboard directly — WiFi setup, your callsign and grid, CQ messages, FT8 filters, memory channels. **Tab** moves between fields, **Enter** saves and **Escape** cancels.

Getting around.

Key	Does
Arrows	Scroll the settings drawer and the manual. Press repeatedly and it scrolls further each time
Page Up / Page Down	Move a whole screen
Enter	The confirming button of whatever is open — Save, Transmit, Yes
Escape	Closes whatever is in front of you, including the settings drawer and the memory page

Shortcuts. Hold **Ctrl** and press a letter:

Ctrl+R Radio menus	**Ctrl+M** User manual	**Ctrl+H** Need guidance?	**Ctrl+L** QSO log
Ctrl+K Memory channels	**Ctrl+P** Panadapter	**Ctrl+F** FT8	**Ctrl+S** Settings drawer
Ctrl+D Display off			

Any key wakes the display again afterwards.

Changing them. The web page has a shortcuts editor — **Miscellaneous ▸ Keyboard shortcuts**. There are 25 actions to choose from, including zoom, brightness, releasing the radio to its own front panel, and every setup window. **Alt** is left completely unused so you can put your own bindings there without disturbing anything. See The web interface.

> **Note - Nothing that transmits can be given a shortcut**
>
> Deliberately. A button you deliberately press is one thing; a two-key combination that a slipped finger can produce is another. Calling CQ, transmitting and tuning stay where you can see them.

This keyboard has no Shift and no Fn key — **Sym** and **Aa** are handled inside the keyboard itself, so **Ctrl** and **Alt** are what shortcuts can use.

12.18. Audio & DSP

IQ Balance — Adaptive I/Q phase correction (usually on). Suppresses mirror-image signals.

12.19. Diagnostic Logging

The diagnostic log is **always on** — there is nothing to enable. All firmware log output is captured to a 5 MB memory ring, with a rolling copy persisted to internal flash (survives a reboot or power-off) and, if a microSD card is inserted, mirrored to the card as well (see microSD Auto-Archive for when the card copy is written — the flash copy is always complete). Download via:

- Web UI: open the **Files** menu in the bottom bar and click **Diagnostic download** ↓ (downloads both the live session log and the flash-persisted copy from before the last reboot)
- microSD card: `/qmx-panadapter/qmx-log.txt`
- USB serial: `tools/capture_serial_log.ps1`

Useful for troubleshooting rare issues.

12.20. microSD Auto-Archive — Station Backup

Insert a microSD card (FAT32 or exFAT, any size — a plain 32 GB FAT32 card is ideal) and it automatically mirrors your whole station to `/qmx-panadapter/` on the card. It's a **grab-and-go backup**: pull the card into a PC (or another Tab5) to back up or move your setup — no computer needed in the field.

File	Contents
`qso.adi`	ADIF QSO log — after each new entry with WiFi off, otherwise at the next start-up
`qso.prev.adi`	The QSO log as it was just before it last got **smaller** — see below
`qmx-config.txt`	All settings + memory channels, as INI text (restore via **Config** upload)
`lotw_cert.b64`, `lotw_key.b64`	Your LoTW signing certificate + private key, so a restored device can sign for LoTW
`qmx-log.txt` (+`.1`)	Diagnostic log, rolling (rotated at 5 MB)
`README.txt`	A plain-text description of every file, written on each mount

After inserting the SD card your Tab5 needs a restart. The Tab5 can only claim the card during a short window early in boot, so a card pushed in while it is running is not used until you restart.

Benefits of a microSD card

The Tab5 works perfectly well without one. With a card in, you also get:

- **A full diagnostic history instead of the last few minutes.** Without a card, the log that survives a restart is held in a small area of internal flash and is overwritten roughly every 11 minutes of busy operating. On the card it is kept whole — so if something goes wrong overnight, or an hour ago, the evidence is still there when you come to report it.
- **A grab-and-go station backup.** Your QSO log, every setting, your WiFi details and your LoTW certificate and key, mirrored automatically. Move the card to another Tab5 and your station comes with it, with no computer involved.
- **A safety net for the QSO log.** The copy from just before the log last got *smaller* is kept beside it as `qso.prev.adi`, so a deletion you only notice two restarts later is still recoverable.
- **Somewhere to put things.** Browse, download, upload and delete everything on the card from any computer at `http://qmx.local/files` — without pulling it out.

A plain 32 GB FAT32 card is ideal. There is no benefit to a fast or expensive one: the Tab5 writes a few kilobytes a minute.

After inserting the SD card your Tab5 needs a restart.

When the mirror runs

The microSD card and the WiFi co-processor share a bus on this hardware and cannot both use it reliably. Rather than fail at an unpredictable moment, the Tab5 picks the behaviour that works:

	What happens
WiFi off (POTA/SOTA)	Continuous mirroring the whole time the card is in
WiFi on	One complete backup within a few seconds of switching on, then file mirroring stops — but the diagnostic log and the CW transcript keep going, every 30 seconds

Either way your QSO log, config, and LoTW certificate and key are backed up. With WiFi on, QSOs made later in that session reach the card at the **next start-up** — so if you have been operating with WiFi up and want them on the card now, restart the Tab5.

What the SD dot means (changed in v1.16.3)

The dot reports whether writes are actually **landing**, not merely whether a card mounted:

Dot	Meaning
Green	A write succeeded in the last 90 seconds
Yellow	A card is mounted, but nothing has been written lately
Grey, struck through	No card in the slot — not an error

Before v1.16.3 the dot went yellow about ten seconds into any WiFi session and stayed yellow for the rest of it, whatever the card was doing — so it read the same on a perfectly healthy unit and on one where every write was failing. It now distinguishes those.

A short spell of yellow is not a fault: on a quiet session there may be nothing to write. Yellow that persists while the Tab5 is clearly busy means writes are failing — see Troubleshooting.

While a browser is watching the spectrum, those 30-second writes wait. The card and the WiFi co-processor share a bus, and writing the card holds the spectrum stream for several seconds — long enough to look like the web page has frozen. The writes go through regardless after three minutes, so a crash still reaches the card; close the browser tab if you want the card fully current sooner.

> ⚠️ **The card holds credentials.** A full backup that can *restore* a station necessarily includes secrets: `qmx-config.txt` stores your WiFi password and QRZ/eQSL logins in clear text, and `lotw_key.b64` is your LoTW **private key**. Keep the card as physically secure as a house key. (The on-card `README.txt` repeats this warning.)

> The diagnostic log is always-on regardless of whether an SD card is present. If no card is inserted, the log still persists to internal flash (see Diagnostic Logging above) and survives a power-off.

Taking the card out, and restarting the Tab5

Two buttons in the settings drawer:

- **Eject microSD** finishes whatever is being written, closes the card and tells you when it is safe to pull it out. The confirmation stays on screen until you close it, so you can look at the card slot without missing it. Pulling the card without ejecting risks losing the last thing written to it.

- **Restart the Tab5** reboots cleanly from the screen instead of the power button. The card is released first.

- **Power off the Tab5** (from v1.20.1) shuts the unit down from the screen, directly under Restart. It is in the **Advanced** section, not Basic, because unlike Restart it does not come back. It asks twice: the first tap changes the button to *Tap again to power off*.

  This is a real shutdown, not a sleep, and it holds with 5 V still present on the USB-C input - measured, because that is how a Tab5 powered from the radio's own battery pack is wired *(Randy N4OPI, whose QMX+ is headless and who had already lost a set of 18650s to undervoltage)*.

  The radio is put back into receive first. The robot is stood down, transmit is disarmed, `TA0;` then `RX;` go to the radio, and settings are flushed - so a transmission in progress cannot be left keyed by shutting the Tab5 down. The same control is in the browser, under Miscellaneous.

  ⚠ The **power button** on the case cuts power in hardware. No firmware runs, so none of the above happens. Use the screen button if the radio might be transmitting.

The card is now also released automatically before a firmware update. Before v1.16.11 a reset caught the card mid-transaction, and because the Tab5 cannot power the card down on its own the card stayed confused until the whole unit was powered off - which is why a card could take several attempts to mount after an update, or not come back at all.

Restoring the log from the card

The card copy used to be one-way: the Tab5 wrote `qso.adi` to it and could never read it back, so a log lost to an erase-and-reinstall needed a computer, a browser, and knowing the file was on the card at all. It now restores from the device itself.

- **On the Tab5:** open the log window (**ADIF Log**) and press **Restore from SD**.
- **In the browser: QSO Logs ▲ -> ↳ Restore from SD card**.

Both merge: contacts already in the log are skipped, nothing is duplicated, and nothing already logged is lost — so it is safe to press twice. You are told what happened, including how many were already there and how many could not be read.

`qso.prev.adi` **— the copy from before.** The card mirrors the *present*, so a QSO deleted before a restart is gone from the card at the next start-up too. Whenever the log about to be written is **smaller** than the one already on the card, the older copy is kept as `qso.prev.adi` first. Normal logging grows the file and never disturbs it, so it holds the last larger version for as long as it takes you to notice.

To use it, copy `qso.prev.adi` off the card (**Files ▲ -> SD Files** in the browser, or a card reader) and restore it with **ADIF restore** ↑.

12.21. Activation (POTA / SOTA)

When you are activating a park or a summit, every contact you make has to be logged as belonging to **that reference** — it is what POTA and SOTA read to credit the activation, for you and for the people who worked you.

Open **Activation (POTA/SOTA)** in the settings drawer, pick the scheme, type the reference, and tap **Start activation**. From then on every logged QSO carries it automatically. There is nothing to remember per contact.

The screen shows your **contact count against the threshold** — ten for POTA, four for SOTA — and turns green when you reach it. That number is read from the log itself, so it survives a reboot and can never disagree with what was actually written.

Stopping matters as much as starting. While a session is running the main button is a red **Stop activation**, and the drawer button itself shows the live reference rather than a generic label. The failure that actually happens is driving home with it still switched on, quietly stamping every later contact with a park you have left.

The session persists across a restart on purpose — a battery change in the field should not lose it — but it is deliberately **not** included in a config backup, so restoring an old backup can never re-activate a park you are nowhere near.

Chasing works without any setup: if you work a station the panadapter has seen spotted at a park or summit, that reference is written into your log entry automatically.

Uploading just one activation: the web UI's ADIF download can be limited to a single reference, so you get that park's log rather than your entire file.

12.22. SWR protection

While transmitting, the QMX reports its SWR back over the control link. If the reading reaches the limit you set here the transmitter is **latched off** — nothing will transmit again until you clear it by tapping the red warning on the FT8 screen.

Choose **Off, 2.0:1, 2.5:1, 3.0:1 or 4.0:1**. The default is **3.0:1**.

This is one station-wide setting and one latch. A bad antenna is a fact about your station, not about the mode you happened to be using, so a trip in WSPR stops FT8 as well and the other way round.

In FT8 the burst is cut short; the SWR is sampled part-way through and the transmission stops there. **In WSPR the burst is cut short too**, from v1.19.0: the SWR is sampled every few seconds through the transmission and a burst that goes bad is stopped where it stands rather than running its full 110 seconds, and the next one is refused the moment you arm it. **In FT4 the check happens after the burst**, so the one burst completes and every later one is blocked. That is not an oversight: FT4 symbols are 48 ms and a control-link query takes about 50 ms, so asking mid-burst would disturb the transmission it is trying to protect. A reading is only acted on if there is real power behind it — the radio can report a meaningless ratio when the amplifier is not actually loaded. That is high enough not to trip on a merely mediocre match, and low enough to stop a burst going into a disconnected antenna, a wrong-band antenna or a damaged feedline — which is how this usually goes wrong in the field. An FT8 transmission is nearly thirteen seconds of continuous key-down, so there is real time to save.

The latch is deliberately sticky. A bad antenna does not repair itself, and resuming automatically would simply transmit into the same fault on the next slot.

> **Note - Set it to Off if you use a tuner**
>
> With a matched antenna the protection never fires, but if you are deliberately loading something unusual you may prefer no interference at all.

12.23. Who has heard me (last 24 h)

This asks PSK Reporter which receivers have copied **your** callsign recently. It is the reverse of the reports the panadapter *sends*, and the two are separate settings — you can have either without the other.

It answers a question no amount of local processing can: **am I actually getting out?** The valuable case is the mismatch. Stations you can hear that cannot hear you is a transmit-side problem — antenna, power, a bad connector — and from the receiving side alone it looks exactly like a dead band.

This one lives in the browser, not on the Tab5. There is no drawer control for it — the answer is a list of stations with distances and bearings, which wants a screen you are already sitting in front of. In the web UI open **Settings -> Spots & reporting** and tick **Who has heard me (last 24 h)**, then **Miscellaneous -> Who has heard me (24 h)** for the list: receiver, country, distance, bearing and the signal report they gave you, sorted by distance.

It is read-only and asks at most once every five minutes, which is the rate PSK Reporter requests.

> **Note - An empty list is the normal answer if you have not transmitted**
>
> Nobody can report hearing you until you have been on the air. Give it a few minutes after a CQ run.

12.24. ADIF & Logging

ADIF Log — View the QSO log on-device. The viewer shows a proper column table:

Column	Content
Call	Callsign
Grid	Their Maidenhead grid square, when they sent one
Country	DXCC entity (looked up from the callsign prefix)
Mode	FT8 or FT4
Band	Band (20m, 40m, …)
Date	UTC date
Time	UTC time
Sent	Your signal report (SNR)
Rcvd	Their signal report (SNR)
Ref	The park or summit *they* were activating, when the contact came from a spot

A sticky header row stays pinned while you scroll. Even-numbered rows are lightly shaded so long logs stay easy to scan.

Search — the field under the title filters the list as you type. It matches **callsign, country, mode, band, date, grid and the park/summit reference**, on any part of a word, and several words must all match: `ha3 20m` finds HA3-prefix contacts on 20 m only. Country is searchable even though the ADIF file does not store it — it comes from the same prefix lookup the Country column shows, so the two can never disagree.

If nothing matches, the list says so in as many words: "*Nothing matches X — so this one has not been worked.*" That is the question the search exists to answer. The decode list already greys out a station you have worked, but only while that station happens to be on the air; this asks the same question whenever you like *(Gyula HA3HZ)*.

Tapping the field brings up the keyboard, and the window moves up and shortens so the matches stay visible while you type. Clear the field to see the whole log again; the search also clears itself each time you open the window.

Today/All filter — the viewer opens on **Today** (falling back to All when nothing was logged today). The toggle button shows the view you *switch to* by pressing it; the title shows the current view with counts.

POTA activation counter — in the Today view the title reads "Today: N (M total)" and turns **green** once today reaches 10 QSOs — a valid POTA activation.

Delete a single record — **long-press** a QSO row: the row highlights red and list scrolling locks. Drag up/down to move the highlight, then release — a Delete/Cancel bar appears at the bottom. **Delete** removes just that one record (useful for duplicates).

Restore from SD — the middle button reads the QSO log back off the microSD card. See Restoring the log from the card below; there is nothing to choose, and contacts already logged are skipped, so pressing it twice does nothing.

Delete all — the red-bordered button at the bottom-left erases the **whole** log. Two-tap confirm: the first tap arms it (the label changes to "ALL *N*?"), a second tap within 5 seconds deletes; wait and it disarms itself. There is no undo — download the ADIF from the web UI first if you want a copy. Handy before a POTA activation: start with an empty log and the ADIF at the end is exactly the file you submit.

Anything this window does — a restore, a delete — reports back in a small panel with an **OK** button rather than a message that fades on its own. The result of something you asked for is worth reading, and dismissing it is how you say you did.

Use the web UI to download the full ADIF file for import into WSJT-X, EQSL, or any other logging software — or view and edit it in the browser (**QSO Logs -> View / edit log**).

Exclude Worked Before — When FT8 filtering, skip stations you've already logged QSOs with (requires you to import your own prior ADIF log first).

12.25. Config Import/Export

Config Download — Export all settings + memory channels + ADIF log as a text file (INI format).

Config Upload — Restore settings from a backup file (settings only; memory channels merge).

Use this to:

- Back up your settings before a factory reset
- Transfer settings to another Tab5
- Recover from accidental changes

12.26. Resets — in the browser, not the drawer

These two live in the browser, not in the drawer — a reset you can trigger by mistake on a touchscreen in the field is a worse idea than one that needs a computer. Both are in the web UI's **Miscellaneous** ▲ menu.

Reset settings — Erases the stored settings and memory channels, returning everything to defaults, and reboots. Use if something is stuck in a state you cannot get out of. Take a **Config** ↓ backup first and you can restore it in seconds.

Reset WiFi — Erases the WiFi credentials and the stored radio/link state. This is the one for a WiFi setup that will not come up no matter what you enter.

Neither of them touches your QSO log. The ADIF log lives in a separate area of flash that is deliberately left alone, so a reset cannot cost you contacts — and neither can a firmware update, which is why a normal flash keeps your log. The only things that erase the log are the **Delete all** button in the ADIF viewer, the same in the web log viewer, and a **clean/erase flash** from the flasher.

Next: Troubleshoot an issue via Troubleshooting or explore the API.

13. WSPR

WSPR (Weak Signal Propagation Reporter, pronounced "whisper") answers one question: *where does my signal actually go?* You transmit a very short, very slow beacon carrying only your callsign, grid and power — and stations all over the world that hear it report the fact. Over a few hours you get a map of what your antenna and your band are really doing, at power levels where nothing else would be heard at all.

It is not a contact mode. Nobody replies, there is no exchange, and nothing goes in your log. That is the point: it measures propagation instead of working people.

Swipe → from the left edge to cycle **Panadapter -> FT8/FT4/JS8 -> WSPR** and back.

13.1. The two-minute rhythm

Everything in WSPR is built on a **two-minute cycle**, aligned to UTC. A transmission starts a second or two after each even minute and lasts about 110 seconds, so one cycle carries exactly one beacon.

```
type: timeline
title: One WSPR cycle - 2 minutes
span: 120
seg: 1-111.6 red transmitting - 110.6 s, 162 symbols of 0.683 s
seg: 111.6-120 dim ~8 s
tick: 0, 30, 60, 90, 120
note: red | a transmission starts one second into an even minute, by the protocol
note: trace | on a RECEIVE cycle the whole two minutes is captured, and the decode has
note: dim | the schedule is two COUNTS of cycles, not a percentage: transmit this many
note: amber | the first burst after you switch transmitting on is always the very next
```

Exactly: 162 symbols of 0.683 s is **110.6 s** of the 120 s cycle. The schedule at the bottom is the part worth studying before you switch transmitting on — it is two counts, not a percentage.

That slow rhythm is why the page looks calm compared with the FT8 screen. There is no countdown urging you on and no QSO furniture, because there is nothing to answer. The countdown at the top left simply tells you where you are in the current window.

Your clock has to be right. WSPR's decoder searches only a narrow slice of time either side of the cycle boundary, so a clock that is out by more than a couple of seconds will hear nothing and be heard by nobody. The Tab5 keeps itself right from SNTP over WiFi, from the QMX's own clock, or from its internal RTC — see Time Sync. If you are running with no network and no GPS, check the clock before you trust an empty screen.

13.2. Reading the page

The browser's WSPR list carries the same columns, and from **v1.20.2** that includes **BAND** - it was on the Tab5 and not in the browser. It matters most with band hopping on, where every spot sits inside the same 200 Hz window and the frequency column alone cannot tell them apart. It is blank for spots recorded before the dial was kept.

The left pane is a log, not a live list of who is on frequency. Spots stay where they are and new cycles are added — an opening band looks different from a closing one only over time, so the page is arranged to let you see that.

Column	Meaning
S	The letter this station is marked with on the waterfall for that cycle — see Which trace is which below. Blank once the cycle it belongs to has scrolled out of the picture
UTC	The cycle this spot came from
CALL	The station heard
GRID	Their Maidenhead locator, as transmitted
COUNTRY	Country from the callsign prefix, spelled out where it fits and shortened where it does not - never a 3-letter code
BND	The band it was heard on, in metres. Blank for spots recorded before v1.10.5, and worth having the moment band hopping is on
PWR	The power **they declared**, not a measurement
SNR	Signal-to-noise, in the WSPR convention (a 2500 Hz reference — figures around −25 dB are entirely normal and perfectly decodable)
TONE	Where in the 200 Hz sub-band they were heard
DR	Drift: how far the sender moved during the transmission. A stable, oven- or GPS-referenced transmitter reads +0; a crystal warming through the burst reads +1 or -2. Whole Hz, which is what wsprnet publishes, and a coarser measurement than TONE beside it - it is the difference between two half-window estimates, so read a trend across several spots rather than one

Column	Meaning
DT	How far into the two-minute cycle that transmission actually started, in seconds. **Nominal is +1.0**, because a WSPR transmission begins one second into its even minute — so a value far from that is the other station's clock rather than anything at your end
KM	Great-circle distance from your grid. The heading reads **MI** if you have chosen miles — see Settings. A leading `~` means the distance came from the station's country rather than its grid

Below the list:

- **DX** — the furthest station of the session, which is usually the number you actually want.
- **WSPRNET** — whether spots are being published, and how many can be. **"N of M publishable"** is the publication gate: a station is only sent to wsprnet once it has been heard **more than once**, so N is how many of the M calls heard so far are eligible. This is also why a site like wspr.rocks can show fewer unique calls than this page reports hearing — it only ever received the publishable ones.
- **Flush** — empties the decode list. It asks first, because the list is also the upload queue: anything not yet sent to wsprnet goes with it, and the publishable count resets.

> **Tip - With a mouse, point at a trace to see whose it is**
>
> Hovering a trace on the waterfall names the station, its tone and its signal report, beside the pointer. It matches within about half a WSPR signal's width, so you have to be on the trace rather than near it. There is no hover on a touchscreen, so this adds to the list rather than replacing it — everything it shows is in the table underneath.

The right pane shows the captured 200 Hz window for the cycle just decoded. WSPR's whole sub-band is narrower than a single FT8 signal, so this is a very close-up view: individual beacons appear as near-horizontal lines, and a sloping line is a drifting transmitter.

On a cycle you transmit, the waterfall does not advance — the receiver is stood down for the whole two minutes, so there is nothing to draw. Rather than leave the previous cycle's picture sitting there looking frozen, the display lays down its cycle-boundary marker and keeps the last received image below it, so you can still see what was there before you transmitted. The status line reads **transmitting** throughout.

The panadapter spectrum is not available while WSPR runs. The receiver takes the IQ stream for the whole cycle, so there is nothing left to draw a live spectrum from. This is expected, not a fault.

Which trace is which

Each time a cycle finishes, the traces the decoder examined are lettered on the waterfall — **A, B, C… from left to right by tone** — and the cycle's UTC time is printed at the left end of the row. The same letter appears in the list's **S** column, so a line of text and a mark on the carpet are the same station, and you can tell at a glance which of five traces produced the report you are reading.

The letters describe the cycle **below** the boundary line they sit under, since that is the cycle that just finished. They scroll down the carpet with it and are gone once it leaves the picture, which is why an older row's **S** column is blank.

- **A `?` is a trace the decoder tried and could not read.** That is worth seeing: it is the difference between "nothing was there" and "something was there and was too weak", which is the whole question when you are judging an antenna or a band. It usually means a signal a few dB below the decode threshold, but a birdie or a carrier sitting in the sub-band will also collect one.
- **Not everything gets a mark.** The candidate finder always returns a full quota of possibilities, most of which are its own noise floor rather than signals, so a `?` is only drawn for a candidate well above the middle of that cycle's crop. Expect a handful, not a row of punctuation.
- **A mark that cannot be placed honestly is left out.** Where several traces are too close together to letter separately, decoded stations are placed first and the rest are dropped rather than nudged aside — a marker pointing at the wrong trace is worse than no marker.
- The letter is drawn a couple of Hz above the tone in the table. A WSPR transmission is four tones about 4.4 Hz wide and the reported frequency is the lowest of them, so the mark sits on the middle of the trace rather than its left-hand edge.

Changing band — tap the button at the top of the left pane and a list drops below it. Touch a line and it lights up; if it is the wrong one, keep your finger down and drag until you are on the one you want, then lift to choose it. Releasing outside the list chooses nothing. The band you are on is named in amber.

> **Warning - Use *this* button, not the top bar**
>
> On the WSPR page the top bar's **Band**, **Mode**, **BW** and **Zoom** are greyed out and do nothing: they belong to the panadapter's idea of the radio, while this page owns the dial. Only **Freq** stays live, so a beacon found off the standard dial frequency can still be chased by typing it in.
>
> On **v1.12.0 only**, the top bar's Band control overlapped this panel and took the touch — picking a band from it wrote an FT8 frequency to the radio while WSPR carried on capturing and labelling its spots with the band it thought it was on. If you used the WSPR page on that release, check which band your radio is actually on.

13.3. Settings

All of WSPR's settings live in the **settings drawer** (swipe ← from the right edge) under **WSPR**, and appear only while the WSPR page is up. The page itself keeps just one control — the **TX** button — because that is the only one you reach while a session is running.

Transmitting is the TX button, and nothing else

There is no separate "allow transmitting" setting. The **TX** button on the page is the only control, so there is nothing that can disagree with it.

It is off every time you open the WSPR page, whatever you did last session. A beacon that resumes on its own because of how you left it a week ago is not a decision you made, so the page always starts as a pure receiver — which is a perfectly good way to use WSPR — and transmitting is something you switch on deliberately each time.

Your callsign and grid must be set, in **Station -> Callsign & Grid square**. Without them there is no transmission at all — the same rule FT8 follows. WSPR sends your callsign to every station that hears you and publishes it to a public database, so it uses the identity you entered and nothing else.

Calibrate Power

Before Declared power can offer anything real, the current band needs calibrating. **Calibrate Power** sweeps the QMX's *Max. PA voltage* on a **dummy load** — not the antenna — and records the real RF output at each step with the radio's own `PC;` readback. The result is saved for that band and used from then on.

The sweep runs the full 1.0–12.0 V range unless you tell it not to, and it puts your Max. PA voltage back when it finishes.

It does not stop at your Max. PA voltage, and it is worth knowing why. Declared power, the Output power slider and WSPR all write that same setting, so the figure in it is usually one this app put there rather than a limit you chose — one station swept a 600 mW ceiling because a WSPR session had left his radio at 6.0 V, and every reconnect re-applied it. Capping the sweep at it measured a fraction of the radio.

If your radio must never see a certain voltage, set "Never exceed PA voltage" in the Output power section. That is a setting of your own that nothing else writes, so the sweep obeys it, and the window tells you the range it will actually cover before you start. Use it for a 9 V build, a radio with a known weakness in the finals, or any PA you do not want driven to 12 V. It is off by default.

The sweep also **stops early once the measured power stops rising**, which is what happens above your supply voltage: on an 8 V supply every step above ~8 V would otherwise re-measure the same watts for minutes of pointless key-down. That is measured rather than assumed, so it works without the app knowing your radio. Bruce N9JCV found it. Note what it does and does not protect: it stops a radio that *cannot* reach a voltage, not one that can reach it but should not — that is what the limit above is for.

Because the sweep can end early, a shorter results table is normal rather than a failure — the window says what ended it.

You reach it two ways: the **"Calibrate this band"** button that appears in place of Declared power (or Output power, on other modes) whenever the current band has no data yet, or **"Recalibrate this band"** in the same place once it does — for redoing a band after a change to your antenna or feedline.

Declared power

This is **a claim, not a measurement** — every station that hears you publishes it worldwide, where other operators use it to reason about propagation. But once the band is calibrated, the dropdown only offers the standard WSPR dBm steps your radio actually reaches, and each one is labelled with the **real measured wattage**, not the textbook figure the step's name implies. Picking "33 dBm" no longer means finding out afterwards that the radio actually did something else.

This is also where you protect the finals now. WSPR transmits for about **110 seconds out of every 120** — nothing else this radio does comes close, and QRP Labs warn about exactly this kind of sustained duty cycle in the QMX manual. There is no separate switch for it any more: picking a low declared level (a few hundred milliwatts is plenty to be heard) keeps the finals cool for the length of a beacon run, the same way choosing any other setting on this dropdown does. Anything above 1 W gets a plain warning label, rather than a guard acting on your behalf underneath you.

During each transmission the Tab5 asks the radio what it actually put out, and shows the answer under the dropdown — "*radio measured 1.6 W last burst = 32 dBm*" — so you can see whether a burst matched what you declared.

Switching transmit off stops a burst that is already on the air. The **TX** button keys the radio down at once rather than waiting for the two-minute cycle to end.

Output power

A separate slider, filed next to Calibrate Power, sets the radio's output for every mode **other** than WSPR — FT8, CW, SSB, and so on. It shares the same per-band calibration but is a genuinely different control: WSPR's Declared power always owns the radio while WSPR is running, so the Output power slider doesn't appear on this page at all. Leaving WSPR hands control back to it immediately.

Running WSPR from a lower supply voltage

A low declared power reduces the heat in the finals. It does not remove it from the radio. The QMX limits PA voltage with a pass transistor, so the difference between your supply and the PA voltage in force is dropped inside the radio as heat instead — measured on a bench QMX, holding the PA at 6 V against a 12 V supply cut the finals' own heat by 76% but total heat by only 18%. That trade is worth making — the PA transistors are the fragile, hard-to-replace part — but it is not the whole answer.

If you intend to beacon on WSPR for hours, feed the QMX from a lower supply. The QMX accepts **6.0 to 12.0 V**, and running it at around 9 V means less voltage to throw away as heat anywhere in the radio. This is the one thing that helps which no firmware setting can do for you.

Transmit schedule *(changed in v1.14.5)*

Two numbers describe one repeating group: **Transmit cycles**, then **Receive cycles**.

Set transmit to 2 and receive to 8, and the beacon transmits for two cycles, listens for eight, and repeats for ever. The period is simply the two added together — ten cycles, twenty minutes — and the Tab5 spells it out under the two controls as you change them:

> Transmit 2, then listen 8 — repeating every 20 min. Transmitting 20% of the time.

Transmit	Receive	Pattern, repeating	Period
1	4	Tx Rx Rx Rx Rx	10 min
1	9	Tx Rx ×9	20 min
2	8	Tx Tx Rx ×8	20 min
2	3	Tx Tx Rx Rx Rx	10 min

Setting transmit to **Receive only** never keys the radio, and greys out the receive count — with no transmission there is no group to space out.

Why two numbers and not a duty cycle. This used to be "1 in N" plus a separate "bursts per transmission", and that pair had no agreed meaning. "1 in 5" plainly means one cycle in five, and with a single burst that is exactly what it did — but ask for two bursts and the phrase stops saying anything. Does the group grow into the listening time, or is the listening time kept? Two people can read it two ways, and did. Two counts cannot be read two ways.

Your existing setting is carried over exactly. A Tab5 upgrading from v1.14.4 or earlier keeps the schedule it was already running, cycle for cycle — the old "1 in 5" becomes 1 transmit and 4 receive, which is the same thing said plainly.

Transmitting more means keying the finals more. Two cycles in a row give a distant receiver a second chance at you when the first falls in a fade, which is why the QMX's own beacon offers it — and it doubles your share of key-down time. Pick the declared power to match; see **Declared power** above.

Band hopping

Tap **Choose bands…** and tick the bands you want. **Ticking two or more turns hopping on**; leaving one ticked keeps you on that band. There is no separate on/off switch, because the list of bands already says what you want.

Only the bands your radio can actually reach are offered. A QMX is built with a fixed set of filters, so the list is asked of the radio rather than assumed — the bench QMX offers 60/40/30/20/17/15 while a QMX+ covers 160–6 m.

Hopping changes band **between** cycles, never during one, and the waterfall's noise floor is reset on each hop so a quiet band is not painted against the last one's noise.

Publish spots to wsprnet

Off by default. When on, the stations you hear are uploaded to **wsprnet.org**, where they join the public database. This is how WSPR is useful to anyone other than you: your receiver becomes one of the reporting stations that lets other operators see where *their* signals went.

It needs WiFi and your callsign and grid.

13.4. Transmitting

With a callsign and grid set and a transmit count other than **Receive only**, the **TX** button on the page arms the station. Which cycles actually transmit is decided by the transmit schedule, and the button shows what is happening.

The countdown on the button is the time until a real transmission. The schedule is worked out in advance, so `TX ON next 6:14` means a burst is coming in six minutes and fourteen seconds — not that a cycle boundary is due and might or might not be used.

A few things worth knowing before you leave it running:

- **Your radio is keyed for real,** for about 110 seconds at a time. Make sure it is connected to an antenna or a dummy load, and that the power it is producing matches what you declared.

- **SWR protection still applies.** If the SWR limit in **Radio -> SWR protection** is exceeded, transmitting stops. From v1.19.0 that means the burst in flight is cut where it stands rather than running to its end, and arming the next one is refused until you clear the latch. Before v1.19.0 this page said so and the firmware did not do it - WSPR measured the SWR, wrote it in the log and carried on.
- **The Tab5 wakes up on the page you left it on, but not transmitting.** It returns to the WSPR page after a power cycle and starts receiving; the **TX** button is off, so it will not resume beaconing on its own. Switch it on again when you are ready.
- **Split stops it, and says so.** WSPR is transmitted on the dial frequency, and the spot you publish names that frequency. If your radio is in split it keys VFO B while still reporting VFO A, so every spot would name somewhere the signal never was. Before each transmission the Tab5 asks the radio, and if split is on it holds the burst and shows **TX held - radio is in SPLIT, clear VFO B**. It will not clear it for you: that is your setting, and on the QMX split cannot be cleared over the cable anyway — use the radio's own menu or power-cycle it.
- **Simulation mode blocks every byte.** If you want to watch the mechanics without keying anything, turn on **FT8 Simulation Mode** in the drawer; it interlocks WSPR TX as well.

13.5. From the browser

The web UI mirrors whatever page the Tab5 is on. Switch it to WSPR with the **Switch to WSPR** link in the bottom bar, and the browser shows the same page the Tab5 does: a **MODE: WSPR** heading, the band selection, the **TX** button with its countdown, the finals-guard voltage, best DX, stations heard per cycle, the wsprnet publishing status — and the table of spots.

The table is shown only while the Tab5 is on the WSPR page, because that is the only time the receiver is running — on any other page it would be a frozen list that looked live.

Transmitting from the browser uses your stored callsign, grid and declared power. They cannot be overridden per request: what goes on the air is the identity you set on the device.

13.6. If nothing is decoded

WSPR is quiet by nature — an empty screen for one cycle means very little. Before assuming a fault:

- **Check the clock.** The bottom bar shows the time source. A clock more than a couple of seconds out will decode nothing.
- **Check the band and the hour.** WSPR follows propagation; 20 m at midnight will be emptier than 40 m.
- **Check the dial.** Each band has exactly one WSPR sub-band, and the picker only offers those — but if you tuned the radio by hand afterwards, you may not be on it.
- **Give it several cycles.** At two minutes each, four cycles is eight minutes. That is a normal amount of patience for this mode.

See also Troubleshooting.

14. Radio menus

The QMX's own menu system, on the Tab5's screen — the same 80×24 terminal you would get from a laptop running PuTTY, without the laptop.

This exists because a QMX+ with no control panel has **no other way in**. If you built a headless radio to save weight and space, this is how you reach Configuration, Band config, System config and everything else.

It is equally useful on a radio that *does* have a panel: two lines of LCD and a rotary encoder is a slow way to move around a deep menu, and the Tab5 shows the whole thing at once.

14.1. First: give the radio a second serial port

The terminal runs on the QMX's **second** USB serial port, and that port is off by default. You only have to do this once — the setting survives a power cycle.

On the radio: **System config -> GPS & Ser. ports -> USB serial ports -> 2**

> **Note - Why a second port, and not the one already in use**
>
> The QMX manual is explicit that leaving a terminal session without choosing *Exit terminal* leaves the radio refusing CAT commands. On the port the panadapter is already using, that would take the whole display down with it.
>
> On its own port it cannot. CAT frequency, mode and the S-meter were measured running normally for the entire length of a terminal session — the panadapter keeps working while you are in the menus.

If the port has not been enabled, the Tab5 tells you so rather than failing quietly — and it puts **the whole menu path on screen**, in both the Tab5 panel and the browser, laid out to be followed while you are looking at the radio. The person who needs that instruction is the one who never read this page, so it belongs where the failure happens rather than in a message that disappears. *(Michael KZ4LY)*

14.2. Opening it

=== "On the Tab5"

```
**Settings drawer -> Radio -> Radio menus**

The keys are along the top, all the same size: **▲ ▼ ◀ ▶**, **Enter**,
**Back**, **BS**, the **keyboard** toggle, and **Close**.

**With the snap-on keyboard attached you can skip all of that** and drive
the radio's menus directly: arrows to move, Enter to select, digits to type
into a numeric field, Backspace to delete, and **Esc to leave the screen**.
Typing a value is far quicker than the on-screen keys.

Or open it without touching the screen at all: **Ctrl+R**.
```

=== "In the browser"

```
**Radio ▸ Radio menus**

The on-screen buttons work, and so does your actual keyboard — arrow keys,
Enter, Esc, and typing. For entering a value into a menu field, the keyboard
is much the faster route.
```

The radio's menus **cascade**: opening a submenu draws a new box over the top of the one you came from, so you can see the path you took. The highlighted row is where you are.

14.3. Leaving it

Use **Close**.

The Tab5 walks the radio back out through its own *Exit terminal* item — it reads the screen to find it, so it still works however deep in the menus you were. That matters: simply dropping the connection is what leaves a QMX ignoring CAT.

Two things happen automatically so you cannot get stuck:

- If you close the browser tab, the session is closed for you.
- If nothing happens for **two minutes**, the Tab5 hands the radio back by itself. You will see "*session timed out — closed*". Open it again to carry on.

You come back to the frequency and mode you left. A trip through the radio's own menus can move the dial on its own — go into Hardware Tests -> Diagnostics to read the supply voltage, come out, and the radio is on 160 m whatever band you started on *(Randy N4OPI)*. On closing, the Tab5 puts the frequency and mode back if they moved, along with re-enabling I/Q mode and re-reading the noise floor.

> **Note - If you changed band on purpose**
>
> That gets put back too — the Tab5 cannot tell a band you chose in Band Configuration from one the menus changed behind your back. Set the band again from the Tab5 afterwards.

> **Tip - Where you land**
>
> A session opens on the **first** menu item, not on *Exit terminal* — even though *Exit terminal* is where the previous session left the highlight. That is deliberate: otherwise your first Enter would drop you straight back out.

14.4. Typing and editing a value

A **block cursor** shows where you are typing — in a field like one of the Messages, that is the difference between editing and guessing. *(Randy N4OPI)*

BS deletes leftward. Land on a value, backspace away the digits you don't want, and type the new ones — the cursor sits on the rightmost digit to begin with. In the browser both your Backspace and Delete keys do this.

> **Note - There is an on-screen keyboard**
>
> Tap the **keyboard button** in the top row and a full QWERTY appears along the bottom. It types straight into whatever field the radio has open — there is no separate text box to fill in first. Tap the keyboard button on the keyboard itself (bottom left) to put it away again.
>
> Useful on a Tab5 without the snap-on keyboard, which otherwise has no way to type a value at all. *(Randy N4OPI, Michael KZ4LY)*

> **Tip - Band config tables: E and D change a setting, and the arrows do not move you**
>
> In the **band config** table the Enable/Disable entries accept either ◀ ▶ *or* the letters **E** and **D** — type them straight in from the on-screen keyboard. Because those fields use the arrows to change the *value*, the arrows will not move you to the next column while you are sitting on one: step onto a numeric column first, then move across.
>
> That is the radio's own behaviour, not a Tab5 limitation, and it explains something that otherwise reads as the arrows being broken in tables. *(Stan Dye KC7XE.)*

> **Tip - Longer values are backspace-and-retype, not arrow-adjust**
>
> Short values — one or two digits — change with ◀ ▶. Anything longer, and anything inside a table, does not: on those the arrows do nothing at all, and in a table they move between columns, which is the radio's own intended behaviour.
>
> For those, use **BS** and type the value instead. `Max. PA voltage`, the band config columns, `CAT Config -> Timeout`, `System config -> TCXO frequency` and the Virtual U3S values all edit this way. *(Randy N4OPI worked this out from PuTTY and told us — it is how the radio has always behaved, not a Tab5 limitation.)*

4b. CW keyer speed

Asked for by Ralph Hellmig.

The radio's **CW keyer speed** is now a slider in the drawer, directly above IF calibration, and the same control is on the web page - so you can change it without walking into the radio's own menus.

It sets the speed over CAT and then **reads back what the radio actually took**. The radio is the authority here: it does not store the keyer speed itself, so the read-back is the only honest answer about where the speed ended up.

⛔ **A speed of 0 means a straight key**, not "no speed set" - the keyer is off and the radio keys straight through. That is a real setting, so the slider treats it as one.

14.5. What it does not do

- **It does not transmit.** Nothing here keys the radio.
- **It does not know what you changed.** If you alter something the panadapter displays — a filter, a band setting — the Tab5 finds out the same way it always does, from its next CAT poll.
- **It is the radio's interface, not ours.** What the menus contain, and what each item does, is documented in the QRP Labs QMX operating manual.

15. Troubleshooting

> **On the Tab5 itself:** swipe in from the right edge and tap **Need guidance?**. It lists these symptoms in plain words, highlights the ones it can see are happening right now, and opens this chapter at the matching section. See Getting Help.

15.1. Common Issues

Spectrum is flat, no signal

Symptoms: Top bar shows `Band: ---`, spectrum shows no activity.

Causes:

1. **Charge-only USB cable** — the #1 issue
2. QMX not powered on
3. QMX not responding to CAT (firmware too old)

Fix:

1. Try a **different USB cable** — use one you know works for data (USB stick, phone file transfer, etc.)
2. Power cycle the QMX (off 5 seconds, back on)
3. Power cycle the Tab5
4. Check QMX firmware version is v1.03.002 or newer, up to v1.04.010 (v1.04.011 is not recommended yet — see Quick Start Step 1)

If still flat after 10 seconds, proceed to Collecting Diagnostics.

QMX won't reconnect after a restart (v1.3.6 changes)

Before v1.3.6, a QMX that vanished mid-session — powered off and on, or present across a Tab5 restart — could stay invisible no matter how many times you restarted the radio, and only a Tab5 reboot recovered. That was two separate bugs, and v1.3.6 handles both:

1. **Fixed on the Tab5:** powering the QMX off while it streamed could jam the Tab5's USB port with the dead connection. The Tab5 now cleans up properly and additionally "replugs" the port by itself if a device sits unrecognized — turning the QMX off and back on reconnects within a few seconds, hands off.

2. **Lives in the QMX's own firmware** (reported to QRP Labs; present in v1.03.002 and v1.04.004 alike): after some Tab5 restarts, the QMX answers USB enumeration incorrectly until *it* is restarted. The Tab5 can't fix this one, but it now detects it and shows **"QMX USB is stuck - power-cycle the QMX to reconnect"** on screen. Do exactly that — radio off for a few seconds, back on — and it connects normally.

If you ever see **"USB stuck - power-cycle the QMX (reboot Tab5 if that fails)"** instead, the port ended up in a rarer state: try the QMX power cycle first; a Tab5 reboot is the fallback.

Spectrum signal is shifted/mirrored, or slides across the whole window as you tune

Symptoms: The signal isn't where it should be — it appears shifted, and turning the QMX's own VFO knob slides it across the *entire* 48 kHz window instead of just nudging it. Audio is often silent until you tune the signal back into the visible range.

Cause: The QMX never confirmed IQ mode for this session. Without it, the radio streams plain (non-IQ) audio instead of a properly centred baseband, which produces exactly this symptom.

Fix (v0.19.3+):

1. The panadapter automatically retries the IQ-mode handshake up to 4 times at connect, so this usually resolves itself within a second of the QMX showing up — no action needed.
2. If it still happens, a **red banner appears across the top of the screen** telling you immediately. When you see it:
    - Power-cycle the QMX (forces a fresh handshake on reconnect), **or**
    - Check the QMX's own **System Config -> IQ Mode** setting is enabled
3. On firmware older than v0.19.3, this failure was silent — only visible in the diagnostic log as `QMX IQ mode NOT confirmed`. Updating is the simplest fix.

The radio jumps to 160 m whenever the Tab5 connects (fixed in v1.16.10)

On v1.16.9 and earlier, every CAT link-up moved your radio. To show you a band list, the Tab5 reads it out of the QMX's own **Band config.** menus — and leaving those menus drops the radio on 160 m, in whatever mode the menus left behind. The Tab5 did not put it back. It happened on every connection, to everyone.

What it looks like: you set up on 20 m, the Tab5 connects, and the radio is on 1.837700 MHz in CW. If you run WSPR or a beacon it is worse than an annoyance — you transmit on a band you did not choose.

Update to v1.16.10, which saves the frequency and mode before the scan and puts them back afterwards, and only if the scan actually moved them.

> **Note - It is easy to blame the wrong thing**
>
> This was reported as a dead CAT link, and then blamed on the radio's Virtual U3S beacon — both wrong, and one operator switched off a beacon that was innocent. The giveaway is in the timing: the band menus flick past in well under two seconds, far faster than any beacon schedule.

QMX loses CAT connection after 1–2 minutes

Symptoms: Frequency/mode/BW stop updating, FT8 can't transmit.

Cause: QMX firmware too old or CDC-ACM driver timeout.

Fix:

1. Verify QMX firmware is v1.03.002 or newer, up to v1.04.010 (see Step 1 in Quick Start)
2. Try a shorter/higher-quality USB cable
3. Restart both devices

Settings will not save - callsign, grid and WiFi are forgotten

The device receives normally and shows stations, but nothing you enter sticks, and every attempt to transmit refuses with "*Set your callsign first*".

Almost always this means the panadapter was installed with the M5Launcher. The panadapter keeps settings in a partition of its own, defined in its own partition table. The launcher writes a different layout with no such partition, so the firmware has nowhere to store anything. Re-installing the same way reproduces it every time.

The cure: download the flasher zip from the Releases page, unpack it on your computer, run `flash.bat` (or `flash.command`) and type **E** at the flash-type prompt. That rewrites the partition table along with the firmware. You will re-enter your callsign, grid and WiFi afterwards.

If it still will not save after a clean flash, the start-up messages say so directly - look for `nvs_open failed` or `user_nvs init failed` in the first page of the serial log, and send it in.

WiFi won't connect

Symptoms: WiFi modal shows "Connecting..." for 30+ seconds, then fails.

Causes:

1. Wrong SSID or password
2. WiFi network uses 5 GHz (Tab5 prefers 2.4 GHz, but 5 GHz works)
3. WiFi network blocks client-to-client traffic (if you're trying to access the web UI from the same network)
4. Tab5 WiFi module issue (rare)

Fix:

1. Double-check SSID spelling and password (copy-paste if possible)
2. Try your **2.4 GHz WiFi network** instead (if you have dual-band)
3. Forget the network and reconnect: Settings -> WiFi -> tap network name -> Forget -> re-add
4. Restart Tab5
5. If still failing, grab the always-on **diagnostic log** — see Collecting Diagnostics

The web page won't change my network, or Preferred network is ignored

Symptoms (v1.16.3): you type a different network into **Network name** and save, and it goes back to the old one. Or you set **Preferred network** and the Tab5 keeps joining whichever network it used last.

Two separate faults, both fixed after v1.16.3.

- **Network name was discarded whenever the password field was blank** — even though that field tells you to leave it blank to keep the stored password. So switching to a network the Tab5 already knew did nothing, with no error message, and the page then showed the old name again.
- **Preferred network was only consulted after the Tab5 had failed to connect twice.** It worked when your usual network was down, and did nothing when both were up and the usual one answered first — which is the normal case. It also needed a reboot to take effect.

Fix: update the firmware. Then leave the password blank when you switch to a network the Tab5 already knows, and pick the preferred network from the list of remembered names now shown under both fields. Only a name on that list works.

If you must stay on v1.16.3: choose the network from the Tab5's own settings drawer instead of the web page.

WiFi connects, then stops working after a few minutes

Symptoms: WiFi (web UI, uploads) worked at first, then went dead after some minutes — while FT8 and radio control kept working normally. On older firmware the only fix was a reboot.

This is fixed in v0.20.0. The cause was a low-level lock-up in the link to the WiFi co-processor; the device now recovers from it automatically (dropping one packet, which is simply re-sent) instead of staying wedged. If you're on **v0.20.0 or later** and still see WiFi die permanently, it's a new issue — please capture the diagnostic log (below) and report it. If you're on an **earlier version**, update the firmware.

FT8 decoding is slow or stops

Symptoms: Decode list isn't updating, or only 1–2 decodes per slot.

First: check your firmware version. The long-standing version of this — the first slots of a session decoding well, then collapsing to a fraction and never recovering — was **fixed in v1.1.0**. The cause was audio being lost at the USB wire itself: the transfer pipeline held only 9 ms of queued audio, so any pause longer than that (a decode burst, redrawing the list) let the stream run dry, and roughly 170–350 ms of every slot went missing with no error reported anywhere. The pipeline now holds 320 ms. If you are on an earlier version, update; nothing else on this page will help.

On v1.1.0 or later, the likely causes are:

1. **The QMX stopped sending audio without disconnecting** — a trip through the radio's own menus can do this, because IQ mode is session state the QMX forgets. CAT keeps answering normally, so everything looks connected while the decode list sits empty. The Tab5 re-asserts IQ mode by itself after 30 s of silence; if it does not recover, power-cycle the QMX.
2. **A cable that is not carrying data properly** — intermittent is worse than dead, because the spectrum can still look alive.

3. **Your clock is wrong.** FT8 needs UTC within about a second; see *Time is wrong* below. Candidates stay high while decodes go to zero — that pattern is a timing problem, not a signal one.

What to do:

1. Check the clock in the bottom bar, and what it says it is synced from
2. Try the QMX on a different USB cable, connected directly — not through a hub
3. Turn WiFi off for a few minutes and see whether decodes change
4. Download the diagnostic log after a session (web UI **Files** menu -> **Diagnostic download** ↓) and post it on GitHub — it records the per-slot decode counts and capture timing, which says immediately which of the above it is

FT8 transmit doesn't key the QMX

Symptoms: You tap "Transmit", the modal closes, but nothing happens (no TX light on QMX, no tone on the air).

Causes:

1. QMX firmware doesn't support Kenwood `TX;` command (v1.03.002+ only)
2. CAT connection is dead (see QMX loses CAT connection above)
3. Tab5 is in simulation mode (deliberate safety feature)
4. QMX is stuck in a menu (rare)

Fix:

1. Check QMX firmware version (see Quick Start)
2. Try a manual CAT command via web API: `curl "http://<ip>/api/cat" -d '{"cmd": "TX;"}' -H "Content-Type: application/json"`
3. If that fails, you have a CAT issue (see above)
4. Check settings -> FT8 -> Simulation Mode is **off**
5. Restart both devices

Battery drains very fast

Symptoms: Battery goes from 100% to 0% in under 1 hour with normal use.

Causes:

1. Display brightness at 100%
2. WiFi on (drains ~30% more than without WiFi)
3. FT8 decoding (uses more CPU than panadapter alone)

4. Battery is old or defective

Fix:

1. Lower display brightness: Settings -> Display -> Brightness -> 50–70%
2. Turn off WiFi if you don't need the web UI: Settings -> WiFi -> off
3. FT8 decoding uses more power; this is expected
4. If battery still drains in <2 hours with low brightness + no WiFi, the battery may be failing

Time is wrong, FT8 doesn't work

Symptoms: Bottom bar shows wrong time, FT8 decoding says "0" and then stops.

Cause: Time is off by >1 second (FT8 is time-critical).

Fix:

1. Check if WiFi is connected (should auto-sync time via SNTP)
2. Manually set time: Settings -> Time Sync -> Set Manual Time -> enter UTC time
3. If no WiFi and no RTC set, the time will be wrong (see Time Sync)
4. Enable FT8-derived sync: Settings -> Time Sync -> Use FT8-Derived Sync (optional, helps if WiFi isn't available)

For POTA/portable operation without WiFi:

- Set the RTC **before you leave home** (Settings -> Time Sync -> Set Manual Time)
- The RTC holds time for 30–40 hours without power

Web UI stops responding, or the Tab5 restarts, after switching the QMX on during start-up

Symptoms: The web interface is unreachable even though the Tab5 is clearly running and on WiFi, and the unit may be more prone to restarting. Often after powering the Tab5 and the QMX together, or switching the radio on immediately after the Tab5.

Cause: In its first few seconds the Tab5 brings up WiFi, the web server and the decoders, all of which need a small pool of internal memory. A radio appearing over USB in the middle of that competes for the same pool, and the Tab5 can end up short of memory for the rest of the session. The web server is usually the first thing to suffer, because it needs memory to accept each new connection.

Fix: Restart the Tab5, wait until the spectrum is running and WiFi shows a network, and only then switch the QMX on. See Quick Start.

This is a firmware limitation, not a fault in your setup, and it is being worked on.

Web UI won't load

Symptoms: Browser says "Connection refused" or "Can't reach this page".

Causes:

1. Tab5 is not on WiFi
2. You're using the wrong IP address
3. WiFi network has client-isolation enabled
4. Web server crashed (rare)

Fix:

1. Check WiFi is **on** (settings drawer)
2. Check the **IP address** shown in settings (e.g., 192.168.1.50)
3. Use that IP in your browser: `http://192.168.1.50`
4. If still failing, try `http://192.168.1.50:80` explicitly
5. Restart the Tab5

The SD dot stays yellow, or the card crosses itself out (fixed in v1.16.9)

Symptoms: A card is in the slot and the **SD** dot is yellow rather than green, and stays that way while the Tab5 is clearly busy.

What the colours mean. From v1.16.3 the dot reports whether writes are actually reaching the card: **green** = something landed within the last 90 seconds, **yellow** = a card is mounted but nothing has lately, **grey with a stroke** = no card.

Causes:

1. **Nothing to write.** On a very quiet session there may genuinely be nothing due, and yellow is then correct and harmless. The diagnostic log normally writes every 30 seconds, so this is uncommon.
2. **The card driver was starved of memory — fixed in v1.16.9.** This was the common one, and on v1.16.8 and earlier it is what you are almost certainly seeing. **Update to v1.16.9.** If you cannot yet, switching **RX audio** off restores the card, at the cost of the speaker and headphone audio.
3. **A genuinely failing card or slot.** Rarer. The log will show mount-level errors (CRC, invalid response) rather than write failures.

Fix:

1. **Update to v1.16.9**, which fixes cause 2. Reseating, reformatting or replacing the card has never helped anybody and will not help here.
2. If you are staying on an older release, either switch **RX audio** off, or run with WiFi off (the normal POTA/SOTA case), where mirroring is continuous and flawless.
3. To tell cause 2 from cause 3, download the diagnostic log and look for `SDFAIL[slowopen] err=0x5` with a low `DMA free=` figure (cause 2) versus mount-time errors (cause 3).

> Cause 2 is fixed in v1.16.9. It was never your card.

What is actually happening (established v1.16.4, fixed v1.16.9). The card is never the problem in cause 2. The diagnostic log now shows the two lines together, in the same millisecond:

```
E dma_utils: esp_dma_capable_malloc(181): Not enough heap memory
W sd_arch: slow diag mirror failed 3 times running - backing off 60s -> 120s
```

The card is mounted and healthy; the write is **refused before it ever reaches it**, because the card driver cannot obtain a memory buffer of the kind it needs. The Tab5 then backs off 30 s → 60 s → 120 s → 240 s rather than hammering a request that cannot succeed, and the dot goes yellow because that is the truth.

The memory is spoken for by the audio path: v1.16.4's start-up ledger (see below) shows the audio output stage alone taking 46.5 KB of it. **This is why reformatting, reseating or replacing the card has never helped anybody.**

v1.16.9 fixes it. 24 KB of that audio total was a buffer sitting in exactly the memory the card driver competes for, with no need to be there; it now lives in external memory, where 14 MB is spare. Measured on the bench: with the old buffer in place and RX audio on, the card failed 40 seconds after boot with **87 bytes** of that memory left. With it moved, the same unit keeps writing.

Reading the start-up memory ledger (v1.16.4)

Every diagnostic log now opens with a table like this, one line per subsystem:

```
memledger: app_main entry          dma=157499 B (lblk 94208)
memledger: rx_audio_preopen (I2S)  dma= 99323 B (lblk 47104)
memledger: dsp_init                dma= 29923 B (lblk 25600)
memledger: rx_audio_init           dma= 15259 B (lblk 13824)
```

`dma` is the memory left that hardware can transfer directly to and from; `lblk` is the largest single unbroken piece of it. **The second number is usually what matters** — several things need one contiguous block, so they can fail while the total still looks adequate.

You do not need to act on it. It is there so that when something declines to start, the reason is already in the file you send.

The Tab5 restarts on its own while you are listening

Symptoms: The Tab5 reboots without warning, often while audio is playing and sometimes several times in a row. Nothing on screen explains it.

Most likely cause: the 5 V supply is dipping, not a firmware crash. The processor has its own brownout detector; when the rail falls below its threshold it stops the chip immediately. Nothing crashed — it lost power for a moment.

From v1.16.4 the Tab5 tells you. If the previous restart was a brownout, a message says so for twelve seconds at start-up, and the diagnostic log records `reset_reason=brownout`. Garbled sections in the log are a corroborating sign: those are writes cut off mid-write by the power loss.

What to check, in this order:

1. **The USB-C supply.** It wants a genuine 5 V 3 A source. A laptop port or a small phone charger is the usual culprit.
2. **The cable.** A thin or long one drops more than people expect at 2–3 A.
3. **That the battery is actually charging** — the Tab5 leans on it for peaks.

Loud audio is the heaviest moment, so brownouts often strike while you are listening or adjusting levels. That is a symptom of a marginal supply, not a reason to keep the volume down: a healthy supply runs the Tab5 at full volume without complaint.

> Raising **AGC Ceiling** to 1500 in v1.16.3 did not cause this, but it does let you reach a louder output than the old 800 limit allowed — which on a marginal supply can be what tips it over *(Samuel W7STF)*.

Bluetooth stays grey although the box is ticked

Symptoms: Bluetooth is enabled in the settings drawer, but the symbol in the bottom bar is grey rather than yellow, and no mouse or keyboard is ever found.

Cause: there was not enough memory to start it. Bluetooth needs a contiguous 24 KB of internal memory at start-up. If it is not there, the Tab5 **declines to start Bluetooth** rather than crash, and your setting is left untouched — which is why the box still reads ticked.

The diagnostic log says so plainly:

```
btmouse: internal heap still 21391 B after 60 s (need 24576)
         - NOT starting BLE this boot
```

What to do: restart the Tab5 **before** switching on RX audio, and let it settle. Audio is the largest consumer of that memory, so a start-up with audio already enabled is the case most likely to squeeze Bluetooth out.

> Known and measured. The margin is genuinely thin, and reducing the audio path's memory footprint is being worked on.

Settings disappear after restart

Symptoms: You set your callsign/grid/WiFi password, but after power cycle they're gone.

Causes:

1. **Not normal** — settings should persist to NVS (non-volatile storage)
2. Recent firmware update may have reset NVS
3. NVS is corrupted (very rare)

Fix:

1. Re-enter your settings
2. If they disappear again, download the diagnostic log (web UI **Files** menu -> **Diagnostic download** ↓) and report on GitHub
3. As a last resort: export your config via the web UI (**Files** menu -> **Config download** ↓), then do a full factory reset, then re-import

A Bluetooth mouse never connects

Symptoms: Bluetooth is enabled, you have restarted the Tab5 and put the mouse into pairing mode, and nothing happens. The Bluetooth symbol in the bottom bar keeps looking and never turns blue.

Most likely cause: the mouse is Bluetooth Classic, not Bluetooth Low Energy. Only Low Energy mice work. The Tab5's Bluetooth comes from a co-processor with no Bluetooth Classic radio in it, so a Classic mouse cannot be made to work by any firmware change, and it never appears to the Tab5 at all — the Tab5 listens only for mice that announce themselves the Low Energy way.

Check the box or the maker's specification for **Bluetooth 4.0 or later**. Most mice sold since about 2014 qualify. A dual-mode mouse works on its Low Energy channel.

If the mouse *does* connect — the symbol turns blue — but the pointer moves oddly or jumps to the edges of the screen, that is a different problem and worth reporting with a diagnostic log, since the Tab5 records what the mouse tells it about its own layout.

Flashing keeps failing, but the COM port is still listed

Symptoms: the flasher runs, the Tab5's COM port is visible in Windows Device Manager, and yet esptool fails again and again — often after the Tab5 has been running for a long time.

Fix: reboot the Tab5 and flash again. Do this before changing cables or reinstalling anything.

Why it happens: unlike most dev boards, the Tab5 has no separate USB-to-serial chip. The serial port you flash over is produced by the **ESP32-P4 itself**, so it only exists while the firmware is running. A busy or unhappy firmware can leave the port enumerated — Windows still lists it — while it no longer responds to the flasher's request to enter download mode. Restarting the Tab5 restarts that serial endpoint with it.

Reported by Samuel W7STF after a seven-hour session; a reboot fixed it immediately. If a reboot does *not* fix it, the usual causes are a charge-only USB-C cable or a serial monitor still holding the port.

Clearing a stuck configuration (web-based reset)

Symptoms: the panadapter behaves as if a setting is wedged — e.g. WiFi won't come up no matter what, or a stored value seems stuck — and re-entering settings doesn't help.

You can now reset from the web page, with no computer or flashing tool needed. In the web UI, open the **Miscellaneous** menu in the bottom bar and pick the scope:

- **Reset settings** — clears the app's stored settings (callsign, grid, filters, preferences) back to defaults. Your memory channels and logs are separate.

- **Reset WiFi** — clears just the Wi-Fi / network state, for when the connection is stuck.

Each choice asks for confirmation first, then the device reboots and clears the selected storage on the way back up. Export your config first (**Files** menu -> **Config download** ↓) if you want to restore it afterward.

15.2. Collecting Diagnostics

If you're stuck, capture a **diagnostic log**:

From the Web UI (Wireless)

The diagnostic log is **always on** — nothing to enable.

1. Reproduce the issue (let it sit for 30 seconds)
2. In the web UI bottom bar, open the **Files** menu and click **Diagnostic download** ↓ — it downloads both the live session log and the copy persisted from before the last reboot (useful if the device crashed or was power-cycled)
3. Alternatively, if a microSD card is inserted, pull `/qmx-panadapter/qmx-log.txt` from the card. With WiFi on it is written every 30 seconds, and those writes wait while a browser is watching the spectrum (never longer than three minutes) — so the card copy can lag by a few minutes. The web download above is always the complete one
4. Post the `.txt` file on GitHub Issues or the QRPLabs Groups.io thread

From Serial Console (Offline)

If WiFi isn't working:

1. Plug Tab5 into your computer with a **USB-C data cable**
2. Run: `tools/capture_serial_log.ps1` (Windows) or equivalent
3. Let it capture for 30 seconds while you reproduce the issue
4. Post the log file on GitHub Issues or the QRPLabs Groups.io thread

The log includes your firmware version, QMX firmware, callsign, grid, hardware revision, and all serial output from the system.

15.3. Still Stuck?

1. **Check the Quick Start** — most issues are covered there
2. **Read Settings** — verify your configuration

3. **Download the diagnostic log** (always on) — it often reveals the root cause
4. **Post on GitHub Issues or the QRPLabs Groups.io thread** — include:
    - Your symptoms (what you saw, what you expected)
    - Your hardware (Tab5 model, QMX/QMX+, antenna)
    - Your firmware versions (read from the About screen)
    - Your diagnostic log (if applicable)

The maintainer (OZ1LAV) and other users are very responsive to well-documented issues.

Next: Explore Build from Source if you want to modify the firmware.

16. Reference

Gestures

Gesture	Where	Effect
Tap	Spectrum / waterfall	Tune to tapped frequency (snapped)
Touch + drag (hold ~250 ms first)	Spectrum / waterfall	Cyan cursor snaps grid-to-grid; tunes on release
Fast one-finger horizontal swipe	Spectrum / waterfall, any zoom	Pan/"stroll" the view; retunes to the new centre on release
Double-tap	Spectrum / waterfall	Reset zoom and pan to ×1.0 / centred
Pinch (two fingers)	Spectrum / waterfall	Zoom ×1.0–×24.0
Two-finger drag	Spectrum / waterfall (zoomed)	Pan the zoomed window
Swipe → from left edge	Left edge strip	Cycle Panadapter → FT8/FT4/JS8 → WSPR
Swipe ← from right edge	Right edge strip	Open settings drawer
Tap right grip handle	Right edge	Open settings drawer (alternative)
Swipe →	Open drawer, or spectrum while drawer open	Close settings drawer
Swipe ↑ from bottom edge	Bottom edge strip	Open memory channel picker
Tap or swipe ↓	Top-bar item (Band/Mode/BW/Freq/Zoom)	Open that item's selector

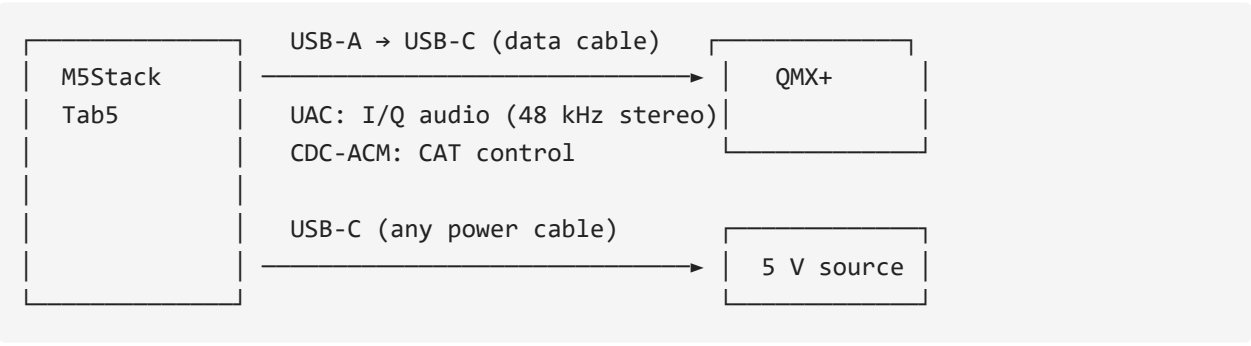
Gesture	Where	Effect
Touch and hold ~250 ms	FT8 decode list row	Dim preview at ~80 ms; full selection at 250 ms (scroll locks, drag moves highlight, lift to confirm)
Quick swipe	FT8 decode list	Scroll the list normally

Per-unit IF calibration

The QMX's +12 kHz IF injection varies slightly between units. If signals appear consistently shifted left or right of where the QMX is actually tuned, open the settings drawer → **IF calibration** slider (±100 Hz, persisted to NVS). Default 0. Reported by Ken KF0AYY, whose unit needed about −55 Hz to centre correctly.

Hardware

- **M5Stack Tab5** — ESP32-P4 v1.3 (ECO2), ST7121 or ST7123 5" 720×1280 MIPI-DSI touch display, 32 MB PSRAM, ESP32-C6 co-processor for WiFi
- **QRP Labs QMX or QMX+** — firmware 1.03.002 up to 1.04.010 (1.04 unlocks AM mode + Antenna Tune; 1.04.011 is not recommended yet)
- **USB-A → USB-C data cable** between Tab5 USB-A host port and QMX (full data, not charge-only)
- **USB-C power supply** for the Tab5 (5 V / 2 A or better, or internal battery)



What works without the QMX connected

Works without QMX	Needs QMX connected
UI, settings drawer, gestures	Spectrum + waterfall (no I/Q = flat line)
WiFi, web UI, screenshots	Top bar Band / Mode / BW
Callsign / grid / WiFi credentials	Tap-to-tune, mode popup (CAT writes)
Brightness, colour map, dB range	S-meter, FT8 decode/TX, QMX firmware readout

If the spectrum is flat and the top bar shows `---`, that is almost always the radio not being seen over USB — which loops back to the cable (or the QMX being off / in flash mode).

Spectrum signal shifted / mirrored, tunable across the whole 48 kHz window

This means the QMX never confirmed IQ mode for the session — without it the radio streams plain (non-IQ) audio instead of a centred baseband, so the signal appears at the wrong place and slides across the full 48 kHz window as you tune. As of v0.19.3 the Tab5 retries the IQ-mode handshake automatically at connect (up to 4 attempts) and this resolves itself almost every time; if it still fails after all retries, a red banner appears across the top of the screen telling you so immediately, instead of leaving you to figure it out from a shifted waterfall. If you see the banner, power-cycle the QMX (forces a fresh handshake on reconnect) or check the QMX's own **System Config** → **IQ Mode** setting.

Reporting hardware issues

Near the top of the boot log you will see:

```
I (xxxx) bsp_info: === TAB5 BSP INFO ===
I (xxxx) bsp_info: chip:     ESP32-P4 rev v1.3
I (xxxx) bsp_info: psram:    30 MB
I (xxxx) bsp_info: panel:    ST7123 (inferred from touch)
I (xxxx) bsp_info: touch:    ST7123 @ 0x55
I (xxxx) bsp_info: heap:     230.5 kB internal free, 28.80 MB PSRAM free
I (xxxx) bsp_info: idf:      v5.4.4
I (xxxx) bsp_info: firmware: v0.21.0
I (xxxx) bsp_info: ====================
```

When opening a hardware issue, paste this block — the `panel` and `touch` lines are the first thing needed to identify which Tab5 hardware revision you have.

> **Note on `reset_reason`:** a deliberate reset-button force-off returns `panic/exception`, not `external-pin/power-on`. A genuine firmware crash always prints a `Guru Meditation` register + backtrace dump immediately *before* the reboot. No backtrace = abrupt power-off, not a crash.

Appendix — Technical Reference

The sections below are the remaining, more technical parts of the same project README (build instructions, internal design notes, roadmap) — included here so the links above resolve, rather than dead-ending. Skip ahead if you only want operating instructions.

Appendix A — Build from source

Requires **ESP-IDF v5.4.4** — pinned; do not upgrade. ESP32-P4 v1.3 silicon requires `CONFIG_ESP32P4_REV_MIN_0=y` and CPU capped at 360 MHz, both baked into `sdkconfig`.

```
# Activate the IDF environment, then:
idf.py build flash monitor
```

Or with the `qmx` PowerShell helper — add to your `$PROFILE` and adjust the COM port:

```
function qmx {
    param([string]$cmd = "fm")
    if (-not $env:IDF_PATH) { & C:\esp\v5.4.4\esp-idf\export.ps1 }
    switch ($cmd) {
        "b"   { idf.py build }
        "f"   { idf.py flash }
        "m"   { python -m esp_idf_monitor -p COM3 build\qmx_panadapter.elf }
        "fm"  { idf.py flash; python -m esp_idf_monitor -p COM3
         build\qmx_panadapter.elf }
        "bfm" { idf.py build flash; python -m esp_idf_monitor -p COM3
         build\qmx_panadapter.elf }
    }
}
```

Exit monitor: `Ctrl+T` then `Ctrl+X`.

Appendix B — Under the hood

I/Q balance correction

The QMX presents I/Q audio with a small but measurable amplitude imbalance and quadrature error between channels. Without correction, every signal produces a mirror image across the centre frequency (~30 dB weaker), cluttering the waterfall on a busy band.

A blind adaptive Gram-Schmidt orthogonaliser runs sample-by-sample in the audio pipeline before the FFT. It maintains running estimates of:

- **DC offset** on each channel (τ = 1 s)
- **Power** in I and Q (τ = 200 ms)
- **Cross-product** I·Q (τ = 1 s) — non-zero cross-product means the channels are not orthogonal

Correction per sample:

```
i_out = i
q_out = (q − K_phi · i) × K_amp
```

No calibration step needed; the estimator converges on ambient band noise within a few seconds of any signal. A two-speed startup runs all alphas at 8× for the first 2 s of real signal after each reset (effective convergence ~125 ms), then drops to steady-state. Toggle in the settings drawer; re-enabling resets the estimator so it reconverges from a clean state.

Quirks and trade-offs

PI4IO expander must be initialised before display bring-up. The Tab5's PI4IO I/O expander holds `LCD_RST` and `TP_RST` low at chip power-on. Without explicit init, on a true cold boot the panel never comes out of reset and the DSI FIFO hangs. Soft resets mask this because the expander retains state across ESP32 resets. `display_init` calls `bsp_i2c_init()` + `bsp_io_expander_pi4ioe_init()` first, then waits 120 ms.

Patched components in `components/`. `espressif__usb_host_uac/` has `create_background_task = true` — required for UAC and CDC-ACM to coexist on the same USB host. `espressif__esp_lcd_touch_st7123/` makes `FW_VERSION_REG` the only mandatory register read (ST7121 NACKs the others and also adds a `max_touches > 10` bounds clamp). Do not replace either with the registry version without re-applying the patches.

LVGL software rotation (~50% FPS cost). The panel is natively portrait; landscape is achieved via `lv_display_set_rotation(LV_DISPLAY_ROTATION_90)`. Every flush goes through `rotate90_rgb565`. ~13 fps landscape vs ~22 fps portrait — acceptable for a panadapter. PPA hardware rotation conflicts with the USB host stack over DW-GDMA channels and silently kills UAC + CDC-ACM; do not set `CONFIG_LVGL_PORT_ENABLE_PPA=y`. A full native-portrait rewrite would recover the lost FPS but isn't planned — accepted as a permanent trade-off.

IDLE watchdog disabled. The LVGL rotation pipeline keeps CPU0 busy past the default watchdog window. `CONFIG_ESP_TASK_WDT_CHECK_IDLE_TASK_CPU0/CPU1` are off; the app-task watchdog (30 s) is still active.

Cross-thread CAT writes must go through the poll task. Writing CAT commands directly from the LVGL thread races the FA/MD/FW poll on the same CDC pipe — commands interleave and the QMX returns `?;`. Pattern: stash the request in a `volatile` and let `poll_task` drain it on its next cycle. See `s_pending_ssb_bw` / `cat_request_ssb_bandwidth()` in `cat.c`.

SSB filter bandwidth needs three coordinated writes. `MMSSB|Filter RX=<hz>;` commits the value (persists, shows in QMX menu); `MMSSB|Bandwidth=<hz>;` applies it live; `FW;` polling must be suspended while the width is pinned because reading the filter makes the QMX revert it. Dead ends: `FW<nnnn>;` CAT set returns `?;`; re-asserting the same mode digit does not reload the filter; `Bandwidth` write alone reverts on the next poll.

HTTPS needs mbedtls allocating from PSRAM, not internal RAM. The QRZ/eQSL upload features (v0.16.2) are this firmware's first-ever outbound HTTPS connections — SNTP, the only prior network client, is UDP. With the default `CONFIG_MBEDTLS_MEM_ALLOC_MODE=MBEDTLS_INTERNAL_MEM_ALLOC`, the TLS handshake's 16 KB+ buffers competed for the ~200 KB internal DRAM already under pressure from USB host/audio/FFT/LVGL and every connection failed with `ESP_ERR_HTTP_CONNECT`. Fixed by switching to `MBEDTLS_EXTERNAL_MEM_ALLOC` in `sdkconfig`, which moves those buffers into the 28 MB of free PSRAM instead.

Project layout

```
main/
  main.c                    app_main, task launch, orchestration
  display/display.c         BSP bring-up (PI4IO + LCD + touch)
  ui/ui.c                   LVGL widgets, touch handler, canvases
  ui/ft8_screen_view.c      FT8 decode list, touch-drag row selection
  ui/ft8_tx_modal.c         TX confirmation modal
  cat/cat.c                 USB CDC-ACM + Kenwood CAT
  audio/audio.c             USB UAC + ring buffer producer
  dsp/dsp.c                 FFT, spectrum mutex, DC blocker
  dsp/iq_balance.c          Gram-Schmidt I/Q correction
  ft8_tx.c                  FT8 TX engine (build/arm/run/abort)
  ft8_test.c                FT8 slot loop (RX decode / TX burst)
  ft8_qso.c                 Auto QSO state machine
  ft8_sim.c                 FT8 simulation mode: phantom-station practice QSOs
  ft8_status.c              Mutex-protected FT8 status string
  adif/adif_log.c           ADIF QSO logging
  adif/qrz_upload.c         QRZ Logbook API upload (one record per request)
  adif/eqsl_upload.c        eQSL.cc upload (batched, multiple records per request)
  rtc/rtc.c                 RX8130CE supercap RTC driver
  time_sync/time_sync.c     Time sync orchestrator
  render/render.c           10 Hz render task, EMA smoothing
  render/render_waterfall.c Waterfall scroll (double-height canvas)
  screenshot/screenshot.c   RGB565 framebuffer capture for /ss.bmp
  storage/settings.c        NVS-backed settings with debounced flush
  wifi/wifi.c               C6 co-processor WiFi + SNTP
  net/webserver.c           HTTP server + WebSocket
  util/cpu_stats.c          Idle %, frame rate and invalidation reporting
  util/diag_log.c           Diagnostic log ring buffer (vprintf hook)
```

Appendix C — Roadmap

The full per-version changelog — every release from v0.1.0 onward — lives in **docs/version-history.md**. This section tracks only what's planned next.

Next up

v1.20.2 is here. Next on the bench:

- **A full two-way JS8 QSO, watched from both ends.** Receiving, sending free text and calling CQ are all proven on air, and the automatic exchange **is implemented** - with automatic replies on, the Tab5 answers JS8 callers by itself and has since v1.17.0. What is missing is a complete exchange observed end to end, not the code to do it. An earlier version of this line said the exchange "is not written yet", which was wrong and was corrected publicly in the v1.18.0 notes after Gyula HA3HZ found four stations answered unattended in his own log.
- **JS8 Slow, Fast and Turbo on the air.** All four speeds are selectable from v1.19.0 and they transmit, but no contact has been made on anything but Normal. The Tab5's own side of Turbo is measured and fast enough; the radio's side is not.
- **CW page.** Canned-message CW TX memories first; decoded-CW display after (the QMX decodes internally — mirroring it over CAT looks cheap).

Longer term

- **CW decoder.** Goertzel-based, text scrolling under the spectrum. The QMX already does this internally — question is whether to mirror its output via CAT or run a parallel decoder on the Tab5.
- **Extended waterfall history.** PSRAM has room for several minutes of scrollback; two-finger drag to scrub through history.
- **QDX / QDX-M support.** Asked for repeatedly. The QDX is I/Q over USB like the QMX, so receive is close; transmit needs a different route, because the QDX has no equivalent of the CAT tone command the Tab5 uses to key FT8. See `docs/qdx-vs-qmx-comparison.md`.
- **RTTY mode.** See `docs/rtty-feasibility.md`. JS8 shipped in v1.17.0.
- **FM audio.** The demodulator handles CW, USB, LSB and - from v1.19.0 - AM. FM is still silent, so an FM signal means listening to it in USB or not at all.
- **DSP polish.** Noise reduction, auto-notch.

Appendix D — Related projects

- **Video demo** by Bruce N9JCV — a hands-on walkthrough of the software, filmed on his phone
- DX-FT8 by Barb (WB2CBA) — open-hardware FT8 tablet transceiver; an inspiring reference for a similar use-case
- `qrp_companion` by Zhenxing Han (N6HAN) — Tab5 companion for QMX with audio + CAT; source of the polling audio task pattern and battery readout approach
- `ft8_lib` by Karlis Goba — FT8 encoder/decoder vendored as `components/ft8_lib`

Glossary

Common terms and acronyms used in this guide:

Term	Meaning
ADIF	Amateur Data Interchange Format — the standard log-file format for QSO records (exported by the panadapter, uploaded to QRZ/eQSL)
AM	Amplitude Modulation — a voice mode. Receive only on the QMX+, and only on firmware 1_04 and later
ARRL	American Radio Relay League — the US national amateur radio society, which runs Logbook of The World
CAT	Computer-Aided Transceiver — radio control protocol (Kenwood-style commands via serial/USB)
CDC-ACM	Communications Device Class / Abstract Control Model — USB standard for serial ports
CQ	General call to any station (not directed at anyone specific)
CW	Continuous Wave — Morse code mode
dBm	Decibel-milliwatts — absolute signal power (0 dBm = 1 mW); used on the spectrum scale and S-meter
Directed message	A JS8 message aimed at one station, shown with the sender's callsign
DSP	Digital Signal Processing — mathematical signal analysis and filtering
DT	In the decode list, how far a station's transmission sits from the slot boundary, in seconds. It measures THEIR clock against yours, so it stays blank until at least two separate stations have been heard
DX	A distant station, or distance worked. **BEST DX** on the WSPR page is the furthest station that heard you, or that you heard, this session
DXCC	DX Century Club — the ARRL award programme, and by extension its list of ~340 "entities" (countries plus separately-counted islands and territories)

Term	Meaning
eQSL	Electronic QSL — online service for confirming and exchanging QSO records
FFT	Fast Fourier Transform — algorithm that converts time-domain audio into a frequency spectrum
Free text	A plain typed message rather than a fixed exchange. JS8 carries it; FT8 and FT4 fit only 13 characters
FT8 / FT4	Digital modes for weak-signal HF contacts, trading a short fixed exchange in 15-second (FT8) or 7.5-second (FT4) slots. Both decode and transmit on the device.
GFSK	Gaussian Frequency-Shift Keying — the modulation FT8, FT4 and JS8 all use
GNSS / GPS	A satellite navigation receiver, which gives the Tab5 UTC and the date with no internet - an M5Stack Unit GPS on PORT.A, a Module GPS on the 30-pin bus, or a GPS-equipped QMX+ answering over CAT
GPIO	General-Purpose Input/Output — microcontroller pins for digital signals
Heartbeat	The periodic `HB <call> <grid>` a JS8 station sends to say it is listening
I2C / SPI	Serial communication protocols for connecting peripherals (sensors, displays, etc.)
IF	Intermediate Frequency — the QMX presents the VFO signal at a +12 kHz offset in baseband
IQ	In-phase / Quadrature — stereo representation of RF signals (real + imaginary parts)
JS8	A keyboard mode built on FT8's modulation that carries free text, so stations hold short conversations. Decoded and transmitted on the device, as a sub-mode of the FT8/FT4 page
LDPC	Low-Density Parity-Check — the error-correcting code that recovers a damaged FT8, FT4 or JS8 message
LoTW	Logbook of The World — ARRL's online QSO-confirmation service

Term	Meaning
LSB / USB (mode)	Lower / Upper Sideband — the two SSB voice modes (note: "USB" also means Universal Serial Bus, below)
LVGL	Light and Versatile Graphics Library — open-source embedded UI toolkit used for the display
Maidenhead / grid	The locator system amateurs use for position, e.g. `JO65`. Four characters is a ~100 km square, six is finer. FT8 and WSPR both carry one
NMEA	The sentence format a GNSS receiver speaks - what the Tab5 reads for time, date and position
NVS	Non-Volatile Storage — persistent memory on the ESP32 (survives power cycles)
PA	Power Amplifier — the radio's final transmit stage. **Max. PA voltage** sets how hard it is driven, and so how much power comes out; see WSPR and Calibrate Power
OTA	Over-The-Air update - the Tab5 fetching and installing new firmware over WiFi, with no cable
Pileup	Everyone calling you at once. The Tab5 collects them in a list so you can work them in turn
POTA	Parks on the Air — portable operating activity from designated parks
PSK Reporter	A worldwide database of who heard whom. The panadapter can send it your decodes, and reads it back to show who has heard **you**
PSRAM	Pseudo-SRAM — extra RAM on the Tab5 (used for large buffers like waterfall history)
QMX / QMX+	QRP Labs HF transceiver — the radio this panadapter controls and receives audio from
QRP	Low-power operating, conventionally 5 W or less. QRP Labs, who make the QMX, are named for it
QRZ	QRZ.com Logbook — online logbook and callsign service for uploading QSOs

Term	Meaning
QSO	Radio contact / conversation between two stations
RBN	Reverse Beacon Network — automated receivers ("skimmers") that continuously report the CW and digital signals they hear. One of SelfSpotter's three sources
RIT	Receiver Incremental Tuning — shifts what you *hear* without moving what you would *transmit* on. The QMX has RIT but no XIT
RTC	Real-Time Clock — battery-backed timer on the Tab5 (keeps time during power-off)
Slot	The fixed window a digital mode transmits in - 15 s for FT8 and JS8, 7.5 s for FT4. Every station starts together, which is why the clock must be right
SNR	Signal-to-Noise Ratio — signal strength relative to the noise floor; the FT8/FT4 signal report
SNTP	Simple Network Time Protocol — synchronizes the system clock via WiFi/internet
SOTA	Summits on the Air — portable operating activity from mountain summits
SSB	Single Sideband — the voice-mode family (USB / LSB)
SSID	The name of a WiFi network
STFT	Short-Time Fourier Transform — sliding-window FFT used to build the waterfall
SWR	Standing Wave Ratio — antenna impedance matching metric (1.0 = perfect)
TX / RX	Transmit / Receive — keying the radio and listening
UAC	USB Audio Class — standard for streaming audio over USB
USB	Universal Serial Bus — physical connector and protocol (carries both audio and CAT commands). In a radio context, "USB" can also mean Upper Sideband — see SSB.
UTC	Coordinated Universal Time — timezone-independent time standard for FT8 slot alignment

Term	Meaning
VFO	Variable Frequency Oscillator — the radio's tuning dial / frequency setting
WSJT-X	The desktop program most FT8 and WSPR operators use. The panadapter does the same job without a PC, and its decodes and reports are meant to be comparable
WSPR	Weak Signal Propagation Reporter, said "whisper" — a beacon mode carrying only callsign, grid and power. Nobody replies; stations worldwide report hearing you, so you learn where your signal actually goes. See WSPR

Contributing

This is a solo project — all coding is done by the author (OZ1LAV) at his own pace, and **pull requests are not being accepted right now**. Bug reports, feature requests, and field reports are very welcome, though, via GitHub Issues or the QRPLabs Groups.io thread. See CONTRIBUTING.md for details.

Appendix E — License

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Relicensed from MIT on 2026-10-09. The reason was JS8: its coding tables exist only inside JS8Call, which is GPL-3, and a permissive licence put them out of reach. Nothing about using or flashing this firmware changes — the obligation falls on anyone DISTRIBUTING a modified build, who must publish their source. Everything this project depends on is one-way compatible: ESP-IDF and the Espressif components are Apache-2.0, ft8_lib and LVGL are MIT, and no proprietary binary is linked (the ESP32-P4 has no radio — WiFi lives on a separate chip over esp_hosted).